

Lecture 1 - Introduction

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Why Study Microeconomics?

Resources are **scarce**, but **wants** are unlimited.

Economics tries to answer how people balance trade-offs

People respond to incentives

People make decisions on the margin

Why Study Microeconomics?

Trade-offs

- How do you chose to spend your (scarce) time?
- How do you spend your (scarce) money?

Incentives

- How will people respond to a gas tax? (rebound-effect)
- Why are B-cycles always broken? (tragedy of the commons)

On the margin

- Do I watch one more episode or go to bed?
- How much pollution should we allow?

Three Tools of Microeconomic Analysis

Nearly all microeconomic models rely on just three key analytical tools:

1. Constrained optimization
2. Equilibrium analysis
3. Comparative statics

Throughout this course, we will use these three tools to analyze microeconomic problems.

Any model requires us to specify which variables will be taken as given and which will be determined by the model.

- An **exogenous** variable is a variable that is taken as given in the model.
- An **endogenous** variable is a variable that is determined by the model.

Try It Yourself

Suppose you create a model to tell you what your estimated grade will be, based on how much you study, and how much you sleep the night before.

What are the **exogenous** and **endogenous** variables?

Solution

Suppose you create a model to tell you what your estimated grade will be, based on how much you study, and how much you sleep the night before.

What are the **exogenous** and **endogenous** variables?

- The amount of studying and amount of sleep are **exogenous** variables.
- Your estimated grade is an **endogenous** variable.

Try It Yourself

Suppose you create a model to tell you how much money you will earn after you graduate, depending on your college major and on the estimated unemployment rate on the day that you graduate. What are the **exogenous** and **endogenous** variables?

Solution

Suppose you create a model to tell you how much money you will earn after you graduate, depending on your college major and on the estimated unemployment rate on the day that you graduate.

What are the **exogenous** and **endogenous** variables?

- Your college major and unemployment rate are **exogenous** variables.
- Your post-graduate earnings are an **endogenous** variable.

Economic models: A formal description of a problem being addressed.

Examples:

- How the drought in California might affect the price of coffee in the United States.
- How an individual makes the decision of whether to attend college, vocational school, or neither.
- How a man or woman chooses their spouse/partner.

Calculus Aside

This class is going to require you to do a lot of algebra and take a lot of partial derivatives. People in this class are coming in with different skill levels, which is okay!

The next lecture will be a calculus review and you'll review problems in your first recitation. So you will have time to practice. If you don't spend the time now remembering how to take them, it will be difficult to succeed in the course.

Constrained optimization

Constrained optimization is used when a decision maker seeks to make the optimal choice, taking into account any possible limitations or restrictions on choices

Example

An individual seeks to maximize, through their choice of Friday night activities, their happiness.

- An **objective function** is the relationship that the decision maker seeks to optimize (e.g. maximize their Friday night).
- The decision maker has to take into consideration their **constraints** (e.g. the individual only has \$20 to spend).

Constrained Optimization

A farmer needs to build a rectangular fence for her sheep. She has F feet of fence, and is able to choose the dimensions (L and W) of her pen. Her goal is to maximize the area of the pen.

What is the objective function and what is the constraint?

Constrained Optimization

A farmer needs to build a rectangular fence for her sheep. She has F feet of fence, and is able to choose the dimensions (L and W) of her pen. Her goal is to maximize the area of the pen.

What is the objective function and what is the constraint?

- The farmer's **objective function** is the function for the area of the pen, $L*W$.
- The farmer's **constraint** is that they only have F feet of fence.

Constrained Optimization

Try It Yourself

A baker has 8 hours in a day, during which he can bake cakes and brownies. If he bakes a cake, he can sell it for \$10, and it will take him 2 hours. If he bakes a tray of brownies, he can sell it for \$8, and it takes him 1 hour. However, the baker only has the supplies to bake at most 4 trays of brownies and 3 cakes in a given day.

How would you set up a problem to solve how many brownies and cakes should the baker bake?

Constrained Optimization

Solution

Let C = cakes and B = brownies.

Objective:

$$\max 10C + 8B$$

Subject to

$$2C + B \leq 8$$

$$C \leq 3$$

$$B \leq 4$$

$$C, B \geq 0$$

Interpretation:

- Objective: maximize the baker's profit.
- Constraints: limited time and supplies.

Marginal Reasoning and Constrained Optimization

Continuing our example of the baker maximizing their profit. Suppose the baker is currently baking 3 cakes and 2 trays of brownies. Their profit is \$46.

What would his profit be if he instead baked 2 cakes, and 4 trays of brownies?

Marginal Reasoning and Constrained Optimization

Continuing our example of the baker maximizing their profit. Suppose the baker is currently baking 3 cakes and 2 trays of brownies. Their profit is \$46.

What would his profit be if he instead baked 2 cakes, and 4 trays of brownies?

- He would gain \$16 from the two additional brownie trays and lose \$10 for the cake.
- The additional profit gained, \$6 reflects the *marginal* impact of the decision.

Margin = “Of Another”

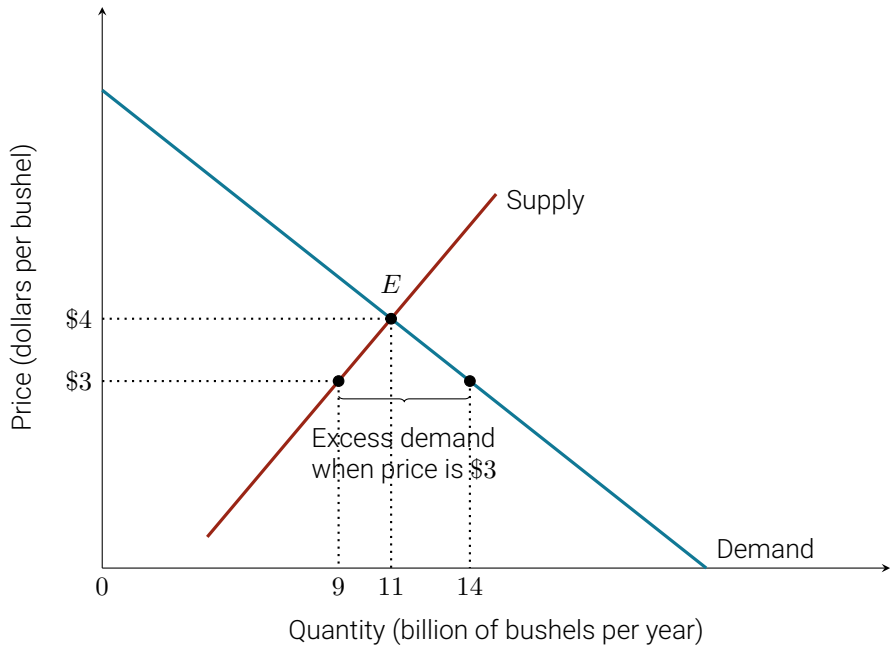
In economics, you will *always* hear the word ‘marginal’. When you hear it, you can always replace it with the phrase ‘of another’

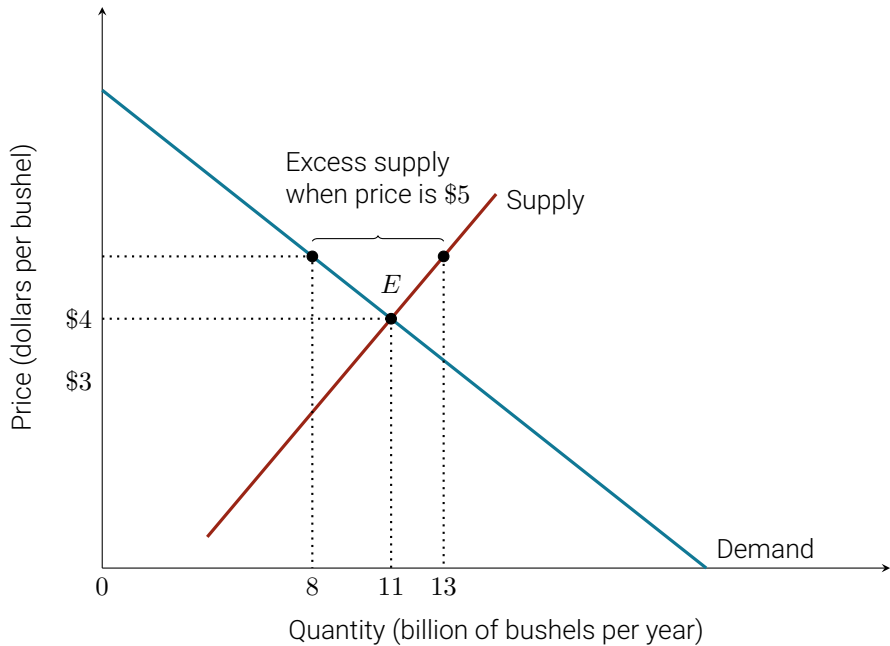
- So for example, if I say “what is the marginal benefit of studying?”, I mean “what is the benefit of studying another hour?”

Equilibrium Analysis

An **equilibrium** in a system is a state or condition that will continue indefinitely as long as exogenous factors remain unchanged.

In a competitive market, equilibrium occurs when the price is such that the quantity supplied by producers is equal to the quantity demanded by consumers.





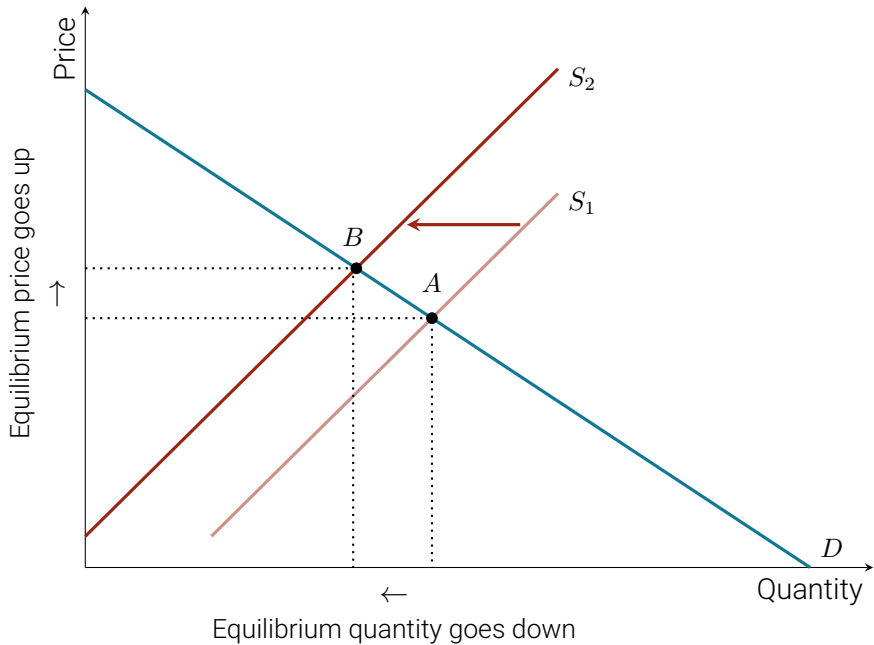
Comparative statics analysis is used to examine how a shock to a system will affect another variable in an economic model.

- This is really useful since it let's you decide whether or not to take an action by predicting how it affects variables you care about

Comparative statics analysis can be applied to constrained optimization or to equilibrium analysis.

Consider the market for pistachios:

- We can see what will happen to the price and quantity of pistachios in the market when the supply curve shifts.
- The shift in the supply curve represents an exogenous shock.



Positive and Normative Analysis

Positive analysis attempts to explain how an economic system works or to predict how it will change over time.

- Explanatory questions, such as "What has happened?"

Normative analysis asks prescriptive questions.

- Involves value judgments.

This class will focus on positive analysis.

Positive and Normative Analysis

Try It Yourself

Which of the following is a positive statement?

- A) Rent controls (a rent ceiling) will cause a housing shortage.
- B) It would be good for the U.S if we stop trading with China.
- C) The minimum wage should be raised to make poor people better off.
- D) Businesses need to pay their employees a living wage.

Positive and Normative Analysis

Solution

Which of the following is a positive statement?

- A) Rent controls (a rent ceiling) will cause a housing shortage.
- B) It would be good for the U.S if we stop trading with China.
- C) The minimum wage should be raised to make poor people better off.
- D) Businesses need to pay their employees a living wage.

Math Review

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Math Review Topics

- Limits
- Derivative Definition
- Derivative Rules
- Partial Derivatives

The limit of $f(x)$ as x approaches a is written as:

$$\lim_{x \rightarrow a} f(x)$$

- The behavior of the function as its input approaches a
- In certain cases, can be evaluated by plugging in $x = a$
- $\lim_{x \rightarrow a} c = c$ for a constant c

Limits

The limit from the left and right are

→ $\lim_{x \rightarrow a^-} f(x) = A$ "as x approaches a from the left ($-\infty$)"

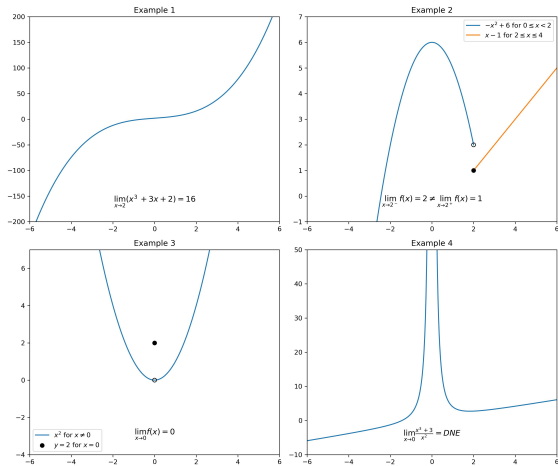
→ $\lim_{x \rightarrow a^+} f(x) = A$ "as x approaches a from the right (∞)"

If $\lim_{x \rightarrow a^-} f(x) = A = \lim_{x \rightarrow a^+} f(x)$ then the limit exists and we write

$$\lim_{x \rightarrow a} f(x) = A$$

- Note: the limit does not exist if $A = \pm\infty$, even if the left and right limit both equal $\pm\infty$

Limits



$$\lim_{x \rightarrow a} f(x) = L \quad \lim_{x \rightarrow a} g(x) = M$$

- **Constant Multiple Rule:** $\lim_{x \rightarrow a} [af(x)] = aL$
- **Sum/Difference Rule:** $\lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm M$
- **Product Rule:** $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M$
- **Quotient Rule:** $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{L}{M}, \quad M \neq 0$
- **Power Rule:** $\lim_{x \rightarrow a} [(f(x))^n] = L^n, \quad n > 0$

Definition of a Derivative

The derivative of f is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

If this limit exists, then we say f is differentiable at x .

Derivative Notation

All of these mean “take the derivative of f with respect to x ”:

→ $f'(x)$

→ $\frac{df(x)}{dx}$

→ $\frac{d}{dx}f(x)$

→ $\frac{dy}{dx}$ if $y = f(x)$

→ y' (sometimes)

→ $\dot{y}$ (particularly in Macro)

Derivative Rules

Let $f(x)$ and $g(x)$ be differentiable functions:

- **Derivative of a Constant:** $f(x) = c, f'(x) = 0$
- **“Power Rule”:** $f(x) = x^n, f'(x) = nx^{n-1}$
- **Sum/Difference Rule:** $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$
- **Product Rule:** $\frac{d}{dx}[f(x) \cdot g(x)] = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
- **Quotient Rule:** $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
- **Chain Rule:** $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

Derivatives Practice Problems

$$f(x) = 2x$$

Derivatives Practice Problems – Solution

$$f(x) = 2x$$

By the power rule ($f(x) = 2x^1$, so $n = 1$):

$$f'(x) = 2$$

Derivatives Practice Problems

$$f(x) = 3x^2$$

Derivatives Practice Problems – Solution

$$f(x) = 3x^2$$

By the power rule ($n = 2$):

$$f'(x) = 3 \cdot 2x^{2-1} = 6x$$

Derivative Rules for Exponents and Natural Log

$$\rightarrow f(x) = e^x, f'(x) = e^x$$

$$\rightarrow f(x) = \ln(x), f'(x) = \frac{1}{x}$$

Useful review of the properties of **exponents** and **logs**.

Derivatives Practice Problems

$$f(x) = \ln(e^x)$$

Derivatives Practice Problems – Solution

$$f(x) = \ln(e^x)$$

Simplify first, using the log property $\ln(e^x) = x$:

$$f(x) = x \implies f'(x) = 1$$

(Or, via the chain rule directly: $f'(x) = \frac{1}{e^x} \cdot e^x = 1$ – same answer either way.)

Derivatives Practice Problems

$$f(x) = x^2 \ln(x)$$

Derivatives Practice Problems – Solution

$$f(x) = x^2 \ln(x)$$

Use the product rule, with $u = x^2$ ($u' = 2x$) and $v = \ln(x)$ ($v' = 1/x$):

$$f'(x) = 2x \ln(x) + x^2 \cdot \frac{1}{x} = 2x \ln(x) + x$$

Derivatives Practice Problems

$$f(x) = e^{2x}$$

Derivatives Practice Problems – Solution

$$f(x) = e^{2x}$$

Use the chain rule, with outer function e^u and inner function $u = 2x$ ($u' = 2$):

$$f'(x) = e^{2x} \cdot 2 = 2e^{2x}$$

Higher-order Derivatives

The second-order derivative is

$$f''(x) = \frac{d}{dx} f'(x)$$

We could keep differentiating as long as the last derivative is differentiable. Notation:

- $f''(x)$
- $\frac{d^2 f(x)}{dx^2}$
- $\frac{d^2}{dx^2} f(x)$
- $\frac{d^2 y}{dx^2}$ if $y = f(x)$
- y'' (sometimes)

Partial Derivatives

The function $z = f(x, y)$ takes two inputs and outputs a third. We can evaluate what happens to that output when we change x or y (but not both).

- $\frac{\partial z}{\partial x} =$ “the derivative of z with respect to x ”
 - The slope of f in the cardinal direction of x (e.g. “north-south”)
 - The tangent of f in the x direction at a point (x_0, y_0)
- $\frac{\partial z}{\partial y} =$ “the derivative of z with respect to y ”
 - The slope of f in the cardinal direction of y (e.g. “east-west”)
 - The tangent of f in the y direction at a point (x_0, y_0)

Partial Derivatives

All of these mean “take the partial derivative of f with respect to x ”:

- $f_x(x, y)$
- $\frac{\partial f(x, y)}{\partial x}$
- $\frac{\partial}{\partial x} f(x, y)$
- $\frac{\partial z}{\partial x}$ if $z = f(x, y)$

Partial Derivatives

Interpret $\frac{\partial z}{\partial x}$ as “what happens to z if I change x , holding y constant.” Likewise for $\frac{\partial z}{\partial y}$.
Suppose $z = x + y + xy$, find $\frac{\partial z}{\partial x}$.

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}[x + y + xy] = \frac{\partial}{\partial x}x + \frac{\partial}{\partial x}y + \frac{\partial}{\partial x}xy = 1 + 0 + y \cdot 1 = 1 + y$$

Partial Derivatives Practice Problems

$$f(x, y) = x^2 + y^2 + x^2y^2$$

Partial Derivatives Practice Problems – Solution

$$f(x, y) = x^2 + y^2 + x^2y^2$$

Holding y constant, differentiate term-by-term with respect to x :

$$\frac{\partial f}{\partial x} = 2x + 0 + 2xy^2 = 2x(1 + y^2)$$

Holding x constant, differentiate term-by-term with respect to y :

$$\frac{\partial f}{\partial y} = 0 + 2y + 2x^2y = 2y(1 + x^2)$$

Partial Derivatives Practice Problems

$$f(x, y) = x^2y + \ln(x)y^3$$

Partial Derivatives Practice Problems – Solution

$$f(x, y) = x^2y + \ln(x)y^3$$

Holding y constant, differentiate with respect to x (treating y, y^3 as constants):

$$\frac{\partial f}{\partial x} = 2xy + \frac{1}{x}y^3$$

Holding x constant, differentiate with respect to y (treating $x^2, \ln(x)$ as constants):

$$\frac{\partial f}{\partial y} = x^2 + 3\ln(x)y^2$$

Properties of Exponents

Let a and b be real numbers and m and n be integers. Then the following properties of exponents hold, provided that all of the expressions appearing in a particular equation are defined:

$$\rightarrow a^m a^n = a^{m+n}$$

$$\rightarrow (a^m)^n = a^{mn}$$

$$\rightarrow (ab)^m = a^m b^m$$

$$\rightarrow \frac{a^m}{a^n} = a^{m-n}, a \neq 0$$

$$\rightarrow \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}, b \neq 0$$

$$\rightarrow a^{-m} = \frac{1}{a^m}, a \neq 0$$

$$\rightarrow a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$\rightarrow a^0 = 1, a \neq 0$$

$$\rightarrow a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

Properties of Natural Log

Note that logs are only defined for positive values of x :

$$\rightarrow \ln(xy) = \ln(x) + \ln(y)$$

$$\rightarrow \ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$

$$\rightarrow \ln(x^y) = y \cdot \ln(x)$$

$$\rightarrow \ln(e^x) = x$$

$$\rightarrow e^{\ln(x)} = x$$

$$\rightarrow \ln(e) = 1$$

$$\rightarrow \ln(1) = 0$$

$$\rightarrow \ln(0) = \textit{undefined}$$

Lecture 2 - Demand and Supply

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Overview

In this lecture, we will review the following:

1. Supply and demand
2. Market equilibrium, both graphically and numerically
3. Elasticities

Demand, Supply and Market Equilibrium

A *perfectly competitive market* comprises a large number of buyers and sellers.

- Buyers and sellers act as *price takers* in these markets.
- Sellers all produce identical products.

Demand, Supply and Market Equilibrium

Try It Yourself

Which of the following would most likely be considered a perfectly competitive market?

- The craft beer industry
- The grill industry
- The soybean industry

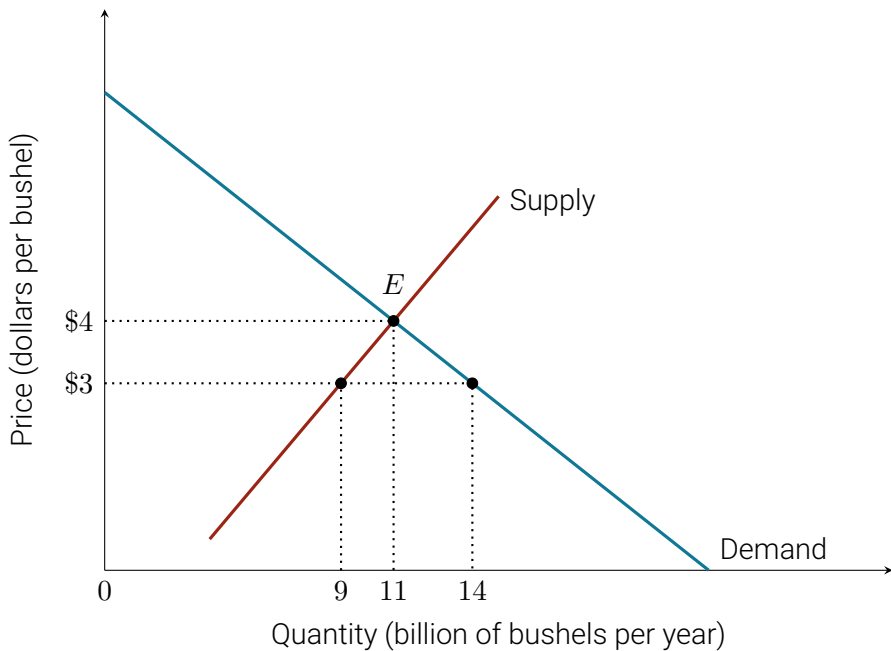
Demand, Supply and Market Equilibrium

Solution

The **soybean industry** is most likely to be perfectly competitive.

- Soybeans are a homogeneous commodity – one farmer's soybeans are essentially identical to another's.
- There are a large number of soybean producers and buyers, none of whom can influence the market price on their own.

Craft beer and grills, on the other hand, are **differentiated products**. Firms compete on brand, quality, and features, so they are not price takers.



Demand Curves

The market demand curve tells us the quantity of corn that buyers are willing to purchase at different prices.

- **Derived demand** is derived from the production and sale of other goods. (e.g. computer chips are not purchased directly, but are used as an input for computers/phones.)
- **Direct demand** is demand that comes directly from consumers.

The **law of demand** is the inverse relationship between the price of a good and the quantity demanded of that good.

Calculating Quantity Demanded

Try It Yourself

Suppose that the demand curve for Chaco's sandals in Boulder is given by $Q_{chaco}^D = 40,000 - 500P_{chaco}$. What is the quantity demanded of Chaco's if the price is \$60?

Calculating Quantity Demanded

Solution

Plug $P_{chaco} = 60$ into the demand curve:

$$Q_{chaco}^D = 40,000 - 500(60) = 40,000 - 30,000 = 10,000$$

At a price of \$60, the quantity demanded of Chaco's sandals is **10,000 pairs**.

Aggregating Demand

Consider if you have two consumers of a good. Each consumer's demand curve tells us *at a given price*, how many units will they buy.

How do we figure out the aggregate demand curve?

- *At a given price*, add up each consumer's quantity demanded

Aggregating Demand

Demand cannot be negative. So when we state demand as

$$Q_{chaco}^D = 40,000 - 500P_{chaco}$$

we are actually saying

$$Q_{chaco}^D = \begin{cases} 40,000 - 500P_{chaco} & \text{if } P_{chaco} \leq 80 \\ 0 & \text{otherwise.} \end{cases}$$

Aggregating Demand

Example

Suppose we have two consumers, A and B. Suppose that

$$Q_A^D = \begin{cases} 20 - 2P & \text{if } P \leq 10 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^D = \begin{cases} 21 - 3P & \text{if } P \leq 7 \\ 0 & \text{otherwise.} \end{cases}$$

What's the aggregate demand curve?

Aggregating Demand

Example

$$Q_A^D = \begin{cases} 20 - 2P & \text{if } P \leq 10 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^D = \begin{cases} 21 - 3P & \text{if } P \leq 7 \\ 0 & \text{otherwise.} \end{cases}$$

At a price of $P \leq 7$, both consumers will purchase good, so aggregate demand is given by

$$Q_{mkt}^D = Q_A^D + Q_B^D = (20 - 2P) + (21 - 3P) = 41 - 5P$$

Aggregating Demand

Example

$$Q_A^D = \begin{cases} 20 - 2P & \text{if } P \leq 10 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^D = \begin{cases} 21 - 3P & \text{if } P \leq 7 \\ 0 & \text{otherwise.} \end{cases}$$

At a price $7 < P \leq 10$, only consumer A will consume ($Q_B^D = 0$), so aggregate demand is given by

$$Q_{mkt}^D = Q_A^D + Q_B^D = (20 - 2P) + 0 = 20 - 2P$$

Aggregating Demand

Example

$$Q_A^D = \begin{cases} 20 - 2P & \text{if } P \leq 10 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^D = \begin{cases} 21 - 3P & \text{if } P \leq 7 \\ 0 & \text{otherwise.} \end{cases}$$

At a price $P > 10$, no one consumes anything, so

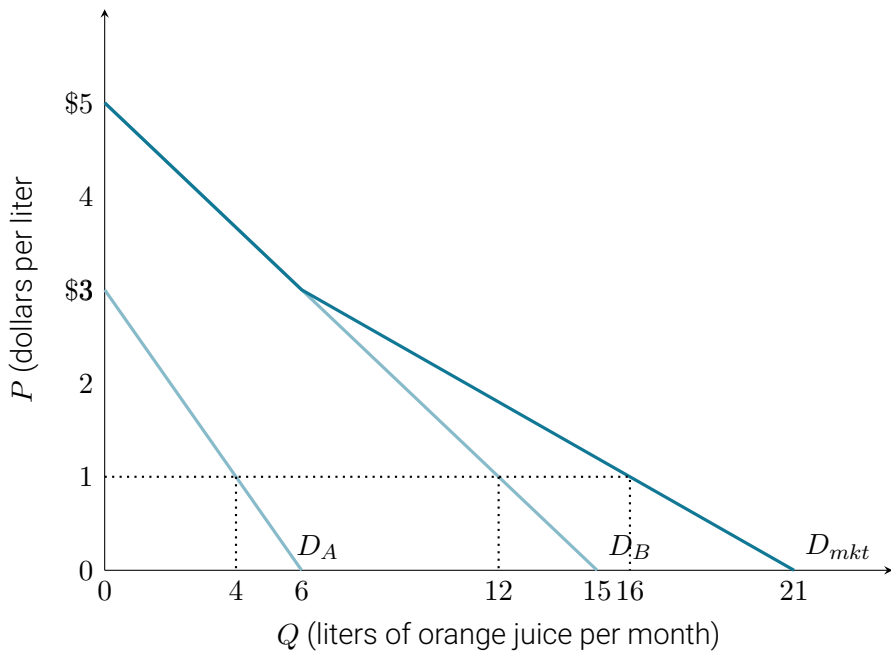
$$Q_{mkt}^D = Q_A^D + Q_B^D = 0 + 0 = 0$$

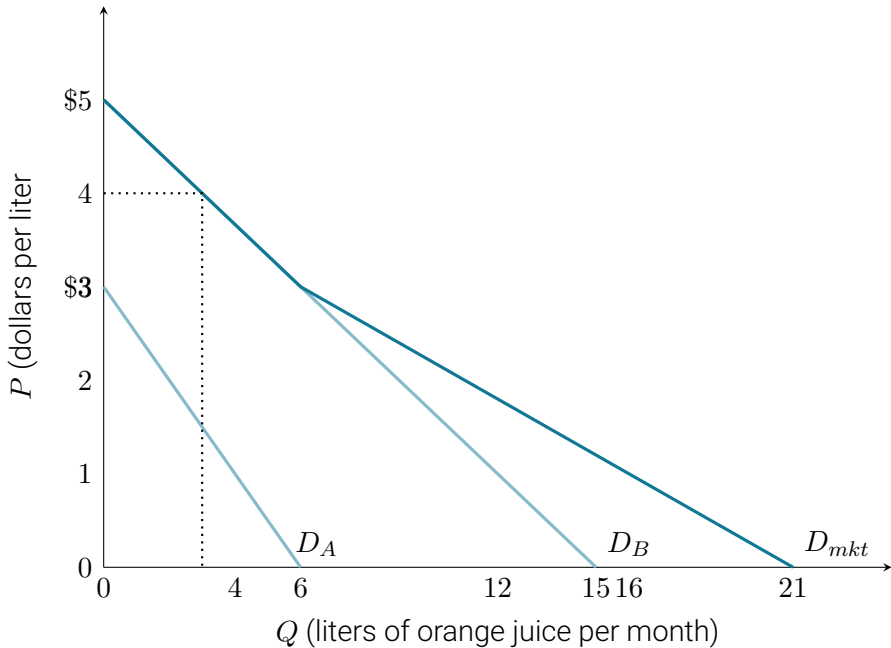
Aggregating Demand

Example

Putting this together

$$Q_{mkt}^D = \begin{cases} 41 - 5P & \text{if } P \leq 7 \\ 20 - 2P & \text{if } 7 < P \leq 10 \\ 0 & \text{otherwise.} \end{cases}$$





Aggregating Demand

Try It Yourself

Find the aggregate demand curve:

$$Q_A^D = \begin{cases} 30 - 6P & \text{if } P \leq 5 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^D = \begin{cases} 32 - 4P & \text{if } P \leq 8 \\ 0 & \text{otherwise.} \end{cases}$$

Solution

At a price of $P \leq 5$, both consumers purchase, so

$$Q_{mkt}^D = (30 - 6P) + (32 - 4P) = 62 - 10P$$

At a price $5 < P \leq 8$, only consumer B purchases ($Q_A^D = 0$), so

$$Q_{mkt}^D = 0 + (32 - 4P) = 32 - 4P$$

At a price $P > 8$, no one purchases, so $Q_{mkt}^D = 0$.

Putting this together:

$$Q_{mkt}^D = \begin{cases} 62 - 10P & \text{if } P \leq 5 \\ 32 - 4P & \text{if } 5 < P \leq 8 \\ 0 & \text{otherwise.} \end{cases}$$

Supply Curves

The market supply curve tells us the quantity of a product that producers are willing to sell at different prices.

- The **law of supply** is the *positive* relationship between the price of a good and the quantity supplied of that good.

Quantity supplied is affected by not just market price. For example, the prices of **factors of production**, or resources used in the production of the good, affect the quantity supplied.

Calculating Quantity Supplied

Try It Yourself

Suppose that the supply curve for Chaco's sandals in Boulder is given by $Q_{chaco}^S = -8,000 + 300P_{chaco}$. What is the quantity supplied of Chaco's if the price is \$50?

Calculating Quantity Supplied

Solution

Plug $P_{chaco} = 50$ into the supply curve:

$$Q_{chaco}^S = -8,000 + 300(50) = -8,000 + 15,000 = 7,000$$

At a price of \$50, the quantity supplied of Chaco's sandals is **7,000 pairs**.

Aggregating Supply

Consider if you have two producers of a good. Each producer's supply curve tells us *at a given price*, how many units will they sell.

How do we figure out the aggregate supply curve?

- *At a given price*, add up each producer's quantity supplied

Aggregating Supply

Suppose we have two producers, A and B. Suppose that

$$Q_A^S = \begin{cases} 4P - 12 & \text{if } P \geq 3 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^S = \begin{cases} 3P - 15 & \text{if } P \geq 5 \\ 0 & \text{otherwise.} \end{cases}$$

At a price of $P \geq 5$, both producers will supply, so supply is given by

$$Q_{mkt}^S = Q_A^S + Q_B^S = (4P - 12) + (3P - 15) = 7P - 27$$

Aggregating Supply

Suppose we have two producers, A and B. Suppose that

$$Q_A^S = \begin{cases} 4P - 12 & \text{if } P \geq 3 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^S = \begin{cases} 3P - 15 & \text{if } P \geq 5 \\ 0 & \text{otherwise.} \end{cases}$$

At a price $3 \leq P < 5$, only producer A will produce ($Q_B^S = 0$), so supply is given by

$$Q_{mkt}^S = Q_A^S + Q_B^S = (4P - 12) + 0 = 4P - 12$$

Aggregating Supply

Suppose we have two producers, A and B. Suppose that

$$Q_A^S = \begin{cases} 4P - 12 & \text{if } P \geq 3 \\ 0 & \text{otherwise.} \end{cases} \quad Q_B^S = \begin{cases} 3P - 15 & \text{if } P \geq 5 \\ 0 & \text{otherwise.} \end{cases}$$

At a price $P < 3$, no one produces anything, so

$$Q_{mkt}^S = Q_A^S + Q_B^S = 0 + 0 = 0$$

Aggregating Supply

Putting this together, we have

$$Q_{mkt}^S = \begin{cases} 7P - 27 & \text{if } P \geq 5 \\ 4P - 12 & \text{if } 3 < P \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

Aggregating Supply

Try It Yourself

Find the aggregate supply curve:

$$Q_A^S = \begin{cases} 5P - 25 & \text{if } P \geq 5 \\ 0 & \text{otherwise.} \end{cases}$$

$$Q_B^S = \begin{cases} 3P - 24 & \text{if } P \geq 8 \\ 0 & \text{otherwise.} \end{cases}$$

Solution

At a price $P \geq 8$, both producers supply, so

$$Q_{mkt}^S = (5P - 25) + (3P - 24) = 8P - 49$$

At a price $5 \leq P < 8$, only producer A supplies ($Q_B^S = 0$), so

$$Q_{mkt}^S = (5P - 25) + 0 = 5P - 25$$

At a price $P < 5$, no one supplies, so $Q_{mkt}^S = 0$.

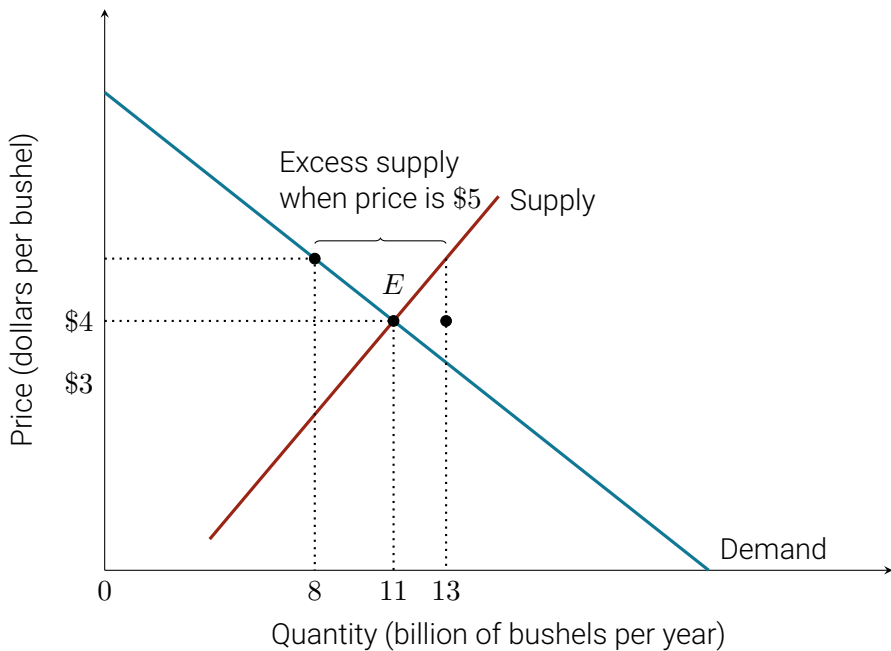
Putting this together:

$$Q_{mkt}^S = \begin{cases} 8P - 49 & \text{if } P \geq 8 \\ 5P - 25 & \text{if } 5 \leq P < 8 \\ 0 & \text{otherwise.} \end{cases}$$

Market Equilibrium

Equilibrium occurs at the price where quantity supplied equals quantity demanded. *Why is this an equilibrium?*

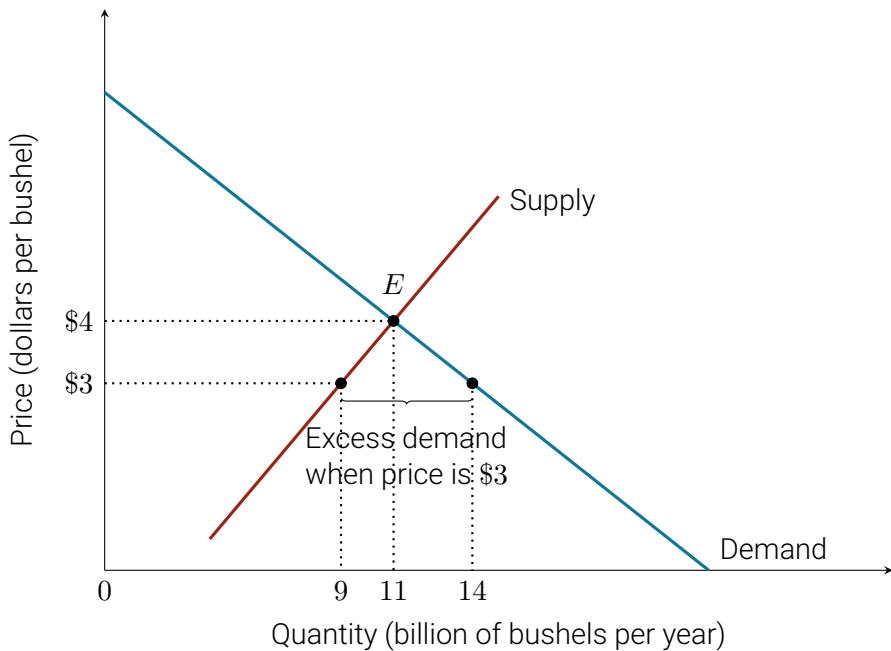
- What would happen if the price is \$5 per bushel?



Market Equilibrium

Equilibrium occurs at the price where quantity supplied equals quantity demanded. *Why is this an equilibrium?*

- What would happen if the price is \$5 per bushel? Excess supply will lead to a price war by suppliers.
 - The price will fall until all units are sold.
- What would happen if the price is \$3 per bushel?



Market Equilibrium

Equilibrium occurs at the price where quantity supplied equals quantity demanded. *Why is this an equilibrium?*

- What would happen if the price is \$5 per bushel? Excess supply will lead to a price war by suppliers.
 - The price will fall until all units are sold.
- What would happen if the price is \$3 per bushel? Excess demand will lead to a bidding war by consumers.
 - The price will rise until all consumers are satisfied.

Therefore, \$4 is an equilibrium because, absent any external forces, the price will not change.

Calculating Market Equilibrium

Try It Yourself

Suppose that the supply curve for Chaco's sandals in Boulder is given by $Q_{chaco}^S = -8,000 + 300P_{chaco}$, and the demand curve is $Q_{chaco}^D = 40,000 - 500P_{chaco}$. What is the equilibrium price of Chaco's?

Calculating Market Equilibrium

Solution

Set quantity supplied equal to quantity demanded:

$$-8,000 + 300P_{chaco} = 40,000 - 500P_{chaco}$$

$$800P_{chaco} = 48,000$$

$$P_{chaco} = 60$$

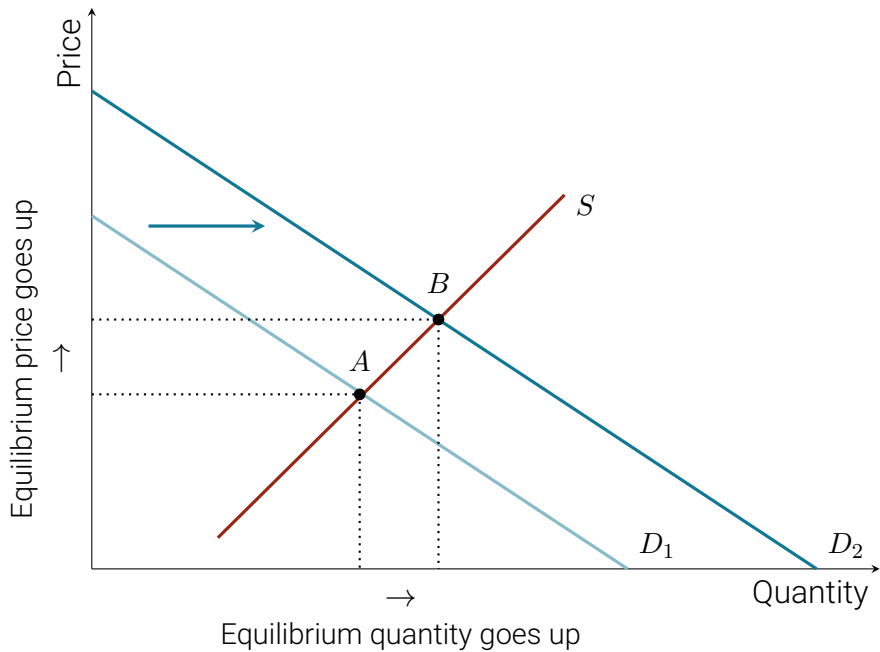
Plugging back in, $Q_{chaco} = -8,000 + 300(60) = 10,000$.

The equilibrium price is **\$60**, with an equilibrium quantity of **10,000 pairs**.

Shifts in Supply and Demand

Previously, we assumed that all factors other than price were fixed. But suppose that consumer incomes increase. *What happens to the demand curve?*

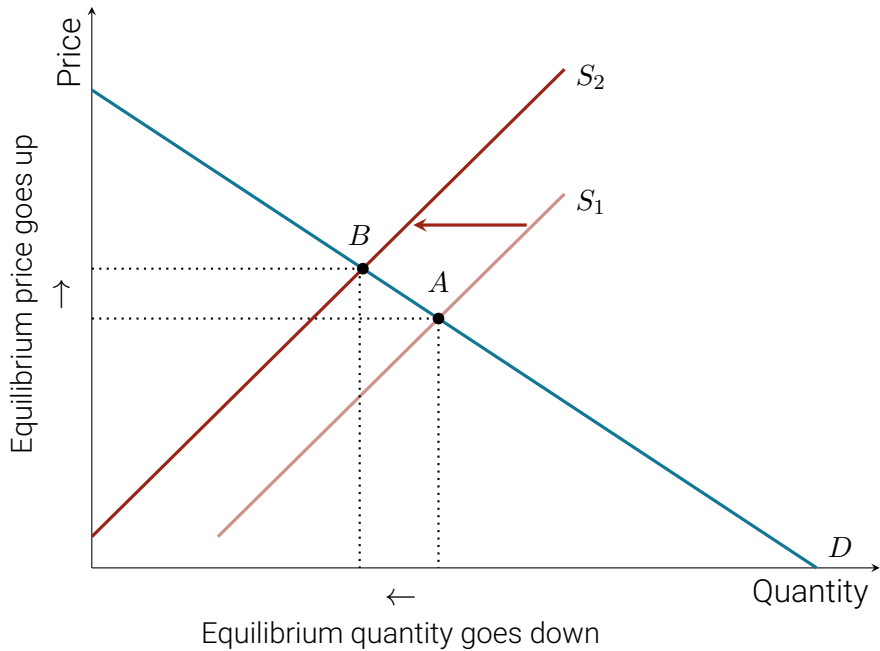
- This causes the demand curve to shift to the right.
- The price will rise, and the quantity sold will rise.
- Other causes of a demand shift are changes in preferences, the number of consumers, and expectations.



Shifts in Supply and Demand

Suppose now that the price of labor increases. *What happens to the supply curve?*

- This causes the supply curve to shift to the left.
- Price increases and quantity sold decreases.
- Other causes of a supply shift are changes in technology, input prices, number of suppliers, and expectations.



Shifts in Supply and Demand

What if *both* curves shift simultaneously? Suppose, for example that demand increases but supply decreases.

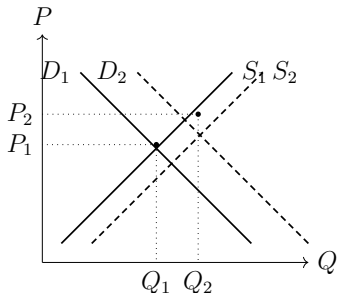
- Both of these shifts result in a higher price.
- But they pull the equilibrium quantity in opposite directions
- We need to know the magnitude to know which direction quantity moves.

Shifts in Supply and Demand

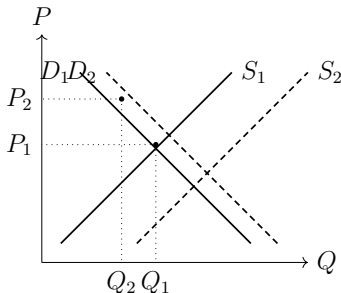
Try It Yourself

Sketch a decrease in supply and an increase in demand where quantity goes up. Sketch a decrease in supply and an increase in demand where quantity goes down.

Quantity Up



Quantity Down



In both cases price unambiguously rises. Quantity depends on which shift dominates:

- **Left panel:** demand's rightward shift is large relative to supply's leftward shift $\Rightarrow Q$ rises.
- **Right panel:** supply's leftward shift is large relative to demand's rightward shift $\Rightarrow Q$ falls.

Shifts in Supply and Demand

In general, when both curves shift, the change in either price or quantity will be obvious...

- ...but not both.
- One of these will always be ambiguous. Need to know magnitude of shifts.

Us professors love to ask questions on this. **When in doubt, draw it out!**

Price Elasticity of Demand

Let's say your boss asks you what will happen to sales if they increase the price of their product. Don't you dare say "sales will go down". *duh!!*

We want to be able to predict **how much** sales go down when we increase prices. *Way more useful!*

This is the **price elasticity of demand**

Price Elasticity of Demand

Price elasticity of demand

Measures the sensitivity of the quantity demanded to changes in the price.

$$\epsilon_{Q,P} = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}$$

Remembering our percent change formula

$$\epsilon_{Q,P} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P} \frac{P}{Q}$$

Classifying the Price Elasticity of Demand

Value of $\epsilon_{Q,P}$	Classification	Meaning
$\epsilon_{Q,P} = 0$	Perfectly inelastic demand	Quantity demanded is completely insensitive to price
$-1 < \epsilon_{Q,P} < 0$	Inelastic demand	Quantity demanded is relatively insensitive to price
$\epsilon_{Q,P} = -1$	Unitary elastic demand	Percentage increase in quantity demanded is equal to percentage decrease in price
$-\infty < \epsilon_{Q,P} < -1$	Elastic demand	Quantity demanded is relatively sensitive to price
$\epsilon_{Q,P} = -\infty$	Perfectly elastic demand	Any increase in price results in quantity demanded decreasing to zero, and any decrease in price results in quantity demanded increasing to infinity

Calculating Elasticity

Try It Yourself

Suppose that when the price of car tires is \$100 per tire, quantity demanded in Detroit is 40,000. Now suppose that the price has fallen to \$90, and the quantity demanded is 50,000. What is the price elasticity of demand?

Calculating Elasticity

Solution

We have $P = 100$, $Q = 40,000$, $\Delta P = 90 - 100 = -10$, and $\Delta Q = 50,000 - 40,000 = 10,000$.

$$\epsilon_{Q,P} = \frac{\Delta Q}{\Delta P} \frac{P}{Q} = \frac{10,000}{-10} \cdot \frac{100}{40,000} = (-1,000)(0.0025) = -2.5$$

Since $\epsilon_{Q,P} = -2.5 < -1$, demand for car tires in Detroit is **elastic** over this range: a 1% decrease in price leads to roughly a 2.5% increase in quantity demanded.

Interpreting Elasticities

$$\epsilon_{Q,P} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}}$$

A 1% increase in price yields a $\epsilon_{Q,P}$ % change in quantity.

Interpreting Elasticities

$$\epsilon = \frac{\frac{\Delta \text{Top Thing}}{\text{Top Thing}}}{\frac{\Delta \text{Bottom Thing}}{\text{Bottom Thing}}}$$

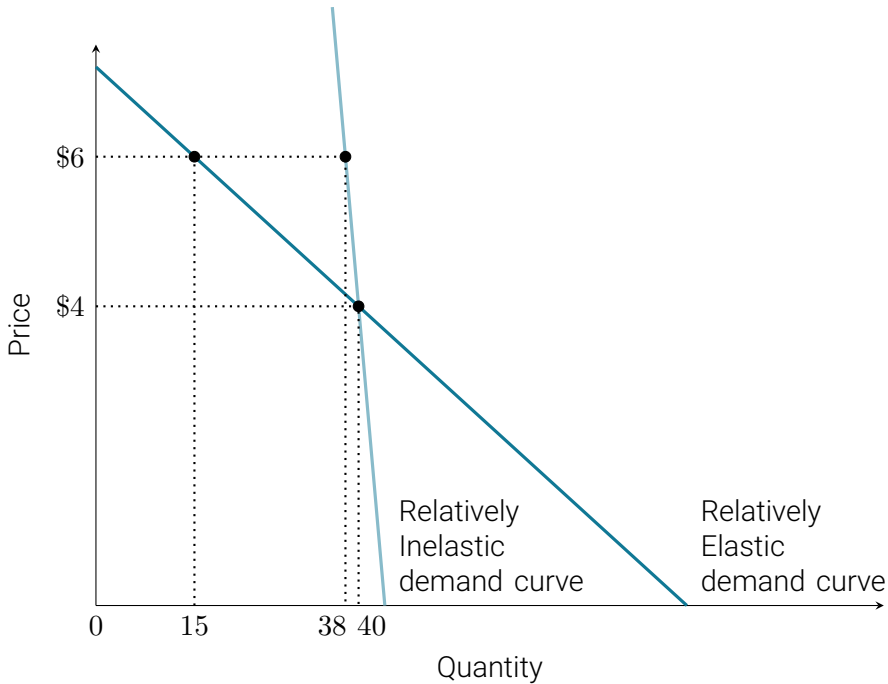
More generally, the way we *always* interpret elasticities is:

A 1% increase in **Bottom Thing** yields a $\epsilon\%$ change in **Top Thing**

Calculating Elasticity

There's all kinds of elasticities we care about in economics.

- The government might care what the price elasticity of demand is for cigarettes if they want to impose a tax.
- Or they might want to know what the cross-price elasticity of demand is for electric vehicles with respect to the price of gas.
- And if the government does impose some tax... firms might want to know how much of that tax they can pass on to consumers via higher prices.



Income elasticity of demand

% change in quantity demanded for every 1% change in income.

$$\epsilon_{Q,I} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta I}{I}} = \frac{\Delta Q}{\Delta I} \frac{I}{Q}$$

Price elasticity of supply

% change in quantity supplied for every 1% change in price of good.

$$\epsilon_{Q^S,P} = \frac{\frac{\Delta Q^S}{Q^S}}{\frac{\Delta P}{P}} = \frac{\Delta Q^S}{\Delta P} \frac{P}{Q^S}$$

Cross-price elasticity of demand

% change in quantity demanded of good i for every 1% change in price of good j .

$$\epsilon_{Q_i, P_j} = \frac{\frac{\Delta Q_i}{Q_i}}{\frac{\Delta P_j}{P_j}} = \frac{\Delta Q_i}{\Delta P_j} \frac{P_j}{Q_i}$$

If $\epsilon_{Q_i, P_j} > 0$, then as P_j increases, Q_i increases.

- Then goods i and j are **substitutes**.

If $\epsilon_{Q_i, P_j} < 0$, then as P_j increases, Q_i decreases.

- Then goods i and j are **compliments**.

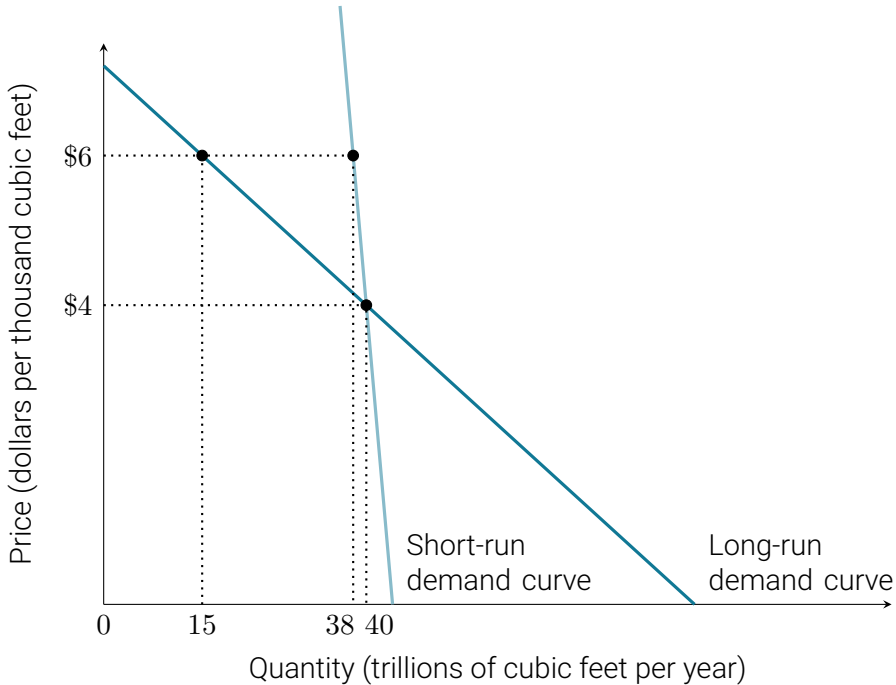
Elasticity in the Long Run vs. the Short Run

Consumers can't always adjust their demand instantly in response to a price change.

- If the price of gasoline doubles, you still have to drive to work.
- But after a while, maybe you will buy a more fuel-efficient car.

Need to distinguish between **long-run demand curve** and **short-run demand curve**.

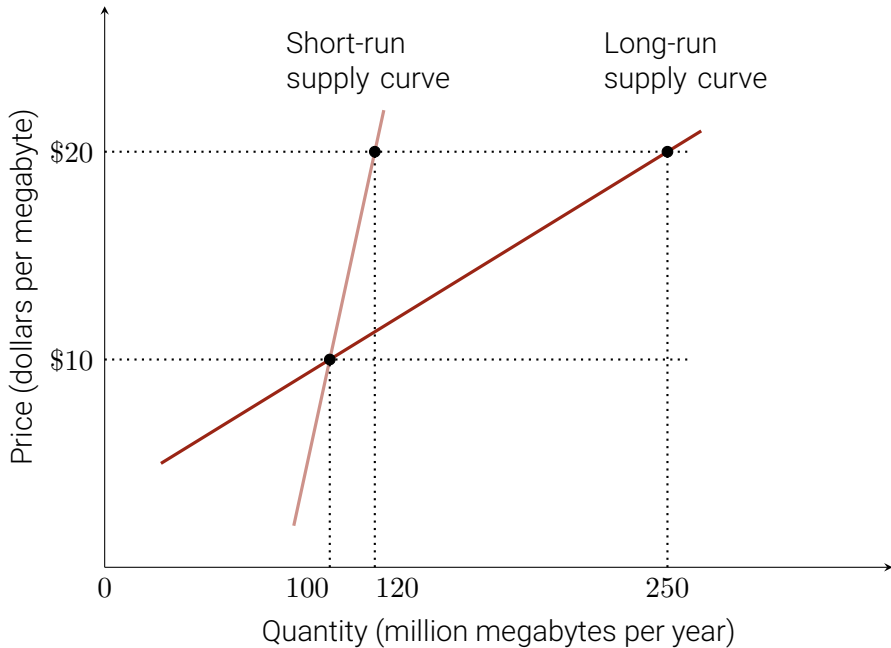
- Long run - Consumers have time to fully adjust purchasing decisions.
- Short run - Consumers do not.



Elasticity in the Long Run vs. the Short Run

The same is true for producers

- May not be able to increase output quickly in response to a price increase. Perhaps they are capacity-constrained.
- But in the long run, they can build another factory, or hire more workers. And quantity supplied increases.



Elasticity in the Long Run vs. the Short Run

In some cases however, the opposite may be true.

If price falls for durable goods such as a new refrigerator, consumers may decide it's a good time to upgrade their old one.

- But in the end, they don't buy *more* refrigerators. They just buy them sooner.
- In this case, demand is more elastic in the short run

The same may be true of producers (such as in markets for used or recycled goods).

Lecture 3 - Consumer Preferences and Utility

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

In this lecture, we will consider the following:

1. How we can represent and analyze consumer preferences.
2. Basic assumptions about consumer preferences.
3. Utility functions - Functions representing an individual's preferences for one or more goods.
4. Indifference curves as a graphical representation of consumer preferences.

Consumer Preferences

We all face trade-offs when it comes to our budget. Most people can't afford to buy everything they want.

- So how do we choose?
- Need a way of deciding between two or more 'baskets' of items.

Consumer Preferences

Consumer preferences tell us how an individual would rank any two baskets of goods (not considering their cost). We make a few assumptions about consumer preferences:

- **Completeness:** Consumer is able to rank any two baskets.
- **Transitivity:** If basket A is preferred to basket B, and B is preferred to C, then A is preferred to C.
- **More is Better:** Having more of a good is better for the consumer.

Consumer Preferences

Ordinal rankings give us information about the *order* in which a consumer ranks baskets. e.g. *1st, 2nd, 3rd, ...*

Cardinal rankings give us information about the intensity of consumer preferences. *I like this one 10, I like this one 25, etc.*

- What do these numbers mean? Units?

Utility Functions

Utility refers to consumer happiness or wellbeing. If a consumer prefers one basket to another, that basket is said to provide greater *utility*.

Utility functions measure the level of satisfaction that a consumer receives from any basket of goods. Examples:

- $U(X, Y) = 3X + 4Y$
- $U(X, Y) = X^{\frac{1}{2}}Y^{\frac{1}{2}}$
- $U(X, Y) = \log(XY)$

Why are we using functions?

Utility cannot actually be measured, so the exact specification is arbitrary.

- What matters is not the level of utility derived from a basket.
- It is how that level of utility *compares* to other baskets.

Utility explains how a person think about trade-offs

Utility cannot be compared *between* individuals, only *within* an individual.

Trade-offs and 'Marginal Utility'

Marginal utility refers to the rate at which total utility changes as the level of consumption of one good rises (holding constant the level of consumption of all other goods), and is given by:

$$MU_x = \frac{\Delta U}{\Delta x}$$

We will see that marginal utility is very helpful since it lets us compare *on the margin* how people compare more of different goods.

- This is also the formula for the slope of the utility function. (U on the y -axis, good x on the x -axis)
- It is also the partial derivative with respect to x : $\frac{\delta U(x)}{\delta x}$.

Diminishing Returns

What is a good you wouldn't want to consume infinite quantities of?

Marginal Utility

Diminishing marginal utility means that as an individual consumes more of a good, their additional (or marginal) utility from that good decreases.

- Think about the additional satisfaction you get from eating more pizza...
- Or watching additional episodes of a tv show on Netflix.

Marginal Utility

Try It Yourself

Suppose that a consumer's preference can be represented by the utility function $U(X, Y) = 4X^{\frac{1}{2}}Y^{\frac{1}{2}}$.

What is the consumer's marginal utility function for good X ?

Solution

Take the partial derivative of $U(X, Y) = 4X^{\frac{1}{2}}Y^{\frac{1}{2}}$ with respect to X , holding Y constant:

$$MU_X = \frac{\partial U}{\partial X} = 4 \cdot \frac{1}{2} X^{-\frac{1}{2}} Y^{\frac{1}{2}} = 2X^{-\frac{1}{2}} Y^{\frac{1}{2}}$$

We can rewrite this as

$$MU_X = 2 \left(\frac{Y}{X} \right)^{\frac{1}{2}}$$

Notice that MU_X falls as X increases (holding Y fixed) – this is diminishing marginal utility.

Tradeoffs and Indifference Curves

In the case where a consumer is choosing quantities of two goods, the consumer's utility function can be represented graphically using indifference curves.

- In the case where a consumer is choosing quantities of two goods, the consumer's utility function can be represented graphically using indifference curves.
- An **indifference curve** represents bundles of two goods which provide a consumer with the same level of utility.
- Consider the following illustration of a utility function with two goods:

Figure: Indifference Curves

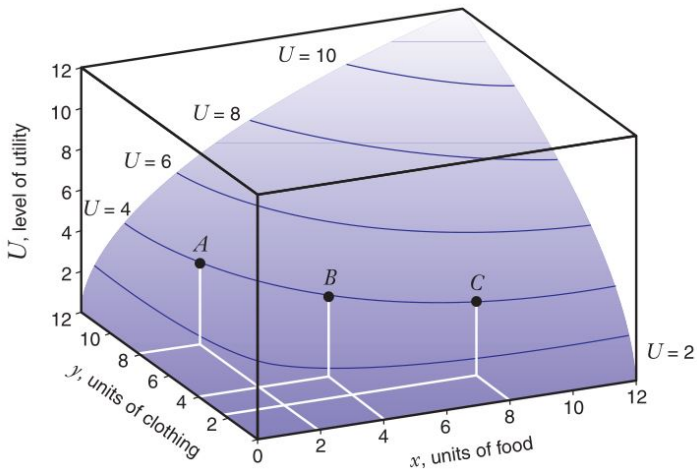
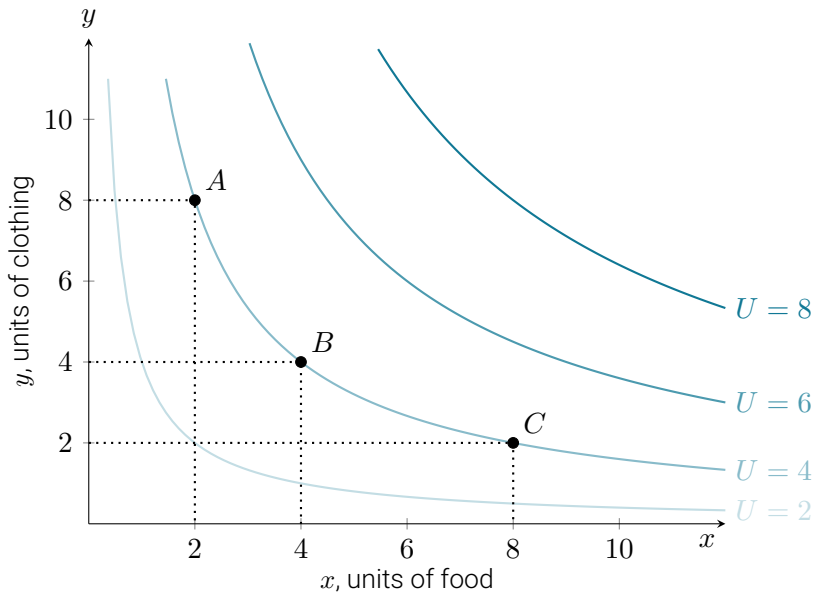


Figure: Indifference Curves in 2D



How do you draw an indifference curve?

Say the utility function is $U(x, y) = x + 2y$. To draw an indifference curve:

1. Pick a value of utility $\bar{U}$. Solve for y .

$$\text{e.g. } \bar{U} = 2 = x + 2y \implies y = 1 - \frac{1}{2x}$$

2. Pick points and 'trace out' indifference curve shape.

In this case, we have a line, so drawing it should be easy.

Tip: Pick a $\bar{U}$ so the math works well.

How do you draw an indifference curve?

Try It Yourself

Draw three indifference curves for the utility function $U(x, y) = 2x + 2y$

Solution

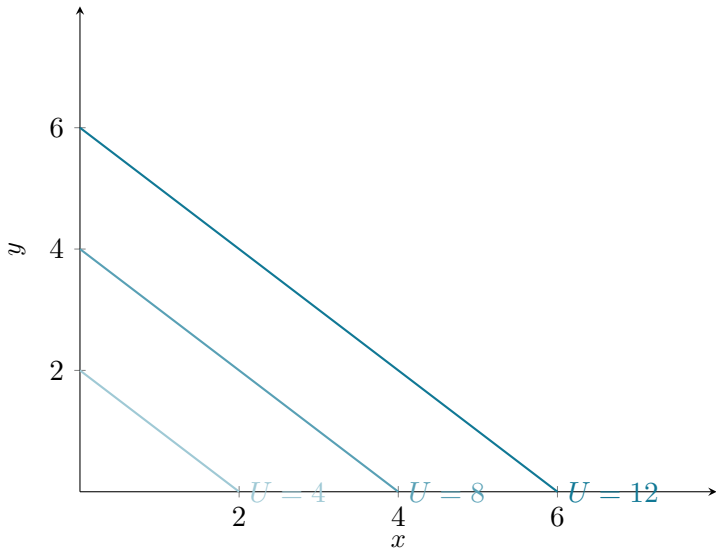
Pick three values of $\bar{U}$, say $\bar{U} = 4, 8, 12$, and solve for y :

$$\bar{U} = 2x + 2y \implies y = \frac{\bar{U}}{2} - x$$

This gives us three parallel lines, each with slope -1 :

$$y = 2 - x \quad (\bar{U} = 4), \quad y = 4 - x \quad (\bar{U} = 8), \quad y = 6 - x \quad (\bar{U} = 12)$$

Figure: Indifference Curves for $U(x, y) = 2x + 2y$



Properties of Indifference Curves

Indifference curves have four properties:

- If both products are 'goods', curves slope downward.
- Indifference curves **cannot** intersect.
- Every consumption basket lies on only one indifference curve.
- Indifference curves are not "thick".

Marginal Rate of Substitution

Note that for a consumer to stay on a given indifference curve (keeping utility constant), if they receive more of one good, they must give up a certain amount of the other. Why is that?

- Just increase one good $\implies$ higher utility
- Just decrease one good $\implies$ lower utility
- Need to increase one and decrease another to keep utility constant

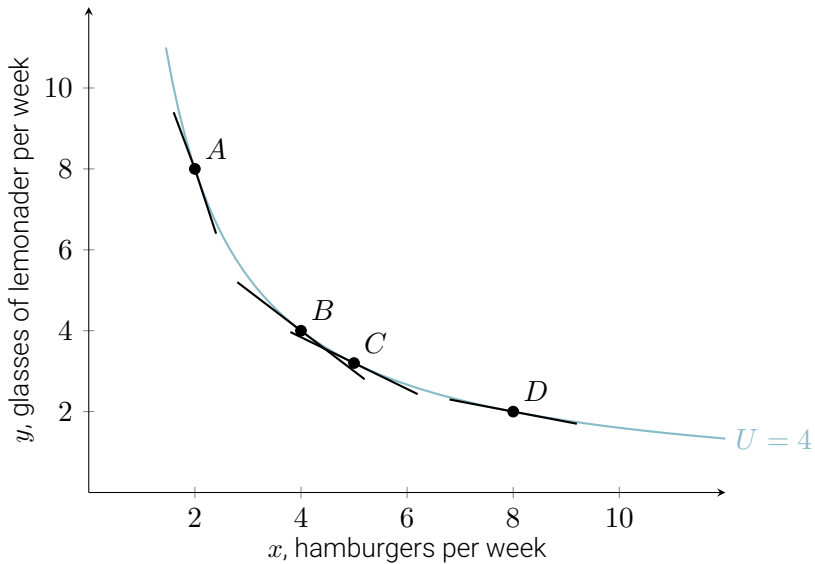
But how much of the other good do we have to give up to stay on the indifference curve?

Marginal Rate of Substitution

But how much of the other good do we have to give up to stay on the indifference curve? This is the **marginal rate of substitution (MRS)**

- The rate at which a consumer must substitute one good for another to stay on the same indifference curve is given by $\Delta y / \Delta x$.
- $MRS_{x,y}$ is therefore also the slope of the indifference curves ($\Delta y / \Delta x$)

Figure: Marginal Rate of Substitution



Marginal Rate of Substitution

Try It Yourself

Suppose that Tom is indifferent between two meals: Meal A contains 3 cheeseburgers (C), and 1 large french fry (F). Meal B contains 2 cheeseburgers and 3 large french fries. What is Tom's $MRS_{C,F}$?

Remember: $MRS_{C,F} = |\frac{\Delta F}{\Delta C}|$

Solution

Since Tom is indifferent between Meal A (3 cheeseburgers, 1 fry) and Meal B (2 cheeseburgers, 3 fries), both baskets lie on the same indifference curve. Moving from A to B:

$$\Delta C = 2 - 3 = -1 \quad \Delta F = 3 - 1 = 2$$

$$MRS_{C,F} = \left| \frac{\Delta F}{\Delta C} \right| = \left| \frac{2}{-1} \right| = 2$$

Tom is willing to give up 2 fries for 1 additional cheeseburger (or equivalently, requires 2 fries to give up 1 cheeseburger).

MRS and Marginal Utility

Can also express MRS as ratio of marginal utilities of two goods

- To stay on the same indifference curve, for any small change in quantity of good x , **the change in utility from good y must offset the change in utility from good x .**

That is,

$$MU_x * \Delta x = -MU_y * \Delta y$$

Rearranging the terms, gives us our formula we will use for the class

$$-\frac{\Delta y}{\Delta x} = \frac{MU_x}{MU_y} = MRS_{x,y}$$

How to interpret MRS

The way you interpret MRS is:

For a 1 unit decrease of x , you would have to get $MRS_{x,y}$ units of y
(and vice-versa)

$MRS_{x,y}$ means you **substitute x for y**

How to interpret MRS

Try It Yourself

Calculate the marginal rate of substitution, $MRS_{x,y}$, for the following utility functions:

$$U(x, y) = 2x + y \quad \text{and} \quad U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}};$$

Solution

For $U(x, y) = 2x + y$:

$$MU_x = 2, \quad MU_y = 1 \quad \implies \quad MRS_{x,y} = \frac{MU_x}{MU_y} = 2$$

For $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$:

$$MU_x = \frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}}, \quad MU_y = \frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}}$$

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{\frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}}}{\frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}}} = \frac{y}{x}$$

Diminishing Marginal Rate of Substitution

Remember that for many goods, as consumption increases, the additional utility gained from consuming the good (MU) decreases.

$$MRS_{x,y} = \frac{MU_x}{MU_y}$$

- More of good $y \implies$ lower $MU_y \implies$ higher $MRS_{x,y}$.
- Therefore to get one unit of good x , the more good y we would give up for .
- Said another way, we would be willing to give less of good x for each additional unit of good y we consume, holding x constant.

The three utility functions in this class

There are three kinds of utility functions we will give you in this course.

If you want to do well in this class, you should practice these three functions a bunch (*calculate MRS, draw indifference curves, and later on calculate demand curves for x and y*)

Preference #1: Perfect Substitutes

If two goods are **perfect substitutes** for a consumer, then the marginal rate of substitution is constant (would always trade them at a constant rate)

These are given by $U(x, y) = ax + by$ where a and b are two numbers that describe the rate you would trade them.

Preference #1: Perfect Substitutes

Try It Yourself

Calculate the $MRS_{x,y}$ for $U(x, y) = ax + by$. Then draw two indifference curves

Preference #1: Perfect Substitutes

Solution

$$MU_x = a, \quad MU_y = b \quad \implies \quad MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{a}{b}$$

This is constant – it doesn't depend on x or y . Solving for two indifference curves ($\bar{U}_1$ and $\bar{U}_2$, with $\bar{U}_2 > \bar{U}_1$):

$$y = \frac{\bar{U}}{b} - \frac{a}{b}x$$

gives two parallel straight lines, both with slope $-a/b$.

Figure: Perfect Substitutes: $MRS_{x,y} = a/b$

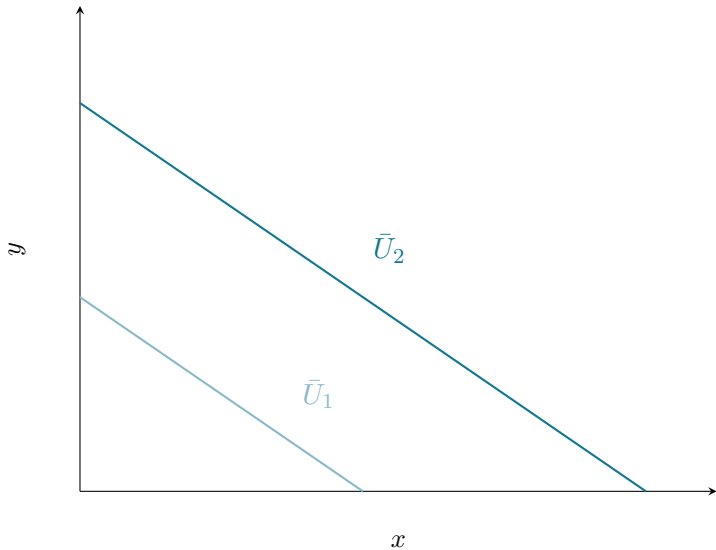
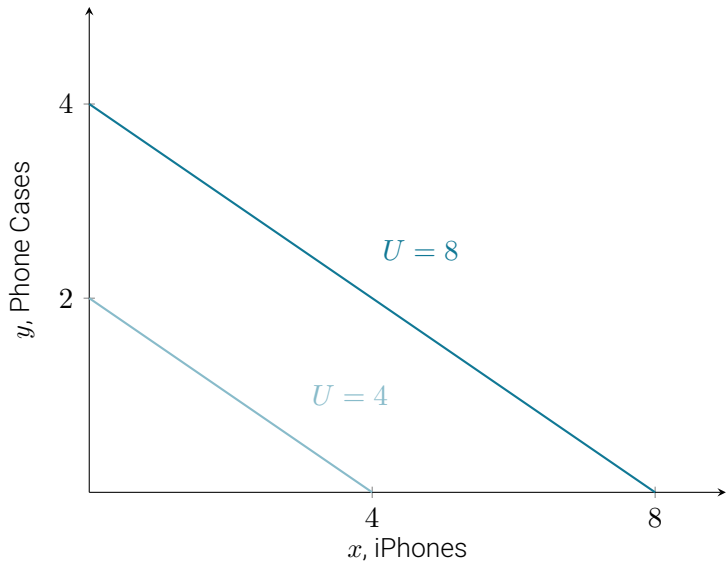


Figure: Perfect Substitutes



Preferences #2: Cobb-Douglas

Utility functions of the form $U = x^\alpha y^\beta$ are known as **Cobb-Douglas Utility Functions**. α and β are numbers (typically between 0 and 1)

Note this utility function displays diminishing marginal rates of substitutions which is the most common case in the real world.

Preference #2: Cobb-Douglas

Try It Yourself

Calculate the $MRS_{x,y}$ for $U(x, y) = x^{\frac{1}{3}}y^{\frac{2}{3}}$. Then draw two indifference curves

Preference #2: Cobb-Douglas

Solution

$$MU_x = \frac{1}{3}x^{-\frac{2}{3}}y^{\frac{2}{3}}, \quad MU_y = \frac{2}{3}x^{\frac{1}{3}}y^{-\frac{1}{3}}$$

$$MRS_{x,y} = \frac{MU_x}{MU_y} = \frac{\frac{1}{3}x^{-\frac{2}{3}}y^{\frac{2}{3}}}{\frac{2}{3}x^{\frac{1}{3}}y^{-\frac{1}{3}}} = \frac{1}{2} \cdot \frac{y}{x} = \frac{y}{2x}$$

Unlike perfect substitutes, $MRS_{x,y}$ here depends on the ratio y/x – it falls as x rises relative to y , giving the characteristic bowed-in (convex) indifference curves.

Figure: Cobb-Douglas Indifference Curves

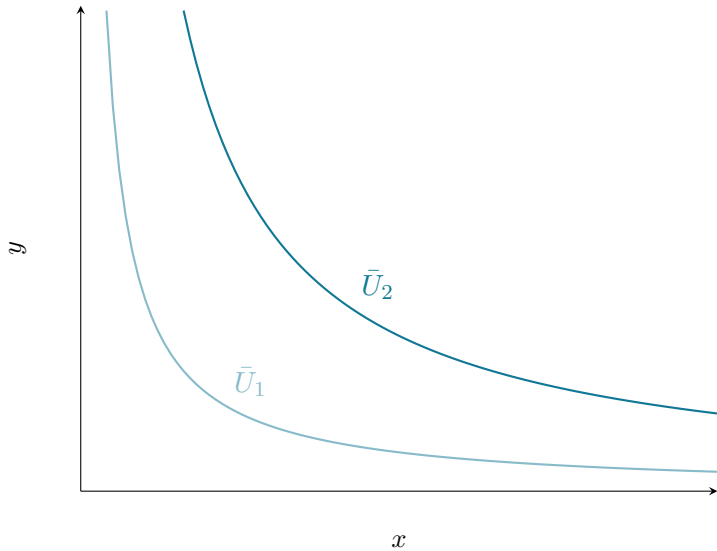
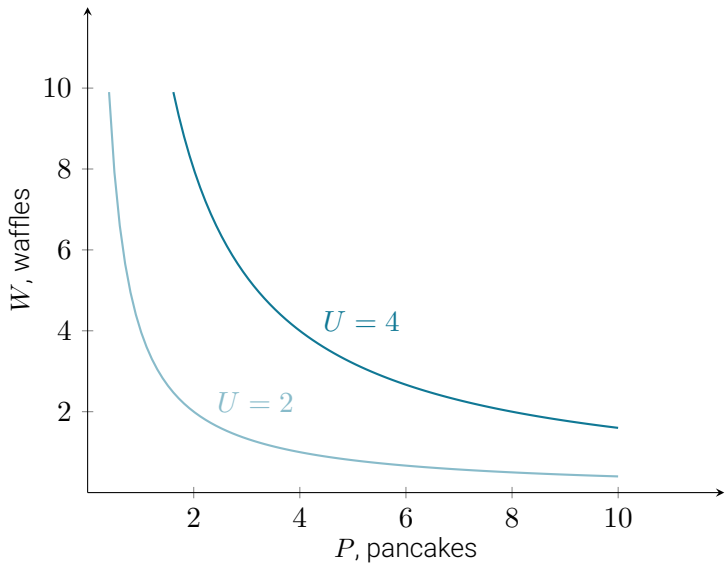


Figure: Cobb-Douglas



Preference #3: Perfect Complements

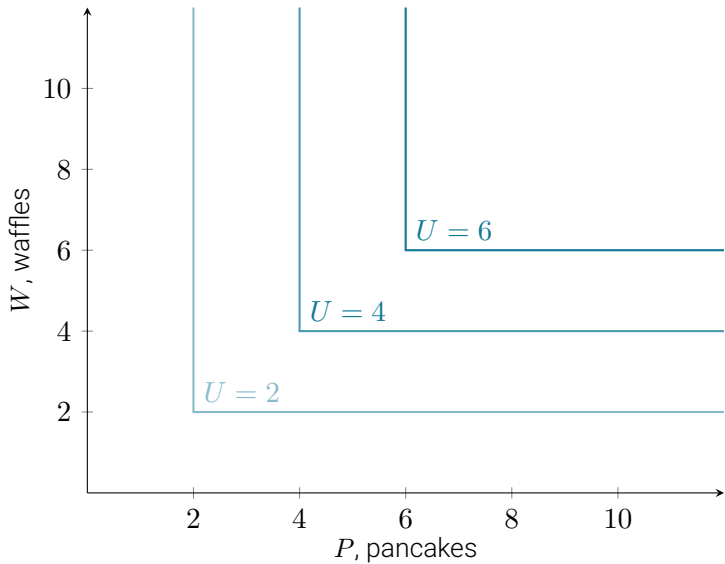
If two goods are **perfect complements** for a consumer, then the consumer only consumes the good in a constant proportion.

They are given by $U(x, y) = \min(ax, by)$. You will always consumer such that $ax = by$. Why?

Holding y fixed:

- if $ax < by$, then you are spending too much money on y since $\min(ax, by) = ax$.
- if $ax > by$, then you are spending too much money on x since $\min(ax, by) = by$.
- Thus, you consume $ax = by$.

Figure: Perfect Complements



Perfect Complements and Marginal Rate of Substitution

Note you can't calculate MRS since you can't take the derivative of the $\min(ax, by)$.

Lecture 4 - Consumer Choice

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Overview

In the previous lecture, we looked at consumer preferences.

- We considered how consumers ranked various bundles of goods.
- We saw how to graphically and numerically represent these preferences.

In the next lecture, we will consider how consumers choose which bundle to consume.

The Budget Constraint

Consumers only have a limited amount of money to spend on goods and services.

- A consumer's **budget constraint** defines the set of baskets that a consumer can purchase.
- If P_x is the price of good x , then $P_x x$ is the amount of money spent on good x
- Similarly, $P_y y$ is the amount of money spent on good y .

Because a consumer can only spend as much as their income, I , their budget constraint can be expressed as:

$$P_x x + P_y y \leq I$$

Budget Constraint

The budget constraint can be generalized to more than two goods:

$$P_{x_1}x_1 + P_{x_2}x_2 + \dots + P_{x_n}x_n \leq I$$

- Because more is better (remember our assumptions), the consumer will always use up their whole budget (I).
- In general, however, a consumer can purchase any bundle which costs I \$ or less.

The Budget Line

A consumer's budget constraint for two goods can also be represented graphically. In the equation $P_x x + P_y y = I$, solve for y .

The rearranged equation is

$$y = \frac{I}{P_y} - \frac{P_x}{P_y}x$$

- This equation in slope-intercept form can be used to plot our budget line.

A **budget line** shows all combinations of goods x and y that can be purchased for a given budget I , at given prices P_x and P_y .

The Budget Line

Try It Yourself

Write the budget constraint with $I = 800$, $P_X = 20$, and $P_Y = 40$. Then, solve for y .

Solution

The budget constraint is

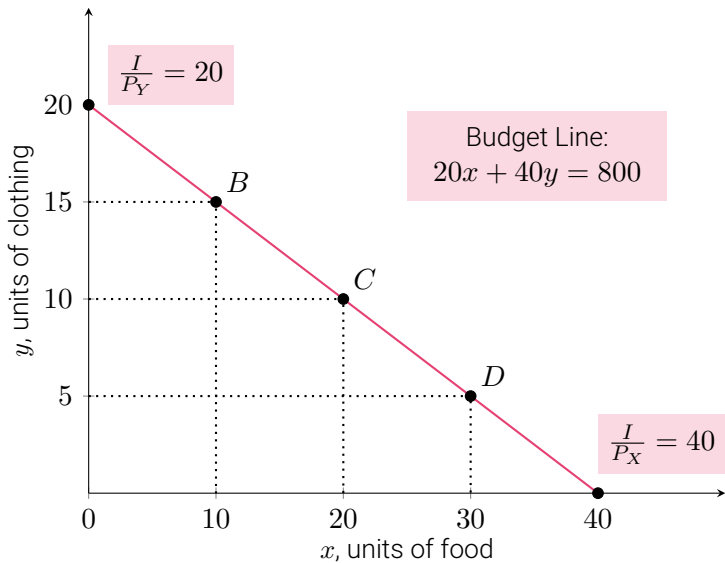
$$20x + 40y = 800$$

Solving for y :

$$40y = 800 - 20x \implies y = 20 - \frac{1}{2}x$$

This is exactly the budget line plotted on the next slide – notice the y -intercept of 20 and slope of $-1/2$.

Figure: The Budget Line



The Budget Line

The y intercept tells us how many units of good y can be purchased if none of good x is bought. Similarly for the x intercept.

The slope tells us how many units of good y must be given up to obtain one more unit of good x .

Conceptual questions

- Remember that a change in income shifts the budget line, without changing the slope (why?)
- What would a change in the price of one (or both) of the goods do?

The Budget Line

Try It Yourself

Suppose that Sammy has \$200 in disposable income to spend each month on concert tickets and football tickets. Concert tickets cost \$50 and football tickets cost \$20. The price of concert tickets increases by \$25, and the price of football tickets increases by \$10. Assume that Sammy's disposable income doesn't change.

Draw the budget line before and after.

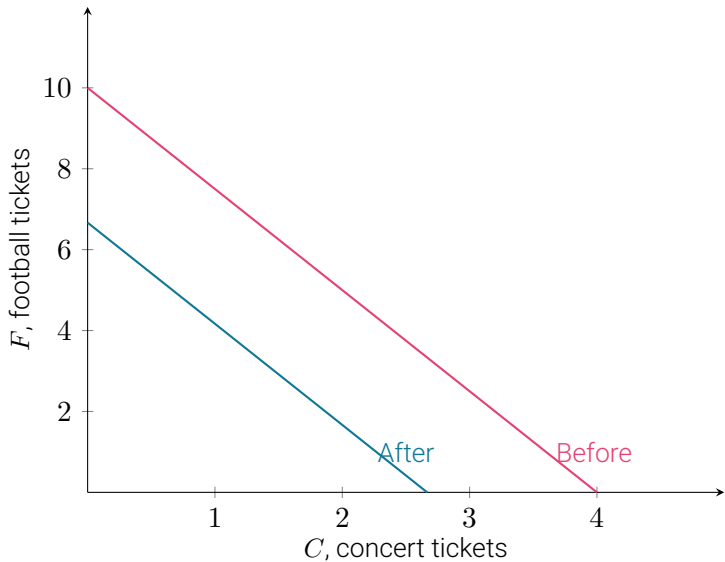
Solution

Before: $50C + 20F = 200$. Intercepts: $C = 4$ (if $F = 0$), $F = 10$ (if $C = 0$). Slope $= -50/20 = -2.5$.

After: concert tickets now cost \$75, football tickets now cost \$30. So $75C + 30F = 200$. Intercepts: $C = 200/75 \approx 2.67$, $F = 200/30 \approx 6.67$. Slope $= -75/30 = -2.5$.

Key insight: both prices rose by exactly 50% ($50 \rightarrow 75$, $20 \rightarrow 30$), so the price *ratio* – and therefore the slope – didn't change. The budget line shifts **inward, parallel** to the original line, just like a decrease in real income.

Figure: Sammy's Budget Line, Before and After



Optimal Choice

Now that we know the consumer's problem (maximize utility), and their constraint (the budget constraint), we can determine the consumer's optimal choice.

- The **optimal choice** is the amount of each good that should be purchased in order to maximize utility.
- This basket of goods must be located on the budget line (why not below?).

Optimal Choice

More formally, we can state the consumer's **utility maximization** problem as:

$$\max_{(x,y)} U(x, y) \text{ subject to: } P_x x + P_y y \leq I$$

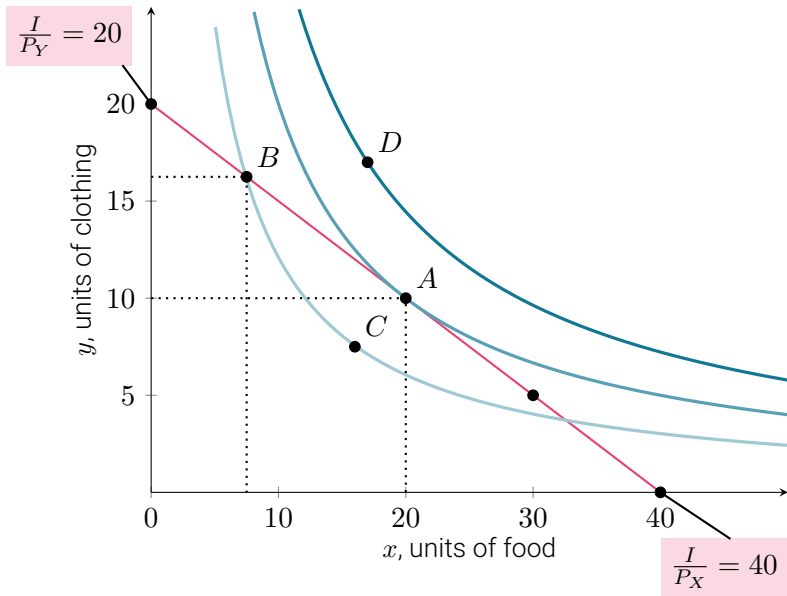
Utility Maximization

How can we interpret this graphically?

- Remember that the consumer wants to get to the highest indifference curve possible.
- That means moving as far up and to the right as possible
- But the consumer can't move beyond their budget line.

The optimal bundle ends up being (in most cases) the point where the budget line is tangent with one of the indifference curves.

Figure: Utility Maximization



Utility Maximization

To find the consumer's optimal level of consumption of x and y , we need some calculus.

We will maximize utility in two-ways:

1. The method of Lagrange
2. 'plugging-in' the constraint to our problem.

Method of Lagrange

Define the Lagrangian ($\mathcal{L}$) as

$$\mathcal{L}(x, y, \lambda) = U(x, y) + \lambda(I - P_x x - P_y y)$$

Maximizing the Lagrangian with respect to x , y , and λ is equivalent to maximizing the original problem.

Method of Lagrange

To maximize the Lagrangian, take derivatives with respect to x , y , and λ , and set them equal to zero. The first order conditions for an optimum are:

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \Rightarrow \frac{\partial U(x, y)}{\partial x} = \lambda P_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0 \Rightarrow \frac{\partial U(x, y)}{\partial y} = \lambda P_y$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow I - P_x x - P_y y = 0$$

Method of Lagrange

Remember that $\frac{\delta U(x,y)}{\delta x} = MU_x$ and $\frac{\delta U(x,y)}{\delta y} = MU_y$. Then, rearranging terms:

$$\frac{MU_x}{P_x} = \lambda \quad \text{and} \quad \frac{MU_y}{P_y} = \lambda$$

We can combine the first two equations and rearrange terms, and the result is:

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

The resulting equality is our [optimality condition](#).

Example

Consider $U(x, y) = 2xy$ and $I = 10, P_x = 2, P_y = 1$. Solve for the Consumers optimal choices of goods x and y using the Lagrangian Method.

Method of Lagrange

Try It Yourself

Consider $U(x, y) = xy$ and $I = 80, P_x = 4, P_y = 2$. Solve for the Consumers optimal choices of goods x and y using the Lagrangian Method.

Solution

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(80 - 4x - 2y)$$

First order conditions:

$$\frac{\delta \mathcal{L}}{\delta x} = y - 4\lambda = 0 \implies y = 4\lambda$$

$$\frac{\delta \mathcal{L}}{\delta y} = x - 2\lambda = 0 \implies x = 2\lambda$$

$$\frac{\delta \mathcal{L}}{\delta \lambda} = 80 - 4x - 2y = 0$$

Solution (continued)

From the first two conditions, $y = 4\lambda$ and $x = 2\lambda$, so $y = 2x$.

Plugging into the budget constraint:

$$4x + 2(2x) = 80 \implies 8x = 80 \implies x^* = 10$$

Then $y^* = 2x^* = 20$.

Optimal bundle: $x^* = 10, y^* = 20$.

Unconstrained Optimization

To solve using unconstrained optimization, first solve for y in the budget constraint $P_x x + P_y y = I$.

$$y = \frac{I - P_x x}{P_y}$$

We then plug this into our utility function, and our problem becomes:

$$\max_x U\left(x, \frac{I - P_x x}{P_y}\right)$$

Unconstrained Optimization

How do we maximize (or minimize) any function?

- Take derivatives, and set them equal to zero.

Taking the total derivative of $U(x, \frac{I - P_x x}{P_y})$ w.r.t x (using the chain-rule!):

$$\begin{aligned}\frac{dU}{dx} &= \frac{\delta U}{\delta x} + \frac{\delta U}{\delta y} \frac{dy}{dx} = 0 \\ \Rightarrow \frac{\delta U}{\delta x} - \frac{\delta U}{\delta y} \frac{P_x}{P_y} &= 0 \\ \Rightarrow \frac{\delta U}{\delta x} / P_x &= \frac{\delta U}{\delta y} / P_y \\ \rightarrow \frac{MU_x}{P_x} &= \frac{MU_y}{P_y}\end{aligned}$$

Unconstrained Optimization

Example

Consider $U(x, y) = 2xy$ and $I = 10, P_x = 2, P_y = 1$. Solve for the Consumers optimal choices of goods x and y using Unconstrained Optimization.

Solution

Solve the budget constraint for y :

$$y = \frac{I - P_x x}{P_y} = \frac{10 - 2x}{1} = 10 - 2x$$

Plug into the utility function:

$$U(x) = 2x(10 - 2x) = 20x - 4x^2$$

Take the derivative and set it to zero:

$$\frac{dU}{dx} = 20 - 8x = 0 \implies x^* = 2.5$$

Then $y^* = 10 - 2(2.5) = 5$.

Unconstrained Optimization

Try It Yourself

Consider $U(x, y) = xy$ and $I = 80, P_x = 4, P_y = 2$. Solve for the Consumers optimal choices of goods x and y using Unconstrained Optimization.

Solution

Solve the budget constraint for y :

$$y = \frac{80 - 4x}{2} = 40 - 2x$$

Plug into the utility function:

$$U(x) = x(40 - 2x) = 40x - 2x^2$$

Take the derivative and set it to zero:

$$\frac{dU}{dx} = 40 - 4x = 0 \implies x^* = 10$$

Then $y^* = 40 - 2(10) = 20$.

The Optimality Condition

We have just found our first **optimality condition**! Now that we have derived this from first principles, we don't have to do that again!

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

- The **optimality condition** tells us that the consumer maximizes utility when the marginal utility per dollar spent on both goods is equal.
- Think about what would happen if it weren't equal for the two goods.

The Optimality Condition

We can also rearrange the **optimality condition** as follows:

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

- This tells us that the slope of our indifference curve should equal the slope of our budget line.
- The result is that the optimal basket is located where the budget line is exactly tangent to one of the indifference curves.
- Compare this to our graphical result.

The Optimality Condition

So far, we only considered the case where baskets contain two goods. However, the **optimality condition** holds for any number of goods.

$$\frac{MU_{x_1}}{P_{x_1}} = \frac{MU_{x_2}}{P_{x_2}} = \dots = \frac{MU_{x_n}}{P_{x_n}}$$

- As before, the utility per dollar spent on each good (or “bang for your buck”), must be equal.

This is a *key insight*! This **optimality condition** is how we think individuals balance **trade-offs** on what to buy

The Optimality Condition

Try It Yourself

Consider a market with only two goods, bread and cheese. Suppose that at Francis' current level of consumption, his marginal utility of bread is 4, and his marginal utility of cheese is 6. Suppose that a loaf of bread costs \$3, and a pound of cheese costs \$6. What should Francis do?

Solution

Compute utility per dollar spent on each good:

$$\frac{MU_{bread}}{P_{bread}} = \frac{4}{3} \approx 1.33 \qquad \frac{MU_{cheese}}{P_{cheese}} = \frac{6}{6} = 1$$

Since $\frac{MU_{bread}}{P_{bread}} > \frac{MU_{cheese}}{P_{cheese}}$, bread gives Francis more utility per dollar than cheese.

Francis should buy more bread and less cheese. As he does, MU_{bread} will fall (diminishing marginal utility) and MU_{cheese} will rise, until the two ratios are equal – at which point he's reached his optimal bundle.

Finding the Optimal Bundle

Now that we know something about the optimal bundle (from the [optimality condition](#)), how do we actually find it?

- Well we also know that the consumer has only $\$I$ to spend.
- This gives us a system of two equations (the [optimality condition](#) and the budget constraint), and two unknowns (quantity consumed of goods x and y).

All that's left is some (sometimes messy) algebra.

Finding the Optimal Bundle

Suppose $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$. Let $P_x = 2$, $P_y = 3$ and $I = 24$.

Calculate MU_x and MU_y and the optimality condition

Finding the Optimal Bundle

Suppose $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$. Let $P_x = 2$, $P_y = 3$ and $I = 24$.

$$MU_x = \frac{y^{\frac{1}{2}}}{2x^{\frac{1}{2}}} \text{ and } MU_y = \frac{x^{\frac{1}{2}}}{2y^{\frac{1}{2}}}$$

Then our optimality condition is $\frac{y^{\frac{1}{2}}}{2x^{\frac{1}{2}}} / \frac{x^{\frac{1}{2}}}{2y^{\frac{1}{2}}} = \frac{2}{3}$

$$\implies y = \frac{2}{3}x$$

Plugging the optimality condition into our budget constraint

$$2x + 3y = 24 \implies 2x + 3\left(\frac{2}{3}x\right) = 24 \implies x^* = 6$$

Plugging x^* back into the optimality condition, $y^* = \frac{2}{3}6 = 4$

Finding the Optimal Bundle

We could also leave out prices and income, and simply solve for x and y in terms of P_x , P_y , and I .

Take the rearranged **optimality condition** $y = x \frac{P_x}{P_y}$. Plug into budget constraint

$$P_x x + P_y x \left(\frac{P_x}{P_y} \right) = I \implies x^* = \frac{I}{2P_x}$$

Plug x back into the **optimality condition**:

$$y^* = \frac{I}{2P_x} \frac{P_x}{P_y} = \frac{I}{2P_y}$$

Demand Functions

The resulting **demand functions** are:

$$x^*(P_x, P_y, I) = \frac{I}{2P_x} \quad \text{and} \quad y^*(P_x, P_y, I) = \frac{I}{2P_y}$$

The **demand function** is useful since we do the work once and then can predict what that person will consumer under any (P_x, P_y, I) combination for comparative statics

- I'll likely do this on the test

Demand Functions

Try It Yourself

Consider $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ and $I = 80, P_x = 4, P_y = 2$. Solve for the Consumers optimal choices of goods x and y and the resulting demand functions.

Solution

Since $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ has equal exponents, we can use the demand functions derived earlier: $x^* = \frac{I}{2P_x}$ and $y^* = \frac{I}{2P_y}$.

Plugging in $I = 80$, $P_x = 4$, $P_y = 2$:

$$x^* = \frac{80}{2(4)} = \frac{80}{8} = 10 \qquad y^* = \frac{80}{2(2)} = \frac{80}{4} = 20$$

Optimal bundle: $x^* = 10$, $y^* = 20$.

Demand functions: $x^*(P_x, P_y, I) = \frac{I}{2P_x}$, $y^*(P_x, P_y, I) = \frac{I}{2P_y}$.

Three Utility Functions

How do you find demand for the three utility functions?

1. **Cobb-Douglas:** $U(x, y) = x^\alpha y^\beta$
2. **Perfect Substitutes:** $U(x, y) = ax + by$
3. **Perfect Complements:** $U(x, y) = \min(ax, by)$

Cobb-Douglas

$U(x, y) = x^\alpha y^\beta$. Our optimality condition is

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \implies \frac{\alpha x^{\alpha-1} y^\beta}{P_x} = \frac{\beta x^\alpha y^{\beta-1}}{P_y} \implies x = \frac{\alpha P_y}{\beta P_x} y$$

Our budget constraint is

$$I = P_x x + P_y y$$

Solve these two equations for x^* and y^* .

Try It Yourself

Consider $U(x, y) = x^{\frac{1}{4}}y^{\frac{3}{4}}$ and $I = 400$, $P_x = 20$, $P_y = 40$. Solve for the Consumers optimal choices of goods x and y and the resulting demand functions.

Solution

Here $\alpha = \frac{1}{4}$ and $\beta = \frac{3}{4}$ (note $\alpha + \beta = 1$). Using the general Cobb-Douglas result:

$$x^*(P_x, P_y, I) = \frac{\alpha I}{P_x} \qquad y^*(P_x, P_y, I) = \frac{\beta I}{P_y}$$

Plugging in $I = 400$, $P_x = 20$, $P_y = 40$:

$$x^* = \frac{\frac{1}{4}(400)}{20} = \frac{100}{20} = 5 \qquad y^* = \frac{\frac{3}{4}(400)}{40} = \frac{300}{40} = 7.5$$

Optimal bundle: $x^* = 5$, $y^* = 7.5$. **Demand functions:** $x^* = \frac{I}{4P_x}$,

$$y^* = \frac{3I}{4P_y}.$$

Perfect Substitutes

$U(x, y) = ax + by$. Our optimality condition is

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \implies \frac{a}{P_x} = \frac{b}{P_y}$$

Wait! This isn't always true!!

- e.g. what if $U(x, y) = x + y$, $P_x = 1$, and $P_y = 2$. This says $1 = 1/2$ which isn't true

Since both goods are perfectly substitutable, you should consume **ONLY** the good that provides marginal utility per dollar (bang for your buck)

Perfect Substitutes

$U(x, y) = ax + by$. For perfect substitutes our demand rule is simple:

1. If $MU_x/P_x > MU_y/P_y$:

$$x^* = I/P_x \quad \text{and} \quad y^* = 0$$

2. If $MU_x/P_x < MU_y/P_y$:

$$x^* = 0 \quad \text{and} \quad y^* = I/P_y$$

3. If $MU_x/P_x = MU_y/P_y$:

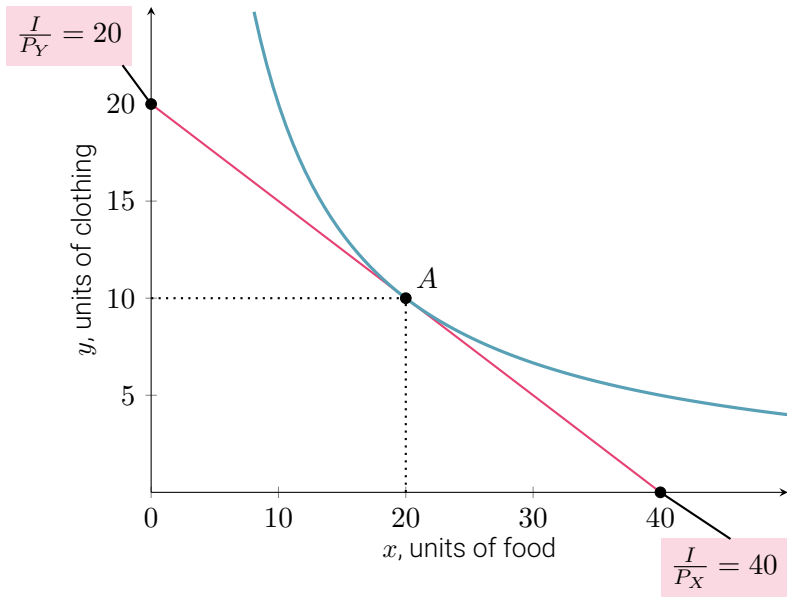
Pick any bundle on your budget line

So far are our maximization problems have had interior solutions where positive quantities of both goods are consumed, e.g. $x^* > 0$ & $y^* > 0$

Key characteristic is that the optimal solution is where the indifference curve is tangent to the budget constraint, which requires preferences to be “well-behaved”:

- Smooth
- Convex (curved in)
- Not touch the two axis

Figure: Utility Maximization

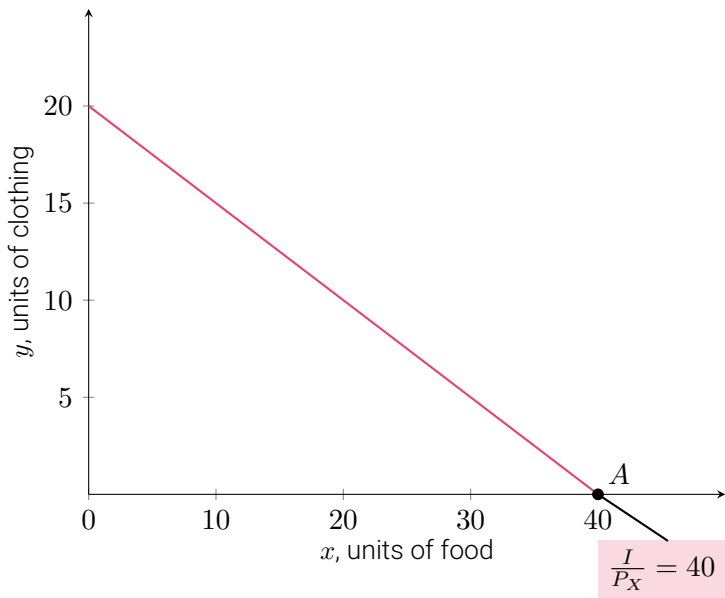


At a corner solution only one good is consumed at a positive quantity,
e.g. $x^* > 0$ & $y^* = 0$

Occurs when preferences are not well-behaved, such as by crossing one of the axis

Optimal solution is to spend all income on one good and none on the other good

Figure: Utility Maximization



Try It Yourself

Consider $U(x, y) = xy + 7y$ and $I = 10, P_x = 1, P_y = 2$. Solve for the Consumers optimal choices of goods x and y and the resulting demand functions.

Solution

This isn't Cobb-Douglas, so we derive marginal utilities directly:

$$MU_x = y \qquad MU_y = x + 7$$

The optimality condition is

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \implies \frac{y}{1} = \frac{x+7}{2} \implies x = 2y - 7$$

Solution (continued)

Plug into the budget constraint $x + 2y = 10$:

$$(2y - 7) + 2y = 10 \implies 4y = 17 \implies y^* = \frac{17}{4} = 4.25$$

$$x^* = 2(4.25) - 7 = 1.5$$

Optimal bundle: $x^* = 1.5$, $y^* = 4.25$ (an interior solution, since both are positive).

Demand functions (solving the same way in terms of P_x , P_y , I):

$$x^*(P_x, P_y, I) = \frac{I - 7P_x}{2P_x}$$

$$y^*(P_x, P_y, I) = \frac{I + 7P_x}{2P_y}$$

Perfect Complements

$U(x, y) = \min(ax, by)$. How do we even take a derivative of this?

Instead, recognize that perfect complements means you have to consume always in the proportion $ax = by$.

- If $ax > by$, you are wasting money on x and vice-versa if $by > ax$

$ax = by$ is therefore our **optimality condition**! Our budget constraint is

$$I = P_x x + P_y y$$

Solve these two equations for x^* and y^* .

Perfect Complements

Try It Yourself

Maria always uses 2 scoops of coffee (x) for every 1 cup of milk (y), so her preferences are given by $U(x, y) = \min(2x, y)$.

She has $I = 100$ to spend. Coffee costs $P_x = 4$ per scoop, and milk costs $P_y = 3$ per cup. Solve for her optimal choices of x and y .

Solution

Here $a = 2$ and $b = 1$, so the optimality condition $ax = by$ becomes

$$2x = y$$

Plug into the budget constraint $P_x x + P_y y = I$:

$$4x + 3(2x) = 100 \implies 4x + 6x = 100 \implies 10x = 100 \implies x^* = 10$$

Then $y^* = 2x^* = 20$.

Optimal bundle: $x^* = 10$ scoops of coffee, $y^* = 20$ cups of milk – exactly a 1:2 ratio, as her preferences require.

Lecture 5 - Theory of Demand

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Overview

In previous lectures, we looked at how consumers rank bundles of goods, and how, given prices and their budget, they choose the utility maximizing bundle.

- We also considered several applications of this theory of consumer choice.

In this lecture, we will consider where a demand curve for a product comes from...

- ...and what happens to that demand curve when exogenous variables, such as prices or income, change.
- We will also extend the analysis to the labor-leisure trade-off.

Optimal Choice and Demand

What happens to the budget line when the price of good x decreases?

- The set of feasible baskets increases
- If p_x decreases, marginal utility per dollar increases for good $x \implies$ consumer more of good

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

The Law of Demand says that as the price of x decreases, the quantity of x consumed increases.

Optimal Choice and Demand

Try It Yourself

Which of the following demand functions violates the law of demand?

A) $x^*(P_x, P_y, I) = 20 - P_x + P_y$

B) $x^*(P_x, P_y, I) = \frac{20 - P_y}{I - P_x}$

C) $x^*(P_x, P_y, I) = \frac{20 * I}{P_x + P_y}$

Solution

The law of demand requires quantity demanded to *decrease* as P_x increases (i.e. $\partial x^* / \partial P_x < 0$).

A) $\frac{\partial x^*}{\partial P_x} = -1 < 0$. ✓ Satisfies the law of demand.

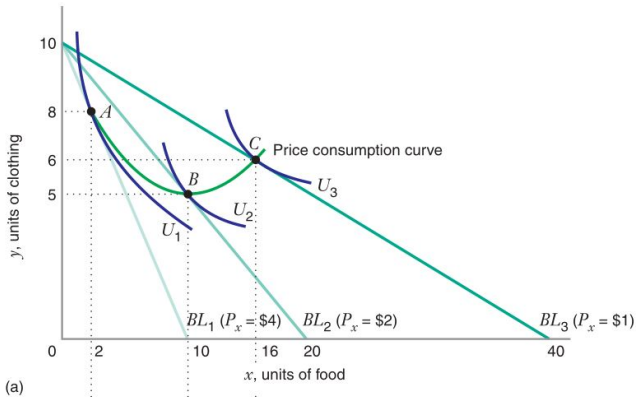
B) $x^* = \frac{20 - P_y}{I - P_x}$. Taking the derivative: $\frac{\partial x^*}{\partial P_x} = \frac{20 - P_y}{(I - P_x)^2} > 0$ (assuming $20 > P_y$ and $I > P_x$, both needed for $x^* > 0$). **This is positive – demand *increases* as price rises! This violates the law of demand.**

C) $\frac{\partial x^*}{\partial P_x} = \frac{-20I}{(P_x + P_y)^2} < 0$. ✓ Satisfies the law of demand.

Answer: B violates the law of demand.

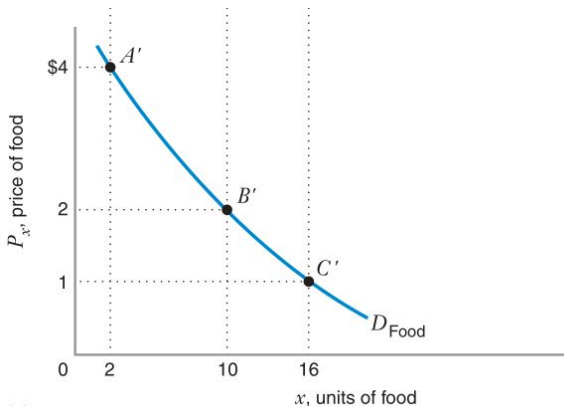
Optimal Choice and Demand

Below is a set of optimal bundles corresponding to various prices for good x . The line connecting these bundles is the **price consumption curve**.



Optimal Choice and Demand

These optimal bundles also map out the consumer's demand curve for good x .



(b)

Optimal Choice and Demand

Note that as we move further *along* the demand curve (i.e. as price falls), we reach higher and higher indifference curves.

Remember that the demand curve represents a consumer's **willingness to pay** for a good. Two ways to think about the demand curve:

1. A consumer's demand curve tells us how much the consumer is willing to pay *for a given quantity*.
2. A consumer's demand curve tells us how much the consumer is willing to purchase *for a given price*.

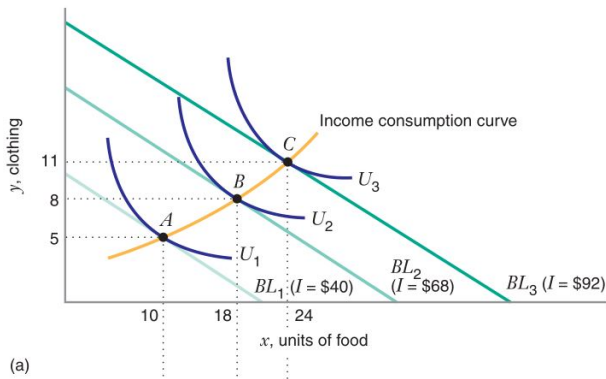
Optimal Choice and Demand

What about a change in income?

- We can also map out (in a similar way to the price-consumption curve) what happens to a consumer's demand when their *income* changes.
- This is called an **income consumption curve**.

Optimal Choice and Demand

Below is the set of optimal bundles corresponding to various income levels. The connecting line is the **income consumption curve**.



Optimal Choice and Demand

Remember that as income increases, the slope of the budget line doesn't change.

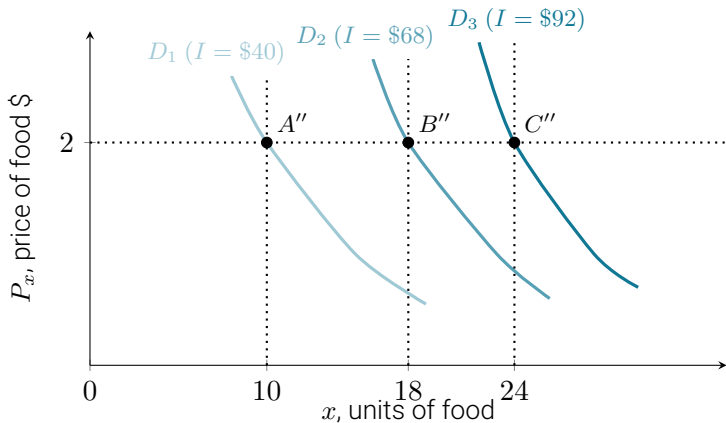
- Ratio of prices stays the same!

The optimal bundles derived from various income levels can be used to map out *shifts* of the demand curve (as in the following graph).

- The income consumption curve is *not always* upward sloping (e.g. stop buying ramen as your income grows)

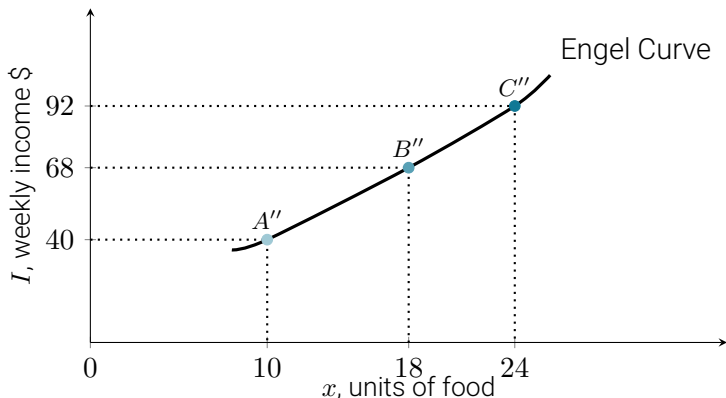
Optimal Choice and Demand

Below you can see how changes in income lead to shifts in the demand curve.



Engel Curves

Another way of showing how a consumer's demand for a particular good varies with income is to draw an **Engel curve** which relates quantity demanded to income



Engel Curves

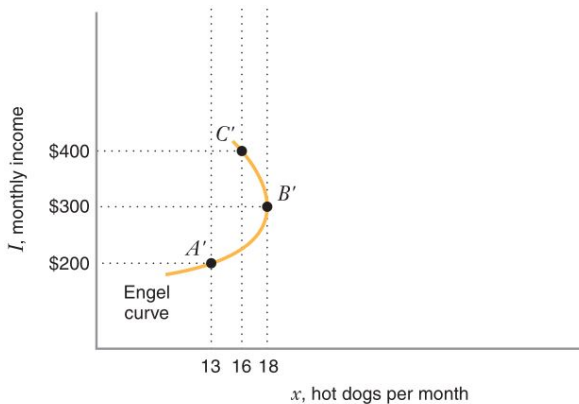
If a consumer's Engel curve for a good is upward sloping (i.e. if as income rises, their consumption of the good increases), then the good is a **normal good**.

If the opposite is true, then the good is an **inferior good**.

- It is possible for a good to be a normal good over some interval, and an inferior good over another interval.
- What are some examples of normal and inferior goods?

Engel Curves

In this example, hot dogs are a normal good when the consumer's income is lower, and an inferior good once their income is higher.



(b)

Engel Curves

Try It Yourself

Which of the following product demand functions indicates that the good is an **inferior** good?

A) $x^*(p_x, I) = 2 \frac{I}{P_x}$

B) $x^*(p_x, I) = 2 * I * P_x$

C) $x^*(p_x, I) = 2 * I * (10 - P_x)$

D) $x^*(p_x, I) = 2 \left(\frac{40 - I}{P_x} \right)$

Solution

A good is **inferior** if quantity demanded *falls* as income rises, i.e.

$$\partial x^* / \partial I < 0.$$

A) $\frac{\partial x^*}{\partial I} = \frac{2}{P_x} > 0$. Normal good.

B) $\frac{\partial x^*}{\partial I} = 2P_x > 0$. Normal good.

C) $\frac{\partial x^*}{\partial I} = 2(10 - P_x) > 0$ (assuming $P_x < 10$, needed for $x^* > 0$).
Normal good.

D) $x^* = \frac{80-2I}{P_x} \implies \frac{\partial x^*}{\partial I} = \frac{-2}{P_x} < 0$. **Inferior good!**

Answer: D

Substitution and Income Effects

Previously, we looked at the overall effect of changes in prices and income on demand for two goods.

Here we will break that overall effect into two parts:

1. One due to the change in marginal utility per dollar of the affected good $\frac{MU_x}{p_x}$
2. The other due to the fact that that $\frac{I}{p_x}$ increases

Substitution and Income Effects

1. The **substitution effect** refers to the change in demand caused by a change in the relative prices of two goods.
2. The **income effect** refers to the change in demand for a good caused by a change in the consumer's *purchasing power* due to the price change.

Substitution and Income Effects

Consider a case where the price of good x decreases

Substitution Effect

At the original bundle, $\frac{MU_x}{p_x}$ is now greater than $\frac{MU_y}{p_y}$ (since before the price changes, the two were equal)

You can now reach the same level of utility for less money by buying less of y and more of x . This is the **substitution effect**

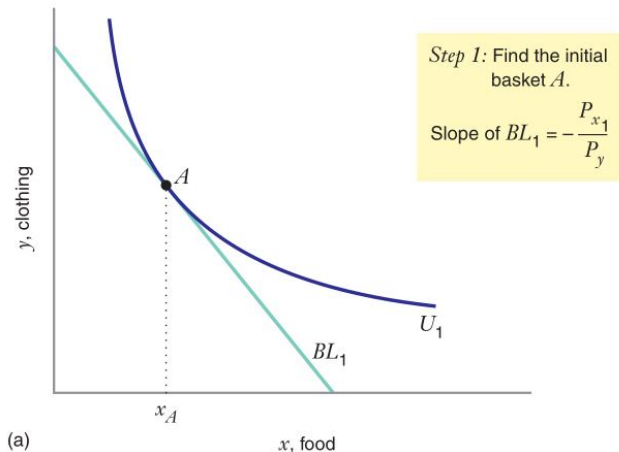
Income Effect

The **substitution effect** told us we reached the original utility for less money. So we have money left to spend.

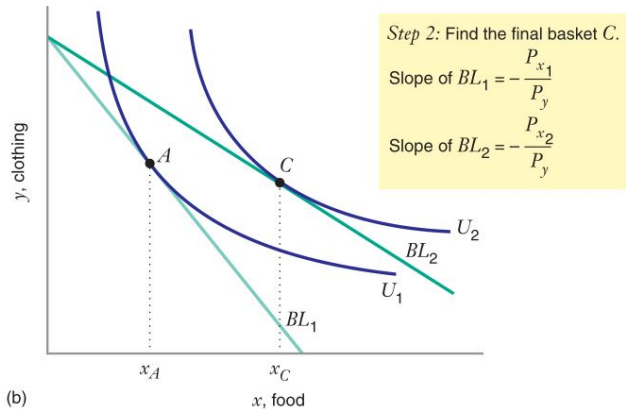
The additional consumption is the **income effect**!

Substitution and Income Effects

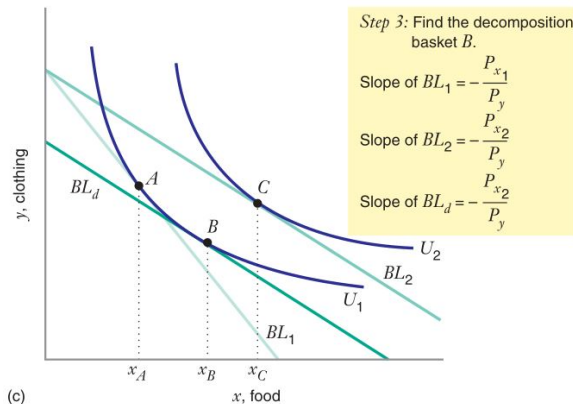
On the following slides, we will look at how to graphically decompose a price change into income and substitution effects.



Substitution and Income Effects

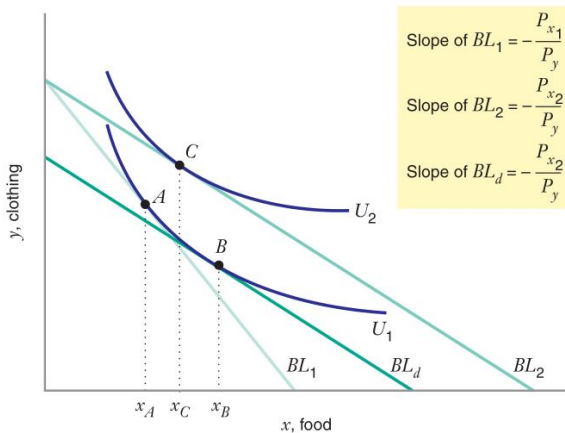


Substitution and Income Effects



Notice that in this case, the income effect leads to an increase in consumption of good x . That is, food is a normal good.

Substitution and Income Effects



This is a case where food is an inferior good. The income effect leads to a decrease in consumption of good x.

Substitution and Income Effects

For both normal and inferior goods, the overall effect of an increase in price is a decrease in demand.

- For normal goods, both the income and the substitution effects are negative.
- For inferior goods, the income effect is positive, but the substitution effect is negative, and the overall effect is negative.

Substitution and Income Effects

Try It Yourself

Suppose that peanut butter is an inferior good for many people.

If the price increases, then would we expect demand for good x to increase or decrease as a result of the substitution effect? What about due to the income effect?

Solution

The price of peanut butter increases. Since it is an **inferior** good:

	Substitution Effect	Income Effect
x	↓	↑

- **Substitution effect:** Consumers substitute away from the relatively more expensive good (↓).
- **Income effect:** Lower purchasing power increases demand for an **inferior good** (↑).
- **Overall effect:** Ambiguous because the two effects work in opposite directions.

Estimating Changes in Consumer Welfare

Suppose the government wants to impose a tax on fast food. How can we measure the impact on consumer welfare?

- As we've talked about, we are not able to measure utility directly.
- But maybe we can measure changes to consumer well-being in monetary terms.
- We can do this in a few different ways.

Consumer Surplus

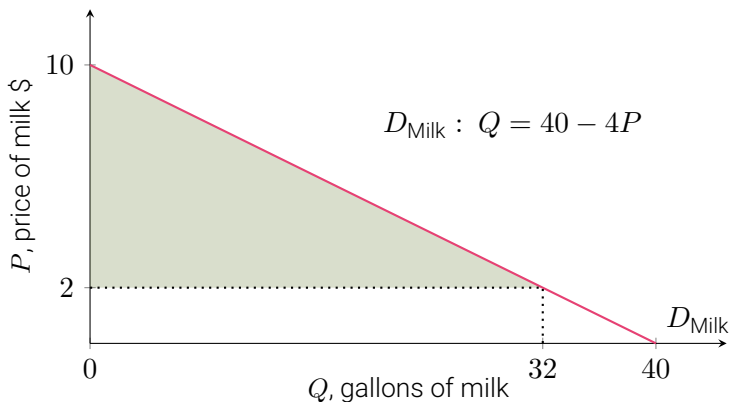
Consumer Surplus is the difference between the maximum amount that a consumer is *willing to pay* for a good and the amount they must *actually pay* in the marketplace.

For example, suppose you want to buy a new pair of Nike running shoes. You would be willing to pay up to \$120 for them, but they are on sale for \$80.

- Your consumer surplus if you buy the shoes is \$40.
- If the government then imposed a tax on Nike shoes, such that the price consumers pay increased by \$15, then your consumer surplus will decrease by \$15.

Consumer Surplus

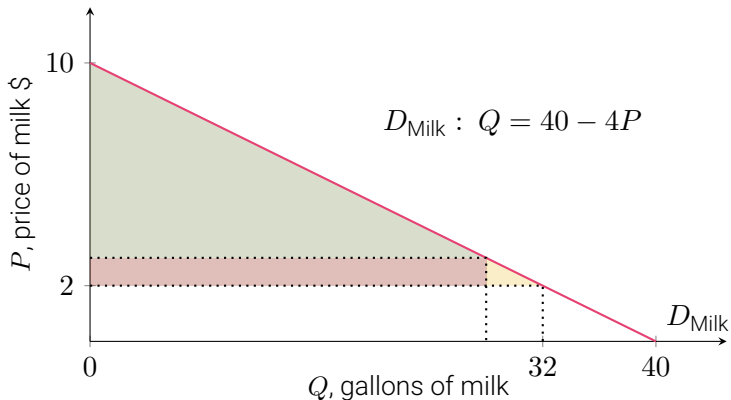
Graphically, consumer surplus is calculated as the area below the demand curve and above the market price.



Consumer Surplus

Suppose that you know the demand function for some market. In order to find the consumer surplus:

1. First, plot the demand function
2. Then, calculate the area of the triangle created by the market price and the demand curve.



When you increase the price, the consumer surplus goes down for two reasons:

- First, people who just barely bought no longer buy items
- Second, people that still buy after the price change have to pay more

Consumer Surplus

Try It Yourself

Suppose that consumer demand for organic, gluten-free dog food in Boulder can be represented by the following equation: $Q_D = 80 - 5P$. If the price of organic, gluten-free dog food is \$10 per pound, what is the consumer surplus?

Solution

Solve the demand equation for P in terms of Q :

$$Q_D = 80 - 5P \implies P = 16 - \frac{Q}{5}$$

The **choke price** (where $Q = 0$) is $P = 16$ – the highest price anyone is willing to pay.

At $P = 10$: $Q_D = 80 - 5(10) = 30$.

Consumer surplus is the area of the triangle above the market price and below the demand curve:

$$CS = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(30)(16 - 10) = \frac{1}{2}(30)(6) = 90$$

Consumer surplus = \$90.

Compensating and Equivalent Variation

When considering the effects of price changes, we have two additional measures that may be useful: compensating variation and equivalent variation.

Compensating variation (CV) measures how much a consumer's income would have to change to keep them at the **same** utility level as before a price change, **after** the price change has occurred.

Equivalent variation (EV) measures how much a consumer's income would have to change **before** a price change has occurred to make them indifferent between the price change and the additional income.

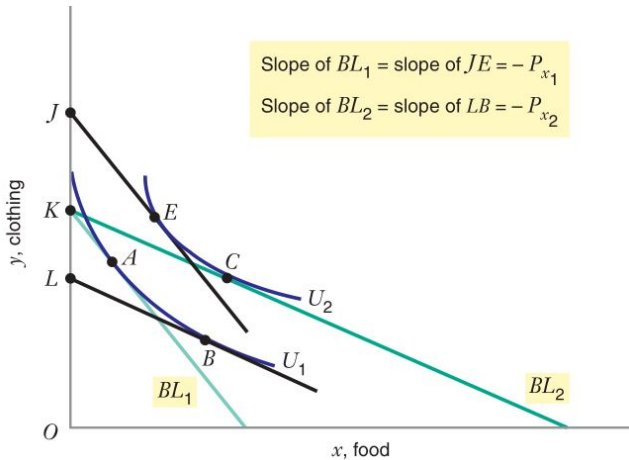
Compensating and Equivalent Variation

Here's another way to remember the difference between EV and CV:

- Compensating variation measures the income that would be needed to **compensate** the consumer for a price change.
- Equivalent variation measures the income change that would be **equivalent** to a hypothetical or proposed price change.

Compensating and Equivalent Variation

Below is a graphical representation of EV and CV.



Compensating and Equivalent Variation

Note that CV and EV are often not the same.

- This is due to income effects.
- EV measures income needed to get to the new level of utility at **old** prices, and CV measures income needed to get back to the original utility level at **new** prices.

Compensating and Equivalent Variation

Which measure you should use depends on the situation:

- If the government wants to impose a larger tax on marijuana to reduce consumption, but doesn't want people to be worse off, CV is appropriate.
- If the government promised a subsidy on electric vehicles, and then decided to give income tax credits instead, EV is appropriate.

Compensating and Equivalent Variation

Try It Yourself

Suppose that a private company offered its workers an employee discount on all products, but then later decided to instead increase their wages. They want to know how much they need to increase wages by to make people as well off as the proposed discount would have. Should you use the Compensating Variation or the Equivalent Variation?

Solution

Use **Equivalent Variation (EV)**.

The employee discount was only **proposed** and never took effect. EV asks:

How much additional income, at the original prices, would make workers as well off as the proposed discount?

Rule of thumb:

- **EV:** Price change is hypothetical.
- **CV:** Price change has already occurred.

The Labor-Leisure Decision

One decision that almost every individual must make is whether to work, and how much. Economists refer to this as the **labor-leisure decision**.

- Leisure refers to all nonwork activities, such as eating, sleeping, recreation, and entertainment.
- Many people would love to allocate **all** of their time toward leisure.
- But why can't they?

The Labor-Leisure Decision

A consumer has to earn money to spend it on goods. So she must give up some leisure in order to earn money.

A consumer has 24 hours in a day. She spends L of those on leisure activities, and $(24 - L)$ on labor activities. Suppose her wage is w

- Then her daily income is therefore $w(24 - L)$
- This can use to buy a composite good at a price of \$1.

The Labor-Leisure Decision

The consumer's budget constraint can be written as

$$Y = w(24 - L)$$

We can also rearrange that, so that it looks more like our typical budget constraint.

$$24w = wL + Y$$

The Labor-Leisure Decision

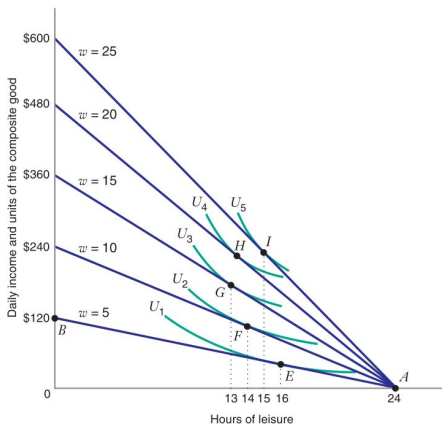
$$24w = wL + Y$$

Written this way, we can think of it in more conventional terms.

- We can think of her income as $24w$.
- She can spend her income on leisure (at a 'price' of w , her wage).
- And she can spend it on the composite good.
- Her utility U depends on the amount of leisure time she has, and on how much of the composite good she can buy.

The Labor-Leisure Decision

Below is an illustration of the budget lines and optimal leisure-composite good bundles for a consumer at various wages.



The Labor-Leisure Decision

Let's consider the **trade-off** at play here:

- If the consumer spends all of their time on leisure, they will earn no money and have nothing to spend on the composite good.
- On the other hand, if they work, they can buy more stuff

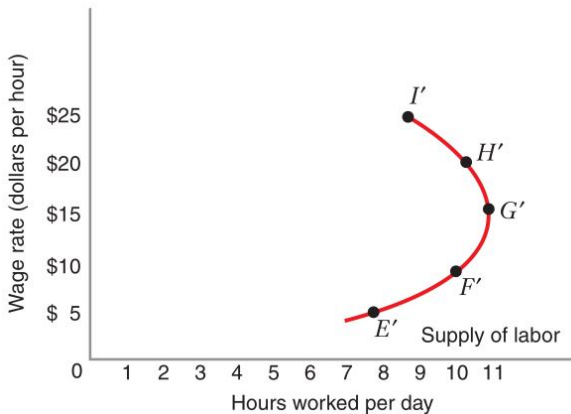
The Labor-Leisure Decision

In general, we assume that the consumer's labor supply curve (the curve tracing out all of the optimal bundles for various wages) is backward bending.

- That is, as their wage increases, they work more up to a certain point.
- Beyond that, as their wage increases they work less.
- This again relates to the *income* and *substitution* effects on labor supply.

The Labor-Leisure Decision

Below is a backward-bending supply curve. At low wages, an increase in the wage results in a greater supply of labor. But eventually, supply diminishes as the wage rises.



The Labor-Leisure Decision

In this model, the substitution effect is negative;

- As the wage increases, the opportunity cost of an hour of leisure, w , increases, and leisure becomes more 'expensive'.
- So the worker will substitute some leisure for more of the composite good.

The income effect is positive.

- As a consumer's wage increases, their income increases, so they 'buy' more leisure (because leisure is a normal good).

The Labor-Leisure Decision

Initially, the substitution effect dominates. As the wage rises, people work more.

But eventually, the income effect dominates. As the wage rises, people work less

The Labor-Leisure Decision

Try It Yourself

Suppose that a worker in the U.S initially earns \$10/hour. Then, due to company-wide pay cuts, the worker is given a pay cut of \$1.50/hour, and as a result, decides to work more. The worker's decision to work more implies that which effect (the income or substitution effect) is dominating?

Solution

The wage *fell* (from \$10 to \$8.50/hour). Consider each effect on labor supply:

Substitution effect: As the wage falls, the opportunity cost of leisure falls – leisure becomes relatively 'cheaper'. The worker should substitute *toward* leisure and away from work. This predicts **less** work.

Income effect: A lower wage means lower income. Since leisure is a normal good, lower income means the worker 'buys' *less* leisure. This predicts **more** work.

Since the worker actually chose to work **more** after the pay cut, this matches the prediction of the income effect.

Answer: the income effect is dominating.

Conclusion

This wraps up the content on consumer theory.

- We discussed how consumers rank various bundles of goods.
- And how consumers make decisions in a variety of contexts.
- We've also discussed how consumer welfare, and changes in it, can be measured.

Lecture 6 - Producer Theory

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

In the next lecture, we begin discussing production functions.

- How do we represent, both graphically and numerically, the relationship between inputs and output?
- What role does technology play in production?
- What is the relationship between different mixes of inputs, and the level of output?

Inputs and the Production Function

Production means transforming things such as labor, materials, and energy, into finished products or services.

- **Inputs** are the capital, labor, materials, and energy used to produce **output**, or final goods and services.
- A **production function** relates the amount of every input used to the amount of output that will be produced.

All of the following slides is going to seem really similar to the consumer setting. The skills you have already learned will help with this a lot

Production Functions

Production functions take the form

$$Q = f(L, K)$$

- Where Q is the quantity of output, L and K are the quantity of labor and capital used, respectively.

This relation tells us that the quantity of output is a function of the amount of inputs used.

Production Functions

Output is not always simply a function of labor and capital.

- It could also be a function of natural resources, such as oil or diamonds, or of raw materials such as wood and computer chips.
- Even labor can be divided up by type (for example, high- and low-skilled labor).

Examples of Production Functions

Example 1: Lawn Care Company

- $Q(K, L) = 1/2 * \min(K, L)$
- K is number of lawn mowers, L is number of workers, Q is number of lawns mowed per hour.
- Each worker needs exactly 1 lawn mower.

Example 2: Grocery Store Checkout

- $Q(K, L) = 10K + 15L$
- K is number of self-checkout machines, L is the number of cashiers, and Q is customers served per hour.
- Cashiers and self-checkout machines are substitutes.

Examples of Production Functions

Example 3: Dog Grooming Company

- $Q(K, L) = 10K - K^2 + 8L - L^2 + KL$
- Q is number of dogs groomed per day, K is number of dog grooming stations, and L is number of dog groomers.
- More workers and more stations mean more dogs groomed, but eventually the grooming room gets too crowded ($-L^2$).

Production Functions

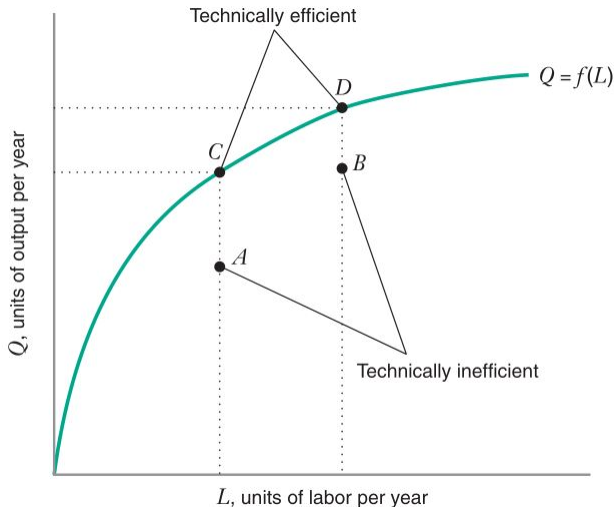
Production functions tell us the maximum output that a firm can produce from a given combination of inputs.

But output could be lower than what is technically feasible. Why?

- All points on or below the production function make up the firm's **production set**.
- Therefore, we say that any point on the production function is considered **technologically efficient**.
- Any point below the production function is **technologically inefficient**.

Production Functions

Below is a graphical representation of a production function $Q = f(L)$.



Marginal and Average Product

For now, we will consider production functions with only one input; labor.

- Firms consider the **marginal product** of each of their inputs when maximizing profit.
- The marginal product of labor measures how output changes if we add an additional unit of labor.

$$MP_L = \frac{\text{change in total product}}{\text{change in quantity of labor}} = \frac{\Delta Q}{\Delta L} = \frac{\partial Q}{\partial L}$$

Marginal and Average Product

Increasing marginal returns to labor - MP_L is increasing

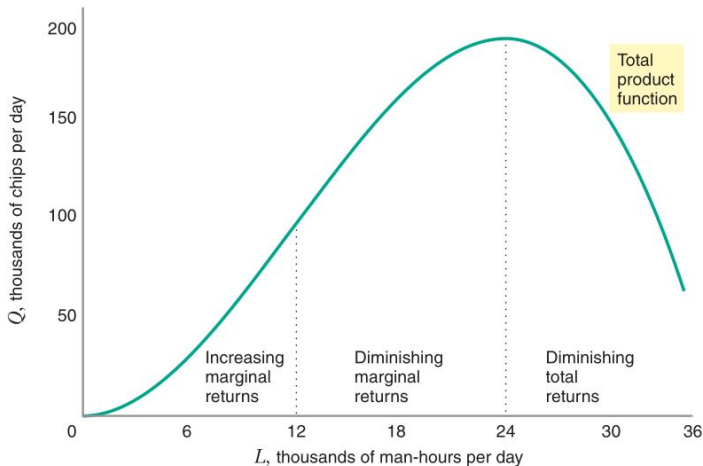
- Each additional unit of labor produces more output than the previous unit

Diminishing marginal returns to labor - MP_L is decreasing.

Diminishing total returns to labor - additional labor results in lower *total* output.

Marginal and Average Product

Below is a production function with labor as its only input.



Marginal and Average Product

Try It Yourself

Suppose a firm uses only labor in the production of its good, and that the firm's production function is $Q(L) = 4\sqrt{L}$. What is the firm's marginal product of labor if $L = 4$?

Marginal and Average Product

Solution

Take the derivative of $Q(L) = 4\sqrt{L} = 4L^{\frac{1}{2}}$ with respect to L :

$$MP_L = \frac{dQ}{dL} = 4 \cdot \frac{1}{2} L^{-\frac{1}{2}} = \frac{2}{\sqrt{L}}$$

Evaluating at $L = 4$:

$$MP_L = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1$$

The marginal product of labor at $L = 4$ is 1.

Marginal and Average Product

In reality, most production processes experience diminishing marginal returns eventually.

- This is what economists call the **law of diminishing marginal returns**.

Another measure of productivity commonly discussed is **average product**. For example, the average product of labor is given by:

$$AP_L = \frac{\text{total product}}{\text{quantity of labor}} = \frac{Q}{L}$$

Marginal and Average Product

Here is a numerical example of calculating marginal and average product:

L	Q	MP_L	AP_L
0	0	-	-
2	12		
4	20		
6	24		
8	24		

Marginal and Average Product

Filling in the table: $MP_L = \Delta Q / \Delta L$ between rows, $AP_L = Q / L$ at each row.

L	Q	MP_L	AP_L
0	0	-	-
2	12	$12/2 = 6$	$12/2 = 6$
4	20	$8/2 = 4$	$20/4 = 5$
6	24	$4/2 = 2$	$24/6 = 4$
8	24	$0/2 = 0$	$24/8 = 3$

Notice MP_L falls throughout (diminishing marginal returns), and once MP_L drops below AP_L (between $L = 2$ and $L = 4$), AP_L starts falling too – exactly the relationship on the next slide. By $L = 8$, $MP_L = 0$:

Marginal and Average Product

Try It Yourself

Given the following production table, find the marginal product of labor and the average product of labor

L	Q	MP_L	AP_L
0	0	-	-
6	15		
9	24		
12	30		
15	33		

Solution

$MP_L = \Delta Q / \Delta L$ between consecutive rows; $AP_L = Q / L$ at each row.

L	Q	MP_L	AP_L
0	0	-	-
6	15	$15/6 = 2.5$	$15/6 \approx 2.5$
9	24	$9/3 = 3$	$24/9 \approx 2.67$
12	30	$6/3 = 2$	$30/12 = 2.5$
15	33	$3/3 = 1$	$33/15 = 2.2$

Notice MP_L is falling throughout (diminishing marginal returns), and AP_L rises while $MP_L > AP_L$, then falls once $MP_L < AP_L$ – exactly the relationship from the next slide.

Relationship Between Marginal and Average Product

There is a systematic relationship between average product and marginal product.

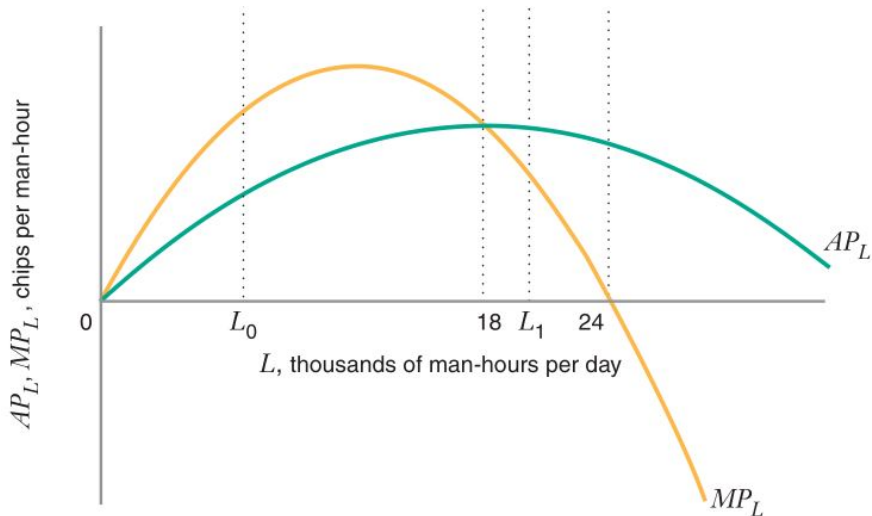
- If AP_L increases in L , then $MP_L > AP_L$.
 - If your last unit of labor produced more than the average, then your average increases
- If AP_L decreases in L , then $MP_L < AP_L$.
 - If your last unit of labor produced less than the average, then your average decreases
- MP_L intersects AP_L at the maximum of AP_L .

Relationship Between Marginal and Average Product

As an analogy, consider calculating the average grade on an exam. Suppose the average is 80%.

- If one additional student takes the exam late, and they score 70%...
- Then the *marginal* grade is lower than the average, and the *average* falls.
- On the other hand, if the *marginal* grade is 90%, then the *average* rises.

Relationship Between Marginal and Average Product



Production with Multiple Inputs

Now, let's consider a firm that uses two goods, labor (L) and capital (K).

- Below is a table of output, similar to the table we saw earlier, but with two inputs:

		K^{**}					
		0	6	12	18	24	30
L^{**}	0	0	0	0	0	0	0
	6	0	5	15	25	30	23
	12	0	15	48	81	96	75
	18	0	25	81	137	162	127
	24	0	30	96	162	192	150
	30	0	23	75	127	150	117

		K^{**}					
		0	6	12	18	24	30
L^{**}	0	0	0	0	0	0	0
	6	0	5	15	25	30	23
	12	0	15	48	81	96	75
	18	0	25	81	137	162	127
	24	0	30	96	162	192	150
	30	0	23	75	127	150	117

- As K increases, for any given level of L , output first increases and then decreases. The same is true for L .
- This illustrates the **law of diminishing marginal product** for both inputs.

Marginal Product

Suppose we want to know how MP_L changes as we increase L .

- Well, that depends on what the level of capital is.
 - For example, if you had one coffee machine, adding a bunch of workers produces no extra coffee

In order to find the marginal product of one input, we have to hold the other input fixed. For example,

$$\begin{aligned} MP_L &= \frac{\text{change in quantity of output } Q}{\text{change in quantity of labor } L} \bigg|_{K \text{ is held constant}} \\ &= \frac{\Delta Q}{\Delta L} \bigg|_{K \text{ is held constant}} = \frac{\partial Q}{\partial L} \end{aligned}$$

Marginal Product

Try It Yourself

Suppose a tutoring company has the production function

$Q(K, L) = 8K - K^2 + 4L - \frac{1}{2}L^2 + KL$. What is the marginal product of capital for this company?

Marginal Product

Solution

Take the partial derivative of $Q(K, L) = 8K - K^2 + 4L - \frac{1}{2}L^2 + KL$ with respect to K , holding L constant:

$$MP_K = \frac{\partial Q}{\partial K} = 8 - 2K + L$$

Isoquants

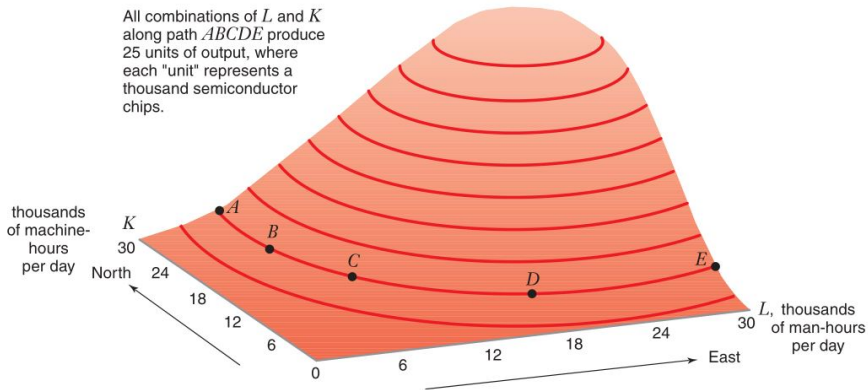
When thinking about consumers, we had a concept of indifference curves. Bundles of goods that created the same level of utility.

For producers, we think of **isoquants** which are bundles of inputs that create the same level of output

- A graph of isoquants and a graph of indifference curves are both just 'contour plots' of their respective functions.
- "Iso" is Greek for same. "quant" is short for quantity.

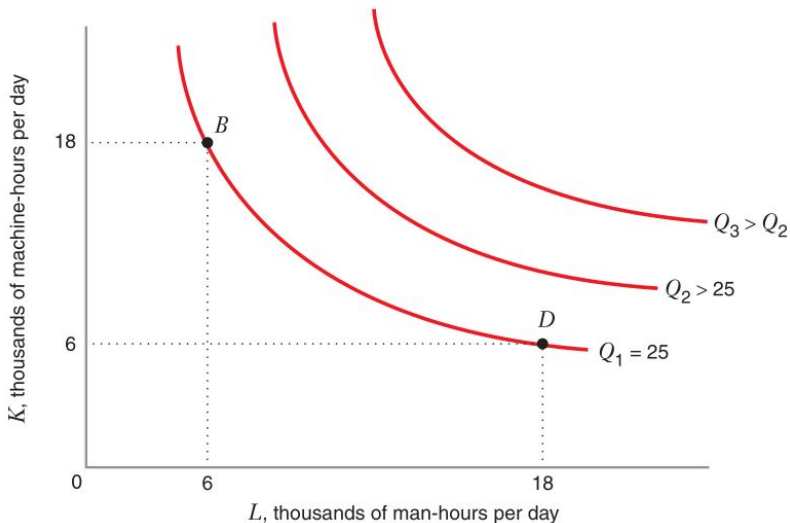
Isoquants

Below is a plot of a production function with countour lines (isoquants) drawn onto it.



Isoquants

And this is a subset of those isoquants plotted in two dimensions.



Marginal Rate of Technical Substitution

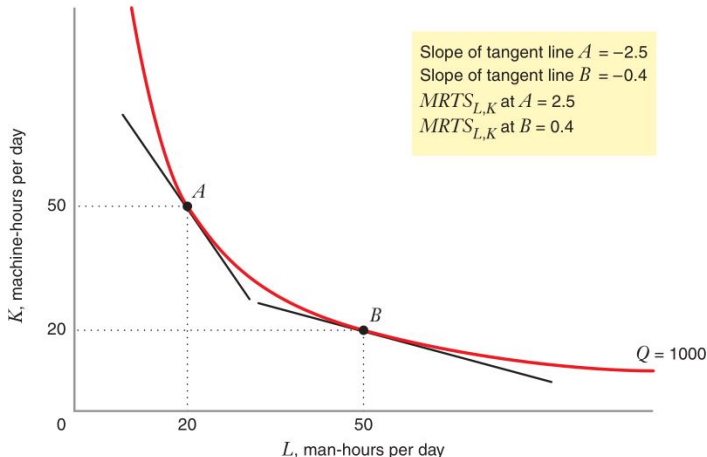
The isoquants illustrate the tradeoff that can be made in production between two inputs (holding quantity of output constant)...

- The slope of these isoquants tells us the rate at which inputs can be substituted, or the **marginal rate of technical substitution**. We call this $MRTS$ for short (similar to MRS !).

For example, the $MRTS_{L,K}$ tells us that if we want to keep output *constant*, then a one-unit *decrease* in labor can be replaced with $MRTS_{L,K}$ additional units of capital.

Marginal Rate of Technical Substitution

Below is a single isoquant for a production function. The slope of this isoquant at any given point is the $MRTS_{L,K}$.



Marginal Rate of Technical Substitution

Note that as we move along the isoquant to the right, the curve becomes flatter.

- This is because for every additional unit of capital you give up, you must compensate with more and more labor.
- This is known as the **diminishing marginal rate of technical substitution**.

Marginal Rate of Technical Substitution

We can show mathematically that the MRTS is the slope of the isoquant with a little bit of math:

$$\Delta Q = (\Delta K)(MP_K) + (\Delta L)(MP_L)$$

In order to stay on the same isoquant, $\Delta Q = 0$.

$$0 = (\Delta K)(MP_K) + (\Delta L)(MP_L)$$

$$\implies (\Delta K)(MP_K) = -(\Delta L)(MP_L)$$

$$\implies -\frac{\Delta K}{\Delta L} = \frac{MP_L}{MP_K}$$

$$\implies \frac{MP_L}{MP_K} = MRTS_{L,K}$$

Marginal Rate of Technical Substitution

Try It Yourself

Suppose a firm has the production function

$Q(K, L) = 20K - 2K^2 + 14L - 2L^2$. Find the firm's $MRTS_{L,K}$ when $L = 2$ and $K = 3$.

Solution

First find the marginal products:

$$MP_L = \frac{\partial Q}{\partial L} = 14 - 4L$$

$$MP_K = \frac{\partial Q}{\partial K} = 20 - 4K$$

Evaluate at $L = 2, K = 3$:

$$MP_L = 14 - 4(2) = 6$$

$$MP_K = 20 - 4(3) = 8$$

$$MRTS_{L,K} = \frac{MP_L}{MP_K} = \frac{6}{8} = \frac{3}{4} = 0.75$$

Economic Region of Production

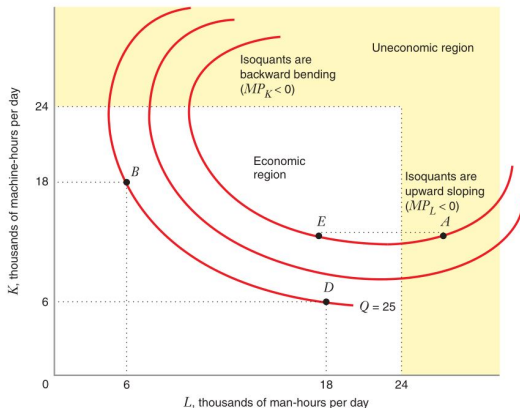
In theory, there could be points where a firm's $MRTS_{L,K}$ is negative, but a firm would never produce at these points.

- Usually this is because there is another input that is fixed (e.g. building space) that suffers from diminishing returns
- These are called **uneconomic regions of production**.

A firm would never produce in this region because they could choose an input combination with both less labor and less capital that produces the same quantity of output.

Economic Region of Production

This graph illustrates economic and uneconomic regions of production. Note that points A and E produce the same amount of output using the same amount of capital, but A uses substantially more labor.



Substitutability of Inputs

For some firms, labor and capital are easily substitutable.

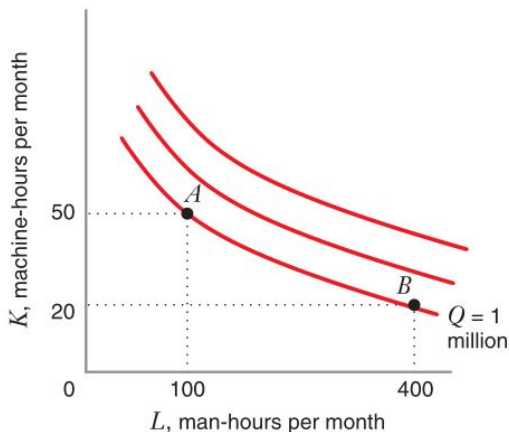
- Think of the example of cashiers and self-checkout machines at the grocery store.
- These firms can more easily switch between high capital-labor ratio and low capital-labor ratio, depending on their needs

But for other firms, labor and capital cannot be easily substituted.

- Such as in a lawn care company. You could substitute a lawnmower for workers who just pluck at the grass with their hands...

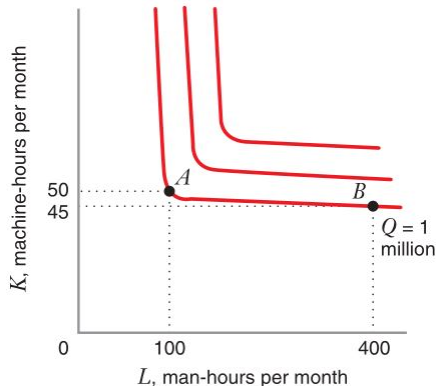
Substitutability of Inputs

The isoquants for the grocery store might look similar to these ones, where labor and capital are easily interchanged at some ratio (lets say 2 self-checkout machines to 1 cashier).



Substitutability of Inputs

The isoquants for a lawn care company might look like these ones, where, if we move away from the corners (lets say a 1-to-1 ratio of workers and lawn mowers), it takes a very large amount of labor to make up for a lost unit of capital, and vice versa.



The three ~~utility~~ production functions in this class

There are three kinds of production functions we will give you in this course.

If you want to do well in this class, you should practice these three functions a bunch (*calculate MRTS, draw isoquant curves, and later on calculate supply curves for K and L*)

Three Production Functions

1. **Cobb-Douglas:** $Q(K, L) = K^\alpha L^\beta$
2. **Perfect Substitutes:** $Q(K, L) = aK + bL$
3. **Perfect Complements:** $Q(K, L) = \min(aK, bL)$

Which functions represent the above two examples (lawncare and grocery store checkout)?

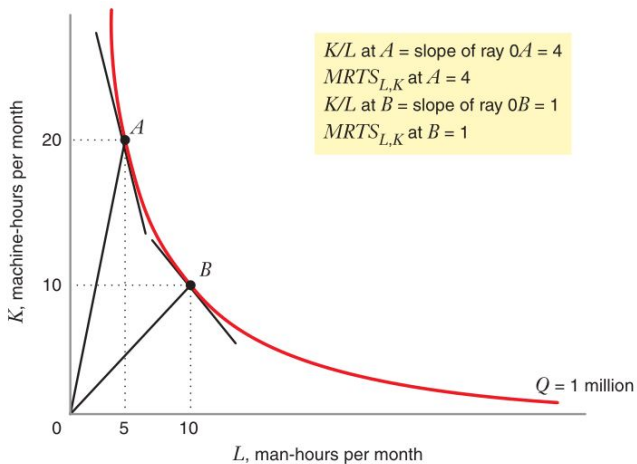
Elasticity of Substitution

We can use the elasticity of substitution to more accurately describe the substitution opportunities in a firm's production function.

- The **elasticity of substitution** measures how quickly the marginal rate of technical substitution changes as we move along the isoquant.

$$\begin{aligned}\sigma &= \frac{\text{percentage change in capital-labor ratio}}{\text{percentage change in } MRTS_{L,K}} \\ &= \frac{\% \Delta \frac{K}{L}}{\% \Delta MRTS_{L,K}}\end{aligned}$$

Elasticity of Substitution

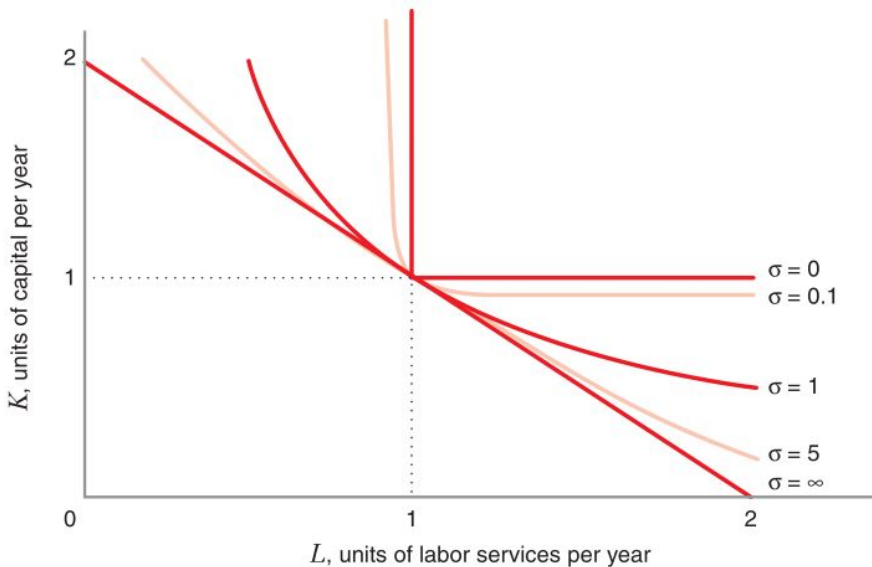


Elasticity of Substitution

What does the elasticity of substitution tell us?

- If σ is close to zero, there is little opportunity to substitute between inputs.
- If σ is large, there is greater opportunity to substitute between inputs.

Elasticity of Substitution



Returns to Scale

We looked at how output responds when a firm increases the quantity of some input, holding all others fixed.

- But what about when a firm increases all input quantities?

Returns to scale tell us the percentage increase in output when a firm increases all of its inputs by a given percentage amount.

Mathematically,

$$\text{Returns to scale} = \frac{\% \Delta \text{ quantity of output}}{\% \Delta \text{ quantity of all inputs}}$$

Returns to Scale

A production function is said to exhibit **constant returns to scale** if multiplying both inputs by a given factor increases output by the same factor. That is...

$$Q(\phi K, \phi L) = \phi Q(K, L)$$

- Suppose ϕ equals 2. Then if we double the amount of both inputs used, our output will double.

Increasing returns to scale implies that scaling both inputs by some factor increases output by *more* than that factor.

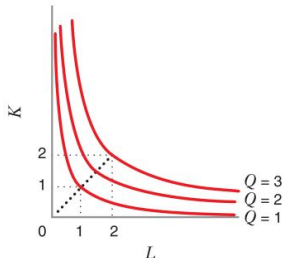
$$Q(\phi K, \phi L) > \phi Q(K, L)$$

Decreasing returns to scale implies that scaling both inputs by some factor increases output by *less* than that factor.

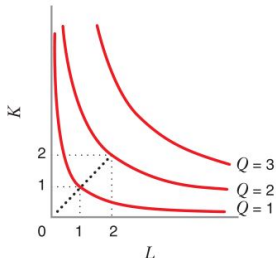
$$Q(\phi K, \phi L) < \phi Q(K, L)$$

Returns to Scale

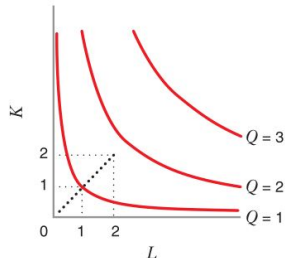
Below are the isoquants for firms with differing returns to scale



(a) Increasing Returns to Scale



(b) Constant Returns to Scale



(c) Decreasing Returns to Scale

Returns to Scale

Why are returns to scale important? Suppose a firm exhibits increasing returns to scale.

- Then they could produce more output than two firms that are each half the size of the single firm.
- Or they could produce the same amount of output at a *lower cost per unit*.

This can mean that a market is more efficiently served by one large firm, rather than two smaller ones.

- This is why governments may often allow certain monopolies to operate (such as utility companies).

Returns to Scale

Try It Yourself

Suppose a firm has the following production function:

$Q(K, L) = 4K^{\frac{1}{2}}L^{\frac{1}{2}}$. Does this production function exhibit increasing, decreasing, or constant returns to scale?

Solution

Scale both inputs by ϕ :

$$Q(\phi K, \phi L) = 4(\phi K)^{\frac{1}{2}}(\phi L)^{\frac{1}{2}} = 4\phi^{\frac{1}{2}}K^{\frac{1}{2}}\phi^{\frac{1}{2}}L^{\frac{1}{2}} = 4\phi K^{\frac{1}{2}}L^{\frac{1}{2}} = \phi Q(K, L)$$

Since $Q(\phi K, \phi L) = \phi Q(K, L)$ exactly, this production function exhibits **constant returns to scale**.

Returns to Scale

Try It Yourself

Which of the following production functions exhibits decreasing returns to scale?

A) $Q(K, L) = 2K + 3L$

B) $Q(K, L) = \sqrt{KL}$

C) $Q(K, L) = K^{1/3}L^{1/3}$

D) $Q(K, L) = 2\sqrt{K} + \sqrt{L}$

Solution

Scale both inputs by ϕ and check the resulting exponent on ϕ :

A) $Q(\phi K, \phi L) = 2\phi K + 3\phi L = \phi(2K + 3L) = \phi Q$. Exponent on ϕ is 1.

Constant returns.

B) $Q(\phi K, \phi L) = (\phi K)^{1/2}(\phi L) = \phi^{3/2}K^{1/2}L$. Exponent is $3/2 > 1$.

Increasing returns.

C) $Q(\phi K, \phi L) = (\phi K)^{1/3}(\phi L)^{1/3} = \phi^{2/3}Q$. Exponent is $2/3 < 1$.

Decreasing returns.

D) $Q(\phi K, \phi L) = 2(\phi K)^{1/2} + (\phi L)^{1/2} = \phi^{1/2}(2K^{1/2} + L^{1/2}) = \phi^{1/2}Q$.

Exponent is $1/2 < 1$. **Decreasing** returns.

Solution (continued)

Both C and D are mathematically homogeneous of degree less than 1, so both technically exhibit decreasing returns to scale.

C is the "textbook" Cobb-Douglas case: the exponents on K and L sum to less than 1 ($1/3 + 1/3 = 2/3$).

D is less obviously so, since it's a *sum* of power terms rather than a product – but because both terms scale by the same power of ϕ ($\phi^{1/2}$), the whole function is homogeneous of degree $1/2$, which is also decreasing returns.

Answer: C and D both exhibit decreasing returns to scale (only B is increasing, and A is constant).

Returns to Scale vs. Marginal Returns

It's important to understand the distinction between returns to scale and marginal returns.

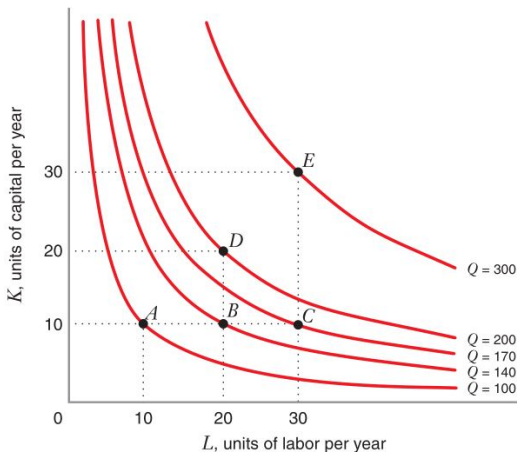
- Returns to scale pertain to the impact of an increase in *all input quantities* simultaneously.
- Marginal returns deal with increasing the quantity of *a single input*.

A firm can have diminishing marginal returns to its inputs, but still experience constant (or even increasing) returns to scale.

- Doubling the number lawnmowers doesn't increase output without additional workers. But doubling the number of both could potentially double output.

Returns to Scale vs. Marginal Returns

This graph illustrates a production function with diminishing marginal product of labor (see points A, B, and C), but constant returns to scale.



Technological Progress

Until now, we've treated a firm's production function as if it were fixed

- The quantity of inputs could vary, but the relationship between inputs and output did not change.

But **technological progress** can allow a firm to produce more output from a given combination of inputs.

- New production techniques can be discovered, worker's can learn new techniques, etc.
- This technological progress can shift the production function.

Technological Progress

As firms devise ways to produce the same amount of output with a lower quantity of inputs, the isoquants of a production function shift inward.

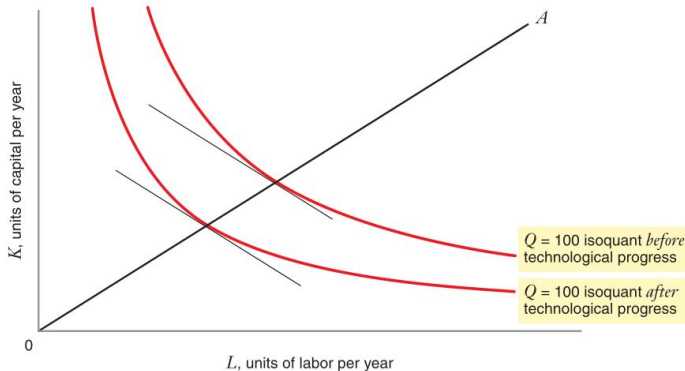
Technological progress can be broadly classified into three categories:

1. Neutral technological progress
2. Labor-saving technological progress
3. Capital-saving technological progress

Neutral technological progress refers to inward shifts in the isoquant (so that a given quantity of output can be produced using fewer inputs), where the $MRTS_{L,K}$ is unchanged.

- Both factors become more productive in equal proportions.

With **netural technological progress**, the same quantity of output can be produced with fewer inputs, but the $MRTS_{L,K}$ remains the same.

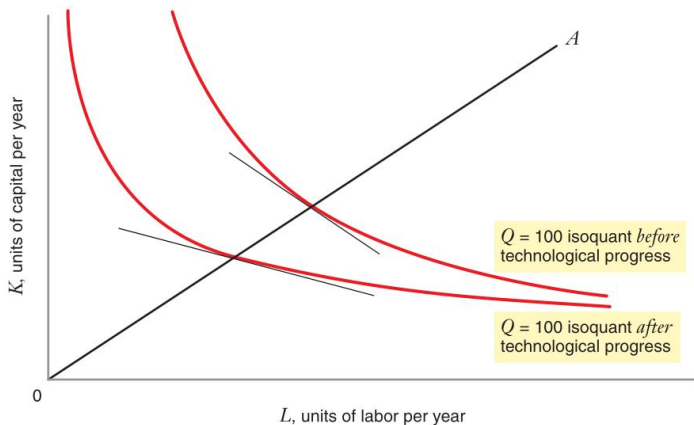


Technological Progress

With **labor-saving** (or **capital-biased**) technological progress, capital becomes more productive relative to labor.

- That means that MP_K increases proportionally more than MP_L , so the $MRTS_{L,K}$ decreases.

With **labor-saving technological progress**, the same quantity of output can be produced with fewer inputs, and the $MRTS_{L,K}$ decreases.



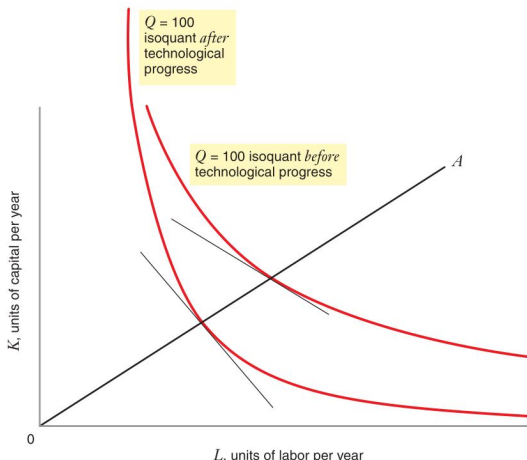
Technological Progress

With **capital-saving** (or **labor-biased**) technological progress, labor becomes more productive relative to capital.

- MP_L increases proportionally more than MP_K , so the $MRTS_{L,K}$ increases.

Technological Progress

With **capital-saving technological progress**, the same quantity of output can be produced with fewer inputs, and the $MRTS_{L,K}$ increases.



Try It Yourself

Suppose that a firm initially has the production function $Q(K, L) = K + L$. After a period of time, the production function changes to $Q(K, L) = 3K + 2L$. What type of technological progress does this represent?

Solution

Before: $Q = K + L \implies MP_K = 1,$
 $MP_L = 1 \implies MRTS_{L,K} = \frac{MP_L}{MP_K} = 1$

After: $Q = 3K + 2L \implies MP_K = 3,$
 $MP_L = 2 \implies MRTS_{L,K} = \frac{MP_L}{MP_K} = \frac{2}{3}$

$MRTS_{L,K}$ **decreased** (from 1 to $2/3$). Notice MP_K tripled while MP_L only doubled – capital became relatively more productive than labor.

This is labor-saving (capital-biased) technological progress.

Lecture 7 - Cost Minimization

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

In the previous lecture,

- We looked at production functions, and how they can be used to represent the relationship between quantity of inputs used, and quantity of output produced.
- We also considered how the level of output varies when we vary the quantity of one or all of the inputs.

In this chapter, we will discuss how a firm might choose the mix of inputs that it uses to produce a given level of output, given the prices of those inputs.

- We will also analyze how changes in the prices of those inputs might impact the firm's decision.
- Finally, we will consider how the firm's problem varies between the short run and the long run.

Cost Minimization

Let's assume for the moment that a firm has decided to produce Q units of output. (We will consider in future lectures how they might choose Q .)

Then a firm which seeks to generate as much profits as possible will be concerned with producing that output at the lowest cost.

- The problem of finding the lowest-cost mix of inputs that produces a given level of output is the **cost-minimization problem**.

Cost Minimization

For now, let's assume that the firm uses only labor (L), and capital (K), in production (just like consumers only choosing between two goods)

- Each input has a price. The price of a unit of labor is called the wage rate (w). The price of a unit of capital services is the rental rate of capital (r).
- These prices will affect the amount of labor and capital that a firm chooses to use in production.

Isocost Lines

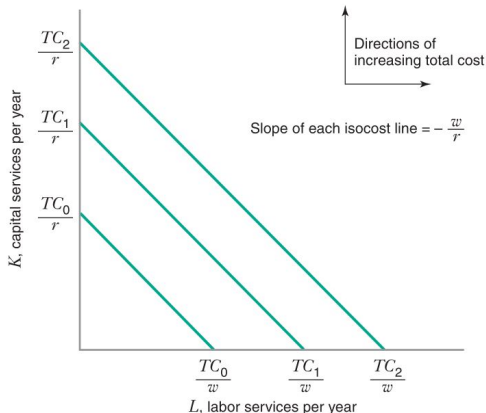
Before we look at the cost minimization problem mathematically, let's think about it graphically.

For this, we will use isocost lines. **Isocost lines** represent combinations of labor and capital that all have the same *total cost* (TC).

- Total cost is given by $TC = wL + rK$ (similar to the budget line for a consumer).

Isocost Lines

Below is a set of isocost lines for a firm, where each line represents a different total cost (TC). Note that as we move northeast in the graph, the isocost lines represent higher and higher total cost levels.

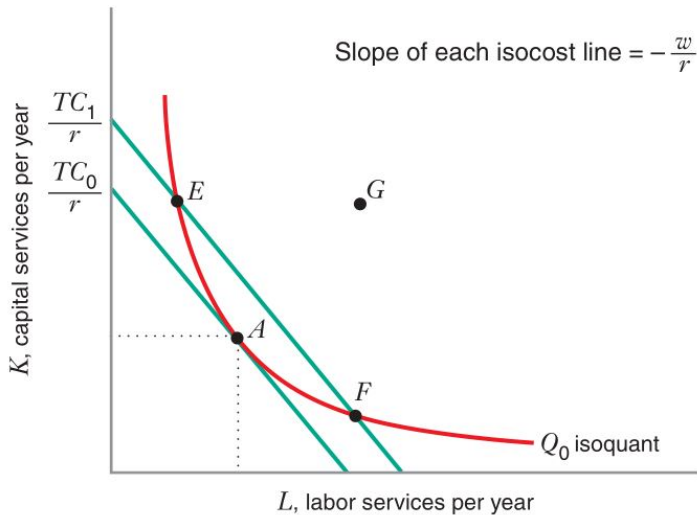


Solving the Cost Minimization Problem

In general, the cost-minimizing choice of inputs will be the point where the isoquant is just tangent to an isocost line. Consider why that is...

- If the firm chooses a point where the isocost and isoquant lines intersect, but are not tangent, then the firm could, by using more of one input, and less of the other, reduce their cost while still being on that isoquant.
- On the other hand, if the firm chooses a point where the isocost and isoquant lines do not touch, then the firm can't feasibly produce the required number of units.

Solving the Cost Minimization Problem



The Optimality Condition

Just like in the consumer choice setting, the above graph gives us an optimality condition.

$$\frac{MP_K}{r} = \frac{MP_L}{w}$$

- The **optimality condition** tells us that the produce minimizes cost when the marginal product per dollar on both inputs is equal.
- Think about what would happen if it weren't equal for the two inputs.

The Optimality Condition

If you were producing the necessary quantity (your constraint!) and

$$\frac{MP_K}{r} > \frac{MP_L}{w},$$

how could you produce the same quantity and lower costs?

Spend \$1 less on labor and $<$ \$1 more on capital!

The Optimality Condition

If you were producing the necessary quantity (your constraint!) and

$$\frac{MP_K}{r} < \frac{MP_L}{w},$$

how could you produce the same quantity and lower costs?

Spend \$1 less on capital and $<$ \$1 more on labor!

The Optimality Condition

Let's double-check that the optimality condition is correct using the Method of Lagrange. Our cost minimization problem can be formally stated as

$$\min_{(K,L)} rK + wL \quad \text{subject to} \quad \bar{Q} = Q(K, L)$$

Our Lagrangian is

$$\mathcal{L}(L, K, \lambda) = rK + wL + \lambda(\bar{Q} - Q(K, L))$$

The Optimality Condition

The first order conditions for an optimum are:

$$\frac{\delta \mathcal{L}}{\delta K} = 0 \Rightarrow r = \lambda(MP_K(K, L))$$

$$\frac{\delta \mathcal{L}}{\delta L} = 0 \Rightarrow w = \lambda(MP_L(K, L))$$

$$\frac{\delta \mathcal{L}}{\delta \lambda} = 0 \Rightarrow \bar{Q} - Q(K, L) = 0$$

If we solve the first two equations for λ , and set them equal to each other, we find

$$\frac{MP_K}{r} = \frac{MP_L}{w}$$

Solving for the Optimal Input Combination

Now we have a system of two equations:

$$\frac{MP_K}{r} = \frac{MP_L}{w} \quad \text{and} \quad \bar{Q} = Q(K, L)$$

We can solve this system to find the cost-minimizing input combination of L and K .

Solving for the Optimal Input Combination

Let's look at an example. Suppose a firm wishes to produce 1,000 units of output, and that the prices of labor and capital are \$5 and \$20, respectively. If the firm has the production function

$$Q = 50\sqrt{LK}$$

Then,

$$MP_L = 25\sqrt{\frac{K}{L}} \quad \text{and} \quad MP_K = 25\sqrt{\frac{L}{K}}$$

Solving for the Optimal Input Combination

Our optimality condition is:

$$\frac{25\sqrt{K/L}}{5} = \frac{25\sqrt{L/K}}{20} \implies 4K = L$$

Our constraint of producing 1,000 units is:

$$1000 = 50\sqrt{LK}$$

Solving for the Optimal Input Combination

Our constraint can be simplified to

$$1000 = 50\sqrt{LK} \implies 400 = LK$$

If we plug in $L = 4K$, then do some algebra

$$K^* = 10 \quad \text{and} \quad L^* = 40$$

Solving for the Optimal Input Combination

Suppose instead we just wanted to find K^* and L^* as functions of $\bar{Q}$, w and r . Then our optimality condition is:

$$\frac{25\sqrt{K/L}}{25\sqrt{L/K}} = \frac{w}{r} \quad \text{or} \quad \frac{K}{L} = \frac{w}{r}$$

If we rearrange this equation, then:

$$K = \frac{w}{r}L$$

Solving for the Optimal Input Combination

Combining this with our output quota:

$$\begin{aligned}\bar{Q} &= 50\sqrt{LK} \\ &= 50\sqrt{\frac{w}{r}L^2} \\ &= 50\sqrt{\frac{w}{r}}L\end{aligned}$$

Solving for the Optimal Input Combination

If we solve the previous equation for L:

$$L^*(w, r, \bar{Q}) = \frac{\bar{Q}}{50} \sqrt{\frac{r}{w}}$$

Then

$$K^*(w, r, \bar{Q}) = \frac{\bar{Q}}{50} \sqrt{\frac{w}{r}}$$

These are the **conditional demand equations** for labor and capital, respectively.

The Total Cost Function

Suppose that, now that we have the conditional demand equations, and we want to find the total cost function.

Remember that the total cost function is $TC = wL + rK$. We can then plug in our functions for L^* and K^* .

$$\begin{aligned} TC &= w * \frac{\bar{Q}}{50} \sqrt{\frac{r}{w}} + r * \frac{\bar{Q}}{50} \sqrt{\frac{w}{r}} \\ &= \frac{\bar{Q}}{50} \sqrt{w^2 * \frac{r}{w}} + \frac{\bar{Q}}{50} \sqrt{r^2 * \frac{w}{r}} \\ &= \frac{\bar{Q}}{50} \sqrt{rw} + \frac{\bar{Q}}{50} \sqrt{rw} \\ &= \frac{\bar{Q}}{25} \sqrt{rw} \end{aligned}$$

Cost Minimization with more than 2 Inputs

As was the case with the consumer's optimality condition, the cost-minimizing producer's optimality condition can be generalized to more than 2 inputs:

$$\frac{MP_{x_1}}{P_{x_1}} = \frac{MP_{x_2}}{P_{x_2}} = \dots = \frac{MP_{x_N}}{P_{x_N}}$$

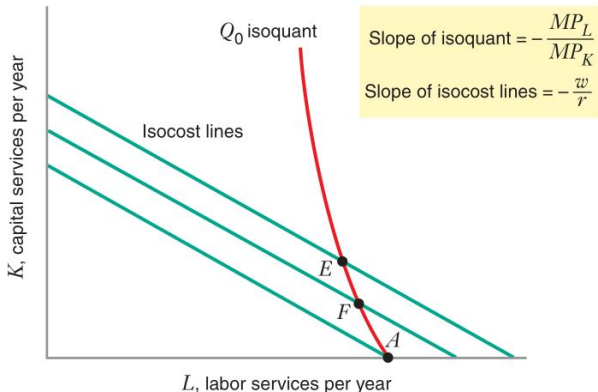
That is, the marginal product per dollar of each output must be equal at the optimal input bundle.

Previously, we considered only cases where there existed a tangency between the isoquant and isocost curves.

- Sometimes, there is no tangency. So how do we find the optimal bundle?
- On the next slide, we'll see a graph where the isocost lines are always flatter than the isoquant.

Corner Solutions

Here, the cost minimizing input bundle is one in which the firm uses only labor, and no capital, to produce the output good.



In these cases, the optimality condition won't hold

- Marginal product per dollar for one of the inputs will be strictly greater than the marginal product per dollar for the other.
- And cost is minimized by using all of that input.

This is just like perfect substitutes in consumer choice!

Try It Yourself

Suppose that for a firm that produces $\bar{Q}$ t-shirts, the cost of an hour of labor is \$5, and the cost of an hour of machine time is \$10. If at the current mix of labor and capital, a worker's output per hour is 15 t-shirts, and a machine's output is 25 t-shirts, which of the following is true?

- A) The firm should use more labor and less capital in order to reduce the cost of producing $\bar{Q}$ t-shirts.
- B) The firm should use more capital and less labor in order to reduce the cost of producing $\bar{Q}$ t-shirts.
- C) The firm is already producing $\bar{Q}$ t-shirts at the lowest cost. It's impossible to tell without knowing what $\bar{Q}$ is.

Solution

Compute the marginal product per dollar for each input:

$$\frac{MP_L}{w} = \frac{15}{5} = 3 \qquad \frac{MP_K}{r} = \frac{25}{10} = 2.5$$

Since $\frac{MP_L}{w} > \frac{MP_K}{r}$, labor generates more output per dollar spent than capital does.

Answer: A. The firm should use more labor and less capital – shifting toward the input with the higher marginal product per dollar lowers cost while keeping output at $\bar{Q}$.

Comparative Statics Analysis

Now that we have found the optimality condition for this problem, let's consider how the solution changes as our input prices and output quantity change.

Suppose that w increases, with r held constant. Then the optimal bundle becomes more expensive, and our optimality condition no longer holds. Now,

$$\frac{MP_L}{w} < \frac{MP_K}{r}$$

- This means the firm gets more output per dollar from additional capital than from additional labor.
- And should use more capital and less labor.

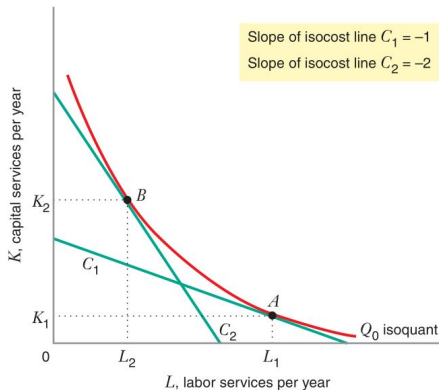
Comparative Statics Analysis

Let's think about it another way. If the wage rate increases, then the slope of the isocost lines, $-\frac{w}{r}$, gets steeper.

- The new point of tangency between the isocost and isoquant lines must be higher up on the isoquant.
- This implies that the firm will use more capital and less labor.

Comparative Statics Analysis

This diagram illustrates how, as the price of labor increases, the firm's optimal input combination is one with relatively more capital and less labor.



Comparative Statics Analysis

Now suppose the output quantity, Q , increases.

- Clearly, for the isoquant and isocost lines to remain tangent, the firm must move to a higher isocost (it costs more money to produce more output!)
- If we find the optimal bundle for a variety of output levels, we can trace what is called an **expansion path**

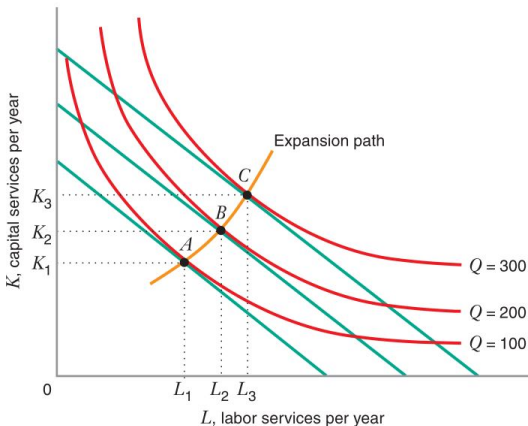
Normal and Inferior inputs

1. If as the output quantity increases, demand for the input increases we say that the input is a **normal input**
2. If as the output quantity increases, demand for the input decreases we say that the input is an **inferior input**

Note the similarities with quantity and income

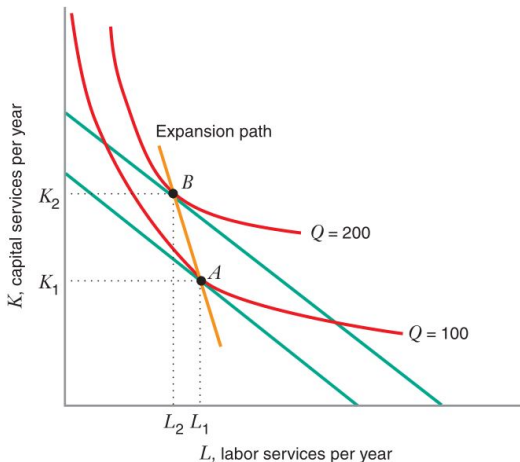
Comparative Statics Analysis

Below is an expansion path where both inputs are normal inputs (as output increases, demand for both inputs increases)



Comparative Statics Analysis

In this diagram, demand for labor decreases as quantity of output increases, implying that labor is an inferior input.



Comparative Statics

Suppose we want to do comparative statics numerically. For example, your boss asks you if they should buy more machines or hire more workers for their expanding business

- We can easily just solve for our conditional input demand functions, and plug in our old and new w , r , and $\bar{Q}$.
- Or, we can see how the functions vary as we change an input using partial derivatives.

Comparative Statics and Partial Derivatives

From above, we had

$$L^*(w, r, \bar{Q}) = \frac{\bar{Q}}{50} \sqrt{\frac{r}{w}}$$

Then

$$\frac{\partial L^*(w, r, \bar{Q})}{\partial w} = -\frac{\bar{Q}}{25} \sqrt{\frac{r}{w^3}}$$

Comparative Statics and Partial Derivatives

Try It Yourself

Which of the following conditional factor demand functions represents demand for an *inferior* input?

A) $L(w, r, Q) = 10 \frac{w}{rQ}$

B) $L(w, r, Q) = 10(w + rQ)$

C) $L(w, r, Q) = 10 \frac{wQ}{r}$

D) $L(w, r, Q) = 10 \frac{Q}{wr}$

Solution

An input is **inferior** if demand for it *falls* as output Q rises, i.e.

$$\partial L / \partial Q < 0.$$

A) $L = \frac{10w}{rQ} \implies \frac{\partial L}{\partial Q} = \frac{-10w}{rQ^2} < 0$. **Inferior input!**

B) $\frac{\partial L}{\partial Q} = 10r > 0$. Normal input.

C) $\frac{\partial L}{\partial Q} = \frac{10w}{r} > 0$. Normal input.

D) $\frac{\partial L}{\partial Q} = \frac{10}{wr} > 0$. Normal input.

Answer: A

Price Elasticity of Input Demand

In chapter 2, we learned how to describe the sensitivity of demand for a good to a change in the price. The same principle can be applied with input demand.

The **price elasticity of demand for labor** is given by

$$\epsilon_{L,w} = \frac{\frac{\Delta L}{L} \times 100\%}{\frac{\Delta w}{w} \times 100\%} = \frac{\Delta L}{\Delta w} \frac{w}{L}$$

The **price elasticity of demand for capital** is given by

$$\epsilon_{K,r} = \frac{\frac{\Delta K}{K} \times 100\%}{\frac{\Delta r}{r} \times 100\%} = \frac{\Delta K}{\Delta r} \frac{r}{K}$$

Price Elasticity of Input Demand

Remember that an elasticity measures the percentage change in demand for an input that results from a 1% change in some price.

- That means that if labor has a price elasticity of 3, then if the wage increases by 1%, demand for labor will decrease by 3%.

If we have input demand functions, then our price elasticity can be calculated using the derivative:

$$\begin{aligned}\epsilon_{L,w} &= \frac{\Delta L}{\Delta w} \frac{w}{L} = \frac{\partial L}{\partial w} \frac{w}{L} \\ \epsilon_{K,r} &= \frac{\Delta K}{\Delta r} \frac{r}{K} = \frac{\partial K}{\partial r} \frac{r}{K}\end{aligned}$$

Price Elasticity of Input Demand

Price elasticity of input demand depends, in part, on the elasticity of substitution.

Suppose that a firm can more easily substitute capital for labor.

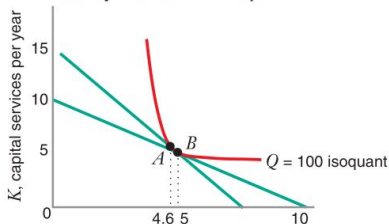
- Then if the price of labor increases slightly, they will use more capital and less labor.

On the other hand, if capital and labor are complements for a firm...

- Their mix of capital and labor may not change much in response to a change in the wage.

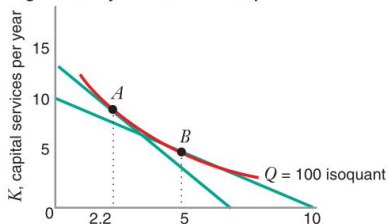
Price Elasticity of Input Demand

Low elasticity of substitution implies . . .



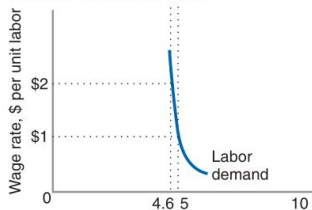
(a) L , labor services per year

High elasticity of substitution implies . . .



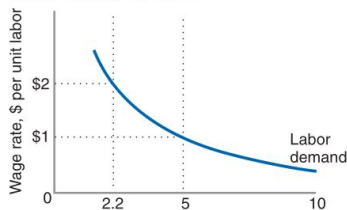
(c) L , labor services per year

inelastic demand for labor.



(b) L , labor services per year

elastic demand for labor.



(d) L , labor services per year

Price Elasticity of Input Demand

These elasticities are important for policy decisions. For example, what will the impact on labor demand be if the government increases the minimum wage?

- If the wage elasticity of labor demand is low, then firms will still hire almost as many workers as before.
- But if the wage elasticity of labor demand is high, they may substitute to capital instead (like self-checkout machines).

Price Elasticity of Input Demand

Try It Yourself

Suppose that the price of low-skill labor increases as a result of an increase in the federal minimum wage. Assume that firms cannot adjust their level of output. For which type of firm will total cost be impacted the most?

- A) A retail store such as Target, for whom cashiers and self-checkout stations are substitutes.
- B) A graphic design firm, for whom designers and computers are complements.

Solution

Answer: B, the graphic design firm.

When inputs are **substitutes** (Target's cashiers and self-checkout machines), a firm has a high elasticity of substitution: it can respond to a wage increase by shifting toward capital (more self-checkout, fewer cashiers), which *cushions* the increase in total cost.

When inputs are **complements** (a designer needs their computer, and vice versa), the firm has little ability to substitute away from labor even as it gets more expensive – since output can't change either, they're largely stuck paying the higher wage on (roughly) the same quantity of labor. This leaves total cost more exposed to the wage increase.

Fixed and Variable Costs

A firm's costs can be divided into two types:

1. **Fixed costs** refer to costs that do not vary with the level of output.
 - For example, a nail salon might have to pay rent every month, regardless of whether they have no customers, or 1,000 customers.
2. **Variable costs** refer to costs that depend on the level of output.
 - For example, a bakery's quantity of output depends on how many hours their bakers work.

Fixed and Variable Costs

The total cost function is a sum of the two costs:

1. **Total variable cost** is made up of all of the firm's expenditures on variable inputs, such as labor and materials.
2. **Total fixed cost** is made up of all expenditures that do not vary with output.

Sunk Costs

When it comes to fixed costs, some can be avoided if the firm chooses not to produce (e.g. lighting for an office building).

But some cannot be recovered even if a firm shuts down (e.g. license fees or a very specialized piece of equipment).

- Fixed costs that cannot be recovered or avoided are known as **sunk costs**.
- Fixed costs that can be recovered, or are avoidable in the event that the firm does not produce, are known as **nonsunk costs**.

Sunk Costs

In general, sunk costs are irrelevant to a firm's decisions. If the cost cannot be recovered, then whether the firm continues producing (and if so, how much), do not depend on that cost.

- However, if the sunk cost has not yet been paid, it is relevant.
- A firm can decide whether or not to incur the cost in the first place.

Short-Run vs. Long-Run

Until this point, we've really only considered the long run.

- The **long run** refers to a time period over which all inputs can vary (capital and labor).
- In the **short run**, at least one input is fixed.

Usually we assume capital is fixed (e.g. rent on a building). But it could also be labor (maybe firms hire employees on a 1-year contract).

Cost Minimization in the Short Run

In the short run, a firm's total cost will always be greater than or equal to a firm's long-run total cost.

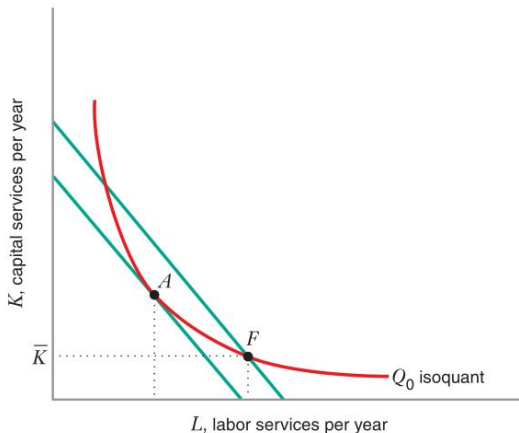
- Because if it were lower in the short run, they would use that same input combination in the long run.

Suppose for example that capital is fixed at $\bar{K}$ units.

- Then in the short run, the firm may not be able to achieve the cost-minimizing input combination, so costs will be higher.

Cost Minimization in the Short Run

Because capital is fixed, the firm's lowest-cost input combination leaves them at a higher isocost line than the long-run cost-minimizing input combination (F in the short-run; A in the long-run)



Cost Minimization in the Short Run

What about our optimality condition?

- You can see from the previous diagram that at the short-run cost minimizing point, our isoquant and isocost lines are no longer tangent.
- Our optimality condition doesn't hold.

Let's try to find the cost-minimizing point using calculus.

Cost Minimization in the Short Run

Suppose a firm's production function is given by $Q = 50\sqrt{KL}$. In the short-run, the firm's problem becomes

$$\min_L wL + r\bar{K} \quad \text{subject to} \quad \bar{Q} = 50\sqrt{\bar{K}L}$$

Note that, because K is fixed, we only have 1 choice variable. In this case, we now simply have 1 equation, $\bar{Q} = 50\sqrt{\bar{K}L}$, and 1 unknown, L .

We can then simply solve for L .

$$L^*(w, r, \bar{Q}) = \frac{\bar{Q}^2}{(2500\bar{K})}$$

Cost Minimization in the Short Run

Try It Yourself

Suppose that a firm has the following production function:

$Q(K, L) = 15KL$. In the short run, suppose that K is fixed at $\bar{K} = 5$. If a cost-minimizing firm decides to produce $Q = 1500$ units, find the cost-minimizing level of labor.

- A) 5 units
- B) 10 units
- C) 15 units
- D) 20 units

Solution

Since K is fixed at $\bar{K} = 5$, the only choice variable is L . Plug into the production function:

$$Q = 15\bar{K}L \implies 1500 = 15(5)L \implies 1500 = 75L$$

$$L^* = \frac{1500}{75} = 20$$

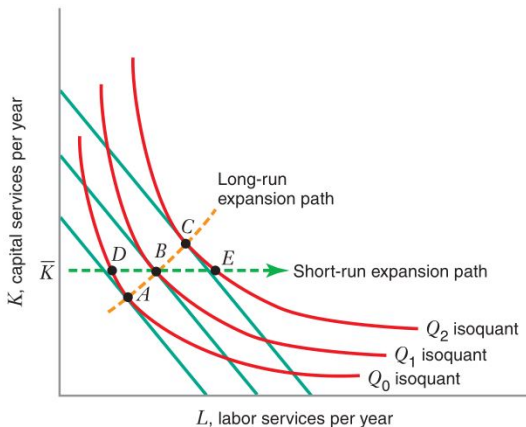
Answer: D, 20 units.

What if the output quantity changes?

- Well, in the short run, K is still fixed, so the firm can only vary the amount of labor used.
- This results in a short-run expansion path that's horizontal.
- In the long run, however, both inputs can vary, resulting in a different long-run expansion path.

Comparative Statics

Below are the short-run and long-run expansion paths for a production function.



Lecture 8 - Cost Curves

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Overview

In the previous lecture, we began talking about costs.

- We considered how output varies as we change the level of inputs (production function).
- And how firms choose the mix of inputs that produces a given quantity at the lowest price (optimal production).
- We considered what happens to the firm's lowest-cost input combination when prices or quantity changed (comparative statics).
- And we looked at how a firm chooses their input combination when the quantity of some input is fixed (short-run production).

Overview

In this section, we will learn how to calculate total, marginal, and average costs, both in the short run and the long run.

- We will look at these costs graphically, and analyze their relationship.
- We will also introduce concepts such as economies of scale, scope, and experience.

Long Run Total Cost Curve

In the previous chapter, we learned how to calculate total costs as a function of Q , w , and r .

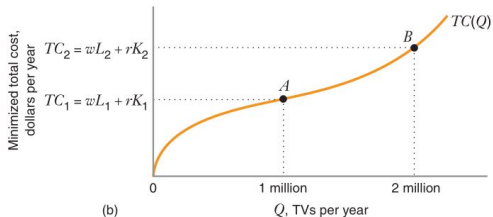
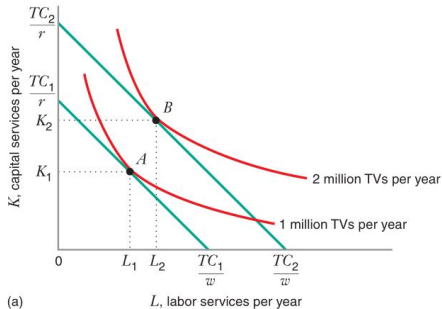
Remember that

$$TC = w * L^*(w, r, Q) + r * K^*(w, r, Q)$$

where L and K are the cost-minimizing levels of labor and capital.

- Optimal L and K are functions of the wage, the rental rate, and the output quantity.

Long Run Total Cost Curve



Long Run Total Cost Curve

Note that the total cost curve is always upward sloping.

- Suppose you could produce 100 units for \$1,000, and 150 units for \$900.
- Then you could also produce 100 units for \$900 ... by producing 150, and throwing some away!

Also note that in the long run, $Q(0) = 0$.

- Since all inputs can vary, the cost-minimizing level of all inputs needed to produce 0 units of output is \$0.

Long Run Total Cost Curve

Try It Yourself

Suppose a firm has the total cost function $TC(Q) = \frac{Q}{25}\sqrt{wr}$. Is the total cost curve increasing with Q ? Do costs increase for the firm when the cost of capital r increases?

Solution

Is TC increasing in Q ? Take the derivative:

$$\frac{\partial TC}{\partial Q} = \frac{\sqrt{wr}}{25} > 0$$

Yes – since $w, r > 0$, this is always positive, so total cost is increasing in Q .

Does TC increase when r increases? Take the partial derivative with respect to r :

$$\frac{\partial TC}{\partial r} = \frac{Q}{25} \cdot \frac{w}{2\sqrt{wr}} = \frac{Qw}{50\sqrt{wr}} > 0$$

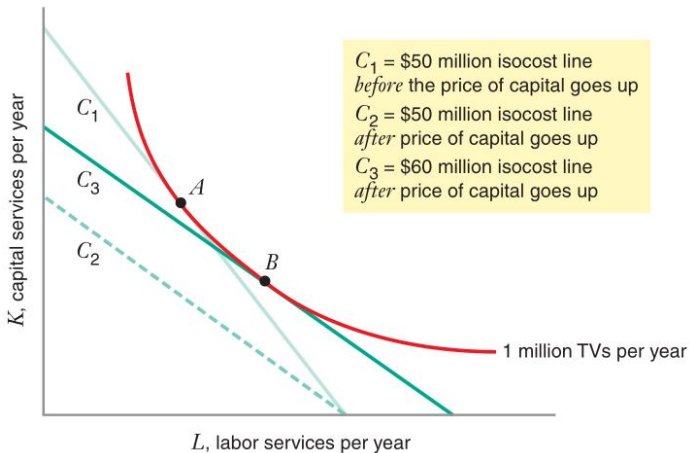
Yes – for any $Q > 0$, this is also positive, so total cost rises when the rental rate of capital increases.

Comparative Statics of Long-run Total Cost

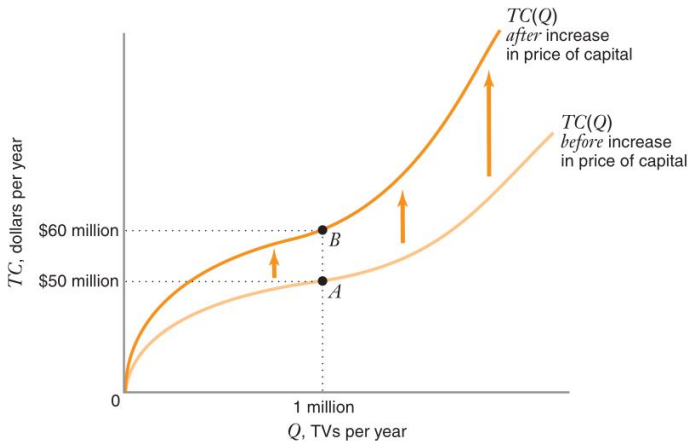
When the price of one input increases, the firm may substitute away from that input, but their TC still increases.

- If TC stayed the same at the higher input price, why wouldn't they have chosen that input combination to start with?
- And in most cases, the increase in cost is greater when a larger quantity is being produced.

Comparative Statics of LRTC



Comparative Statics of LRTC



Comparative Statics of LRTC

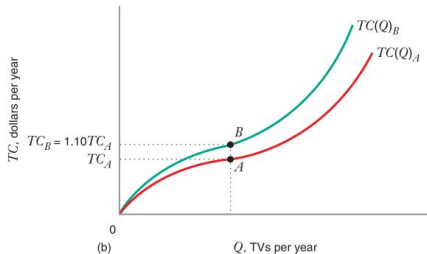
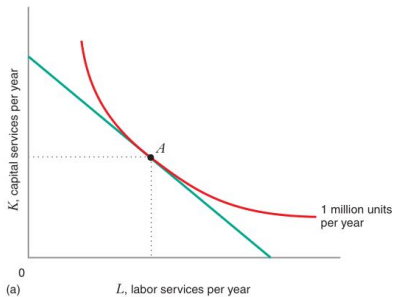
What if both input prices increase proportionally?

Then the slope of the isoquants ($-w/r$) doesn't change.

- So the point of tangency doesn't change.
- In other words, the optimality condition still holds.

$\implies$ The input combination doesn't change. It just gets proportionally more expensive to make.

Comparative Statics of LRTC



Long-Run Average and Marginal Costs

There are two other important costs we need to consider: *marginal cost* and *average cost*.

- Marginal cost is important in deciding *how many* units to produce.

How much should the firm produce?

- Average cost is useful in determining a firm's overall profitability (should they produce anything at all).

Should they be producing at all?

Long-Run Average and Marginal Costs

Long-run average cost is the firm's cost per unit of output.

$$AC(Q) = \frac{TC(Q)}{Q}$$

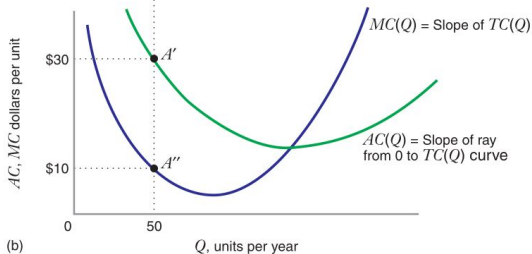
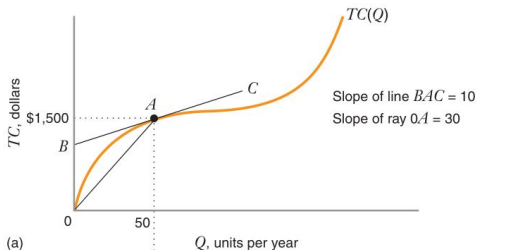
Also the slope of a ray from the origin to the point on the total cost curve.

Long-run marginal cost is the rate at which long-run total cost changes with respect to a change in output.

$$MC(Q) = \frac{\Delta TC}{\Delta Q}$$

Also the slope of the tangent line at that quantity.

Long-Run Average and Marginal Costs



Long-Run Average and Marginal Costs

Try It Yourself Suppose that the total cost function for a cost-minimizing firm is given by $TC(Q) = 4Q$. What is the firm's marginal and total cost?

Solution The total cost function is simply $TC(Q) = 4Q$ – so total cost at any quantity Q is $4Q$.

Marginal cost is the derivative of TC with respect to Q :

$$MC(Q) = \frac{dTC}{dQ} = 4$$

Marginal cost is **constant at \$4**, regardless of quantity produced.

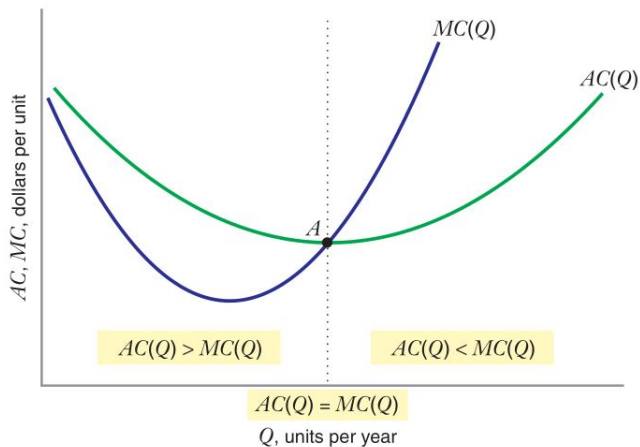
Note that average cost is $AC(Q) = TC(Q)/Q = 4$ as well – since TC is linear through the origin (no fixed costs), $AC = MC = 4$ at every quantity.

Relationship Between AC and MC Curves

As we saw with marginal product and average product, there is a systematic relationship between marginal cost and average cost.

- If average cost is decreasing as quantity is increasing, then average cost is greater than marginal cost.
- If average cost is increasing as quantity is increasing, then average cost is less than marginal cost.
- If average cost is constant as quantity is increasing, then average cost equals marginal cost.

Relationship Between AC and MC Curves



Economies of Scale

It's important to consider what will happen to a firm's production costs as they produce more output.

- Perhaps as a firm produces more output, employees can specialize more...
- ... or maybe their production process uses *indivisible inputs*, which cannot be scaled down easily...
- ...and their cost per unit falls.

A firm is said to experience **economies of scale** when average cost of production falls as output rises.

Or perhaps as a firm grows, managerial needs outpace output growth increase the per-unit cost.

- When costs arise due to an increase in managerial needs, this is known as **managerial diseconomies**.
- When average cost rises as output rises, a firm is said to experience **diseconomies of scale**.

Economies of Scale

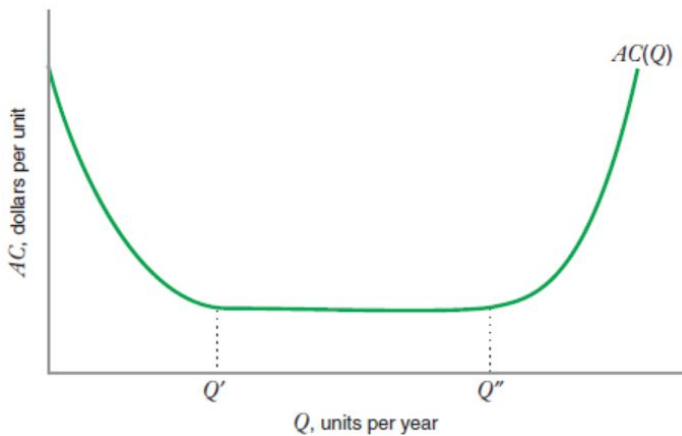
A firm is said to be producing at its **minimum efficient scale (MES)** when long-run average cost is at its minimum.

- This is an important concept since it tells us, *at best*, how efficiently firms can produce the good

Firms with a large minimum efficient scale relative to market size have greater economies of scale.

- In these markets, it is more difficult for smaller firms to compete.

Economies of Scale



Economies of Scale and Returns to Scale

If average cost *decreases* as output *increases*, then we have *economies of scale* and *increasing returns to scale*.

- E.g. $Q = L^2$

If average cost *increases* as output *increases*, then we have *diseconomies of scale* and *decreasing returns to scale*.

- E.g. $Q = \sqrt{L}$

If average cost *stays the same* as output *increases*, then we have *neither economies nor diseconomies of scale* and *constant returns to scale*.

- E.g. $Q = L$

Economies of Scale and Returns to Scale

Note that the previous slide assumed that input price didn't depend on quantity of production.

A firm could experience constant returns to scale in technology...

- ...but if quantity discounts are available for inputs, they may experience economies of scale.
- Or if an input is scarce, the price may rise as output increases and that same firm may experience diseconomies of scale.

Measuring Economies of Scale

Can use elasticities to tell us how sensitive total cost is to output.

The **output elasticity of total cost** is given by:

$$\epsilon_{TC,Q} = \frac{\frac{\Delta TC}{TC}}{\frac{\Delta Q}{Q}} = \frac{\frac{\Delta TC}{\Delta Q}}{\frac{TC}{Q}} = \frac{MC}{AC}$$

Value of $\epsilon_{TC,Q}$	MC Versus AC	How AC Varies as Q Increases	Economies/ Diseconomies of Scale
$\epsilon_{TC,Q} < 1$	$MC < AC$	Decreases	Economies of scale
$\epsilon_{TC,Q} > 1$	$MC > AC$	Increases	Diseconomies of scale
$\epsilon_{TC,Q} = 1$	$MC = AC$	Constant	Neither

Measuring Economies of Scale

Try It Yourself

Which of the following is a possible reason why a firm might experience increasing returns to scale, and diseconomies of scale?

- A) The firm uses indivisible inputs in its production process.
- B) The firm experiences managerial diseconomies when output rises.
- C) The firm receives quantity discounts on its inputs when output rises.
- D) The firm's employees are able to specialize more when output rises.

Solution

- **A)** False. Indivisible inputs create *economies* of scale.
- **B) True.** Managerial diseconomies can raise average cost even if the production technology exhibits increasing returns to scale.
- **C)** False. Quantity discounts reinforce economies of scale.
- **D)** False. Specialization also creates economies of scale.

Key idea: Returns to scale describe the *production technology*, while economies of scale depend on *average cost*, which can also be affected by managerial and organizational costs.

Answer: B

Short-Run Cost Curves

In the previous lecture, we learned the difference between the short run and the long run.

- In the short run, the levels some inputs cannot be varied.
- In the long run, the levels of *all* inputs can.
- In this section, we will assume that capital (K) is the fixed input.

Short-Run Cost Curves

The short-run total cost curve $STC(Q)$ tells us the minimized total cost of producing Q units of output when at least one input is fixed.

The total cost curve can be divided into a **total variable cost curve (TVC)** and a **total fixed cost curve (TFC)**.

- $TVC(Q)$ is the sum of expenditures on variable inputs at the cost-minimizing input combination.
- $TFC(Q)$ is the total cost of fixed inputs.

$$STC(Q) = TVC(Q) + TFC(Q)$$

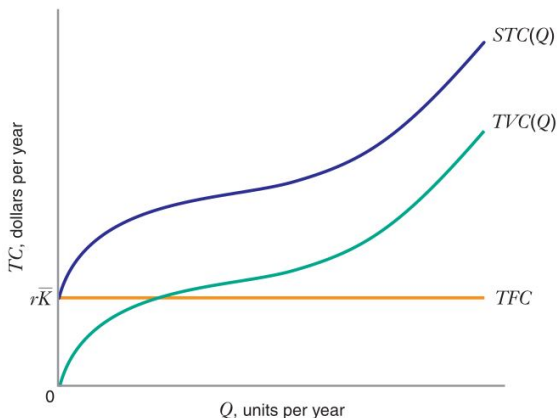
Short-Run Cost Curves

Because we let capital be our fixed input, TFC is simply the amount of money spent on $\bar{K}$ units of capital. In other words, $TFC = r\bar{K}$.

Thus, $STC(Q) = TVC(Q) + r\bar{K}$.

Short-Run Cost Curves

Note that the vertical distance between the $STC(Q)$ curve and the $TVC(Q)$ curve is equal to $r\bar{K}$.



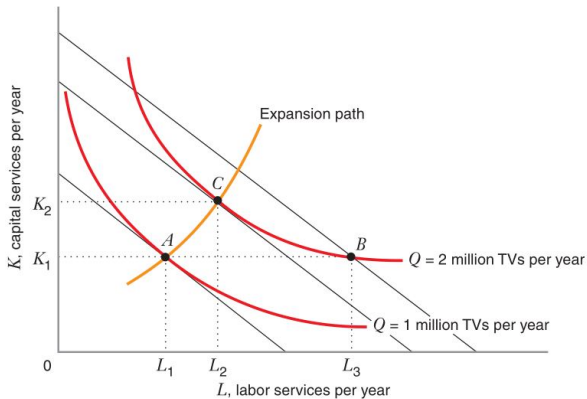
Relationship Between LTC and STC

Consider a firm that produces televisions, and wants to expand production.

- In the short run, they can only adjust their level of labor.
- But in the long run, they can adjust the level of capital as well.

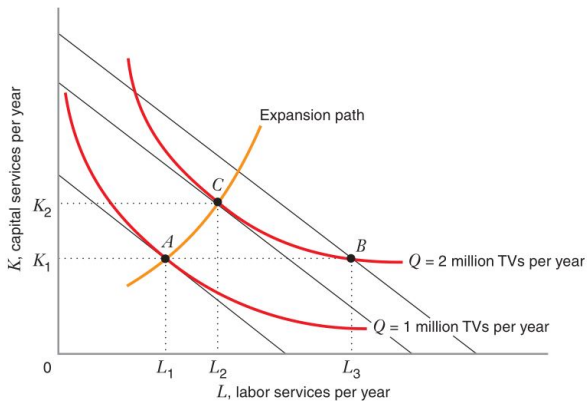
Relationship Between LTC and STC

Going from 1 million to 2 million TVs. In the short run, you have to go to B .



Relationship Between LTC and STC

Going from 1 million to 2 million TVs. In the short run, you have to go to B . In the long run, you can go to C at a lower cost.



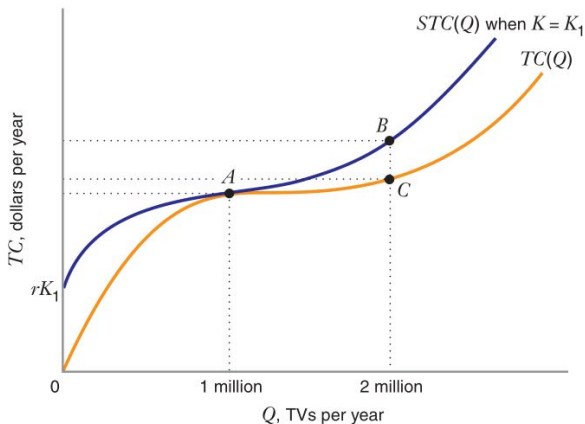
Relationship Between LTC and STC

Note that total cost will **never** be lower in the short run than in the long run.

- If you can use that input combination in the short-run, you can use it in the long run!

Relationship Between LTC and STC

Below are the STC and LTC curves for some firm. the $STC(Q)$ always lies above the $LTC(Q)$.



Short-Run Average and Marginal Cost Curves

We can define **short-run average cost**, SAC , and **short-run marginal cost**, SMC , similarly to their long-run counterparts.

$$SAC(Q) = \frac{STC(Q)}{Q} \text{ and } SMC(Q) = \frac{\Delta STC}{\Delta Q}$$

We can also break $SAC(Q)$ into average variable costs and average fixed costs.

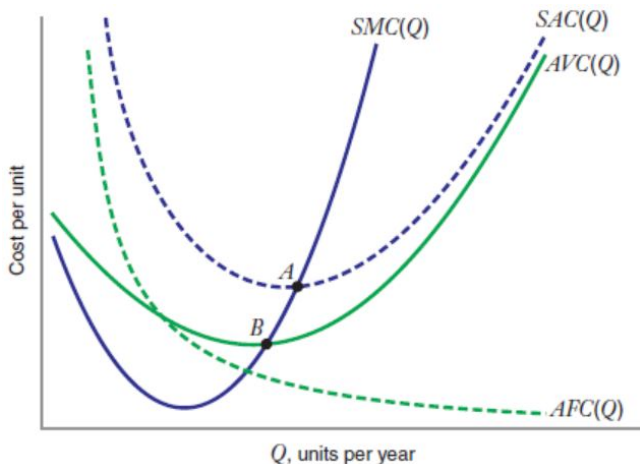
$$SAC(Q) = AVC(Q) + AFC(Q)$$

Short-Run Average and Marginal Cost Curves

If we plot these curves, we can see that SAC is found by vertically summing the AFC and ATC curves.

- The AFC curve slopes downward because fixed costs do not change as output increases. The fixed cost is spread over more and more units of output.
- Note that the marginal cost curve intersects the average variable and average total cost curves at their minima.

Short-Run Average and Marginal Cost Curves



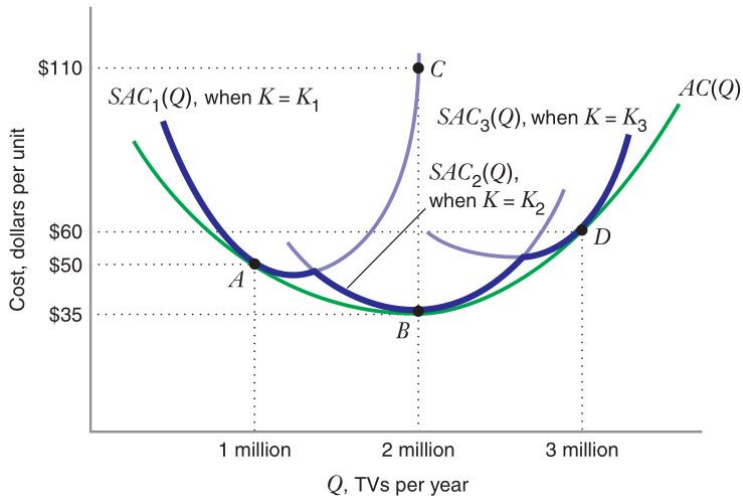
Relationship Between Long-Run and Short-Run AC and MC Curves

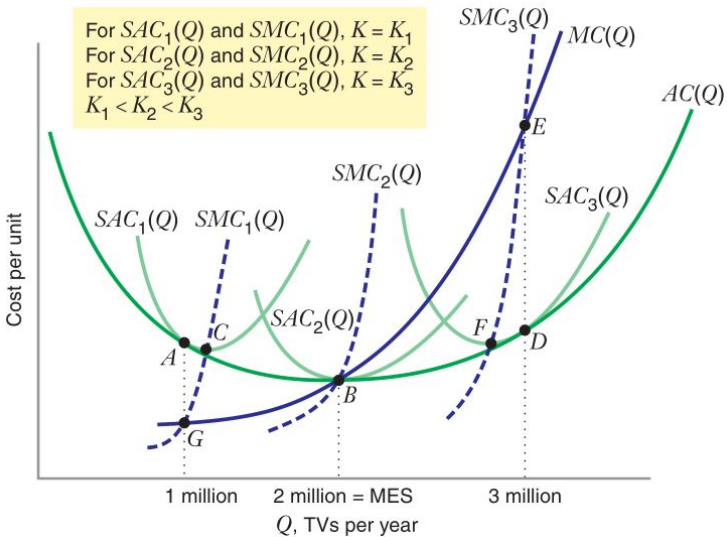
A firm can have many short-run average cost curves depending on the level of capital.

In the long-run, capital can vary

- And the firm will choose the level of capital that minimizes average cost for a given output level.
- Thus, the long-run average cost curve is the boundary of all of the short-run average cost curves.
- It traces the minimum SAC for every level of output.

The long-run curve can be thought of as the lower envelope of the set of short-run curves.





Relationship Between Long-Run and Short-Run AC and MC Curves

Try It Yourself

Which of the following relationships is always true?

- A) $MC(Q) > SMC(Q)$
- B) $AC(Q) > MC(Q)$
- C) $SAC(Q) > MC(Q)$
- D) $SAC(Q) > AC(Q)$

Solution

- **A)** False. MC and SMC cross, so either can be larger.
- **B)** False. $MC < AC$ when AC is falling, but $MC > AC$ when AC is rising.
- **C)** False. MC and SAC also cross.
- **D) True.** The long-run average cost curve is the *lower envelope* of all short-run average cost curves, so

$$SAC(Q) \geq AC(Q),$$

with equality only at the output where the fixed capital level is optimal.

Answer: D

Other Determinants of Cost

So far, we have focused on firms that only produce 1 good or service.

- But many firms sell a large variety of goods (Samsung, Johnson & Johnson, General Electric).

Why do some firms choose to produce many different products?

- Often, because fixed costs can be spread among many more units.
- For example, a satellite TV company broadcasts many channels...
Not just one.

Other Determinants of Cost

Or Budweiser can use their machinery to bottle many different types of beer.

- Firms can also spread administrative costs over more units of output
 - E.g. the accounting department or human resources department.

Can also receive discounts on inputs (if their products use a common input) or shipping costs.

Other Determinants of Cost

Mathematically, a firm is said to experience **economies of scope** if:

$$TC(Q_1, Q_2) < TC(Q_1, 0) + TC(0, Q_2)$$

In other words, the total cost of 1 firm producing both goods (Q_1 and Q_2) is less than the sum of the total costs of each firm producing separately

The opposite could be true, implying **diseconomies of scope**

Other Determinants of Cost

Sometimes costs can fall over time as a firm learns more efficient production methods.

This is known as **economies of experience** or **learning-by-doing**.

Think of the solar industry.

- The price of solar power has fallen 60% in the last decade.
- This is at least partly a result of accumulated experience.

Lecture 9 - Perfect Competition

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Overview

In this chapter, we will:

- Define a perfectly competitive market.
- Discuss the difference between economic profit and accounting profit.
- Learn how to calculate a firm's optimal input combination for profit maximization.
- Consider both short-run and long-run profit-maximization

Perfectly Competitive Markets

Perfectly competitive markets have 4 defining characteristics:

1. Fragmented market - many buyers and sellers
2. Undifferentiated products (e.g. oil, corn, soybeans).
3. Consumers have perfect information about prices of all sellers
4. Equal access to resources - same technology, access to same inputs

Perfectly Competitive Markets

These characteristics have 3 implications:

1. Buyers and sellers are price takers
2. Law of one price - demand is perfectly elastic
3. Free entry and exit - If market is profitable, firms enter. If unprofitable, firms exit.

This final implication implies that profits equal zero

Economic & Accounting Profits

What do I mean by zero profits? Why would anyone start a company in a perfectly competitive market if they're guaranteed to make 0 profits!!

Remember that economic costs = explicit costs + implicit costs (such as the opportunity cost of not working a job)

- **Economic profit** = Sales Revenue - Economic Costs
- **Accounting profit** = Sales Revenue - Accounting Costs

When we discuss profit maximization, we mean economic profit.

Economic & Accounting Profits

Try It Yourself

Suppose that Tom, a lawyer quits his job to start a lawn care company. Before he quit, the lawyer was earning \$120,000 per year (which he could return to at any time). With the lawn care company, Tom expects his revenue to be \$300,000 per year. He expects his annual cost of wages for workers to be \$80,000, and additional operating costs to be \$20,000. Additionally, Tom invested \$80,000 in equipment, money which he could have otherwise invested in the stock market, with a return of \$5,000 per year. Find Tom's economic profit.

Solution

Explicit costs: wages (\$80,000) + operating costs (\$20,000) = \$100,000

Implicit costs: forgone salary (\$120,000) + forgone stock market return (\$5,000) = \$125,000

Economic costs = explicit costs + implicit costs
= 100,000 + 125,000 = \$225,000

Economic profit = Revenue – Economic Costs
= 300,000 – 225,000 = \$75,000

Note the \$80,000 spent on equipment itself isn't an annual cost – only the *opportunity cost* of that money (the \$5,000 forgone return) counts as an implicit cost each year.

Profit Maximization

Remember that a firm's profit function looks like

$$\pi = P * Q - TC(Q, w, r)$$

where $P * Q$ is equal to total revenue, $TR(Q) = PQ$.

- $TC(Q, w, r)$ is the total cost function that we looked at in chapters 7 and 8.
- Because the firm is a price taker, P doesn't depend on how many units the firm sells.

Profit Maximization

A profit-maximizing perfectly competitive firm chooses the quantity Q which maximizes profit.

In other words, the firm's problem is

$$\max_Q \pi(Q) = PQ - TC(Q, w, r)$$

If we take the partial derivative of this equation with respect to Q and set it to zero, we find the quantity that maximizes profit.

$$P - \frac{\partial TC(Q, w, r)}{\partial Q} = 0 \Rightarrow P = MC(Q, w, r)$$

Profit Maximization

In words, the firm will produce output until the revenue it receives from an additional unit (P) equals the marginal cost of producing that unit ($MC(Q)$).

- The revenue that it receives from an additional unit of output is called **marginal revenue**.
- That is, Q is chosen such that $P = MR = MC$.

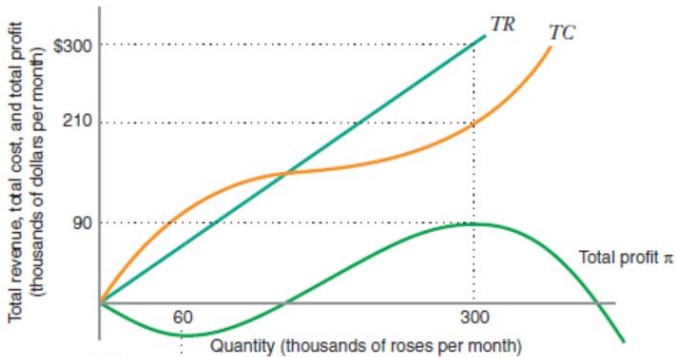
Profit Maximization

Here's the intuition: After some point, MC is increasing. As long as a firm can sell a unit for more than its marginal cost, $P > MC(Q)$, it should keep selling more units.

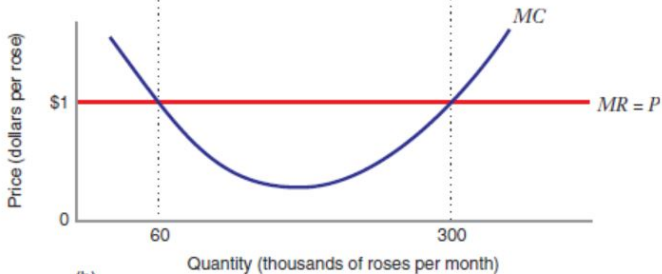
Eventually, marginal cost will rise enough that price is below marginal cost.

- At that point, a firm would lose money on each additional unit sold.

$\implies$ profit is maximized where $P = MC(Q)$.



(a)



(b)

Profit Maximization

Once you have found the marginal cost function, all you need to do is solve $P = MC(Q, w, r)$ for Q .

Either you will be given

1. $TC(Q)$, or
2. will have to derive it from a production function $Q(K, L)$

Example

Suppose a firm's total cost function is given by

$$TC(Q, w, r) = 2wQ^2 + 8r$$

The firm's profit maximizing quantity of output must satisfy $P = MC$:

$$MC(Q, w, r) = \frac{\partial TC(Q, w, r)}{\partial Q} = 4wQ$$

Solving for Q :

$$P = 4wQ \Rightarrow Q^* = \frac{P}{4w}$$

Now that we know $Q^*(P, r, w)$, we can plug that back into our conditional input demand functions for capital and labor and find our **unconditional input demand functions**.

- These functions do not condition on a given level of quantity $\bar{Q}$.
- Instead, they implicitly reflect how Q will change as P changes, in order to maximize profit.

Profit Maximization

Another way that we can find the firm's profit maximizing output quantity is by writing total cost as a function of L and K :

$$Q = f(K, L) \text{ and } TC(Q, w, r) = wL + rK$$

Plugging those into the firm's profit function, the firm's problem then becomes:

$$\max_{L, K} \pi = P * f(K, L) - wL - rK$$

Profit Maximization

$$\max_{L,K} \pi = P * f(K, L) - wL - rK$$

Again using unconstrained optimization, we can find the two first-order conditions:

$$\begin{aligned}\frac{\partial \pi}{\partial L} &= P \frac{\partial f}{\partial L} - w = 0 \Rightarrow P = \frac{w}{MP_L} \\ \frac{\partial \pi}{\partial K} &= P \frac{\partial f}{\partial K} - r = 0 \Rightarrow P = \frac{r}{MP_k}\end{aligned}$$

Profit Maximization

If we combine the two first-order conditions, we get the following **optimality condition** for price maximization:

$$P = \frac{w}{MP_L} = \frac{r}{MP_k}$$

The second part of this looks a lot like our optimality condition for cost minimization.

That's because maximizing profit requires producing at the lowest cost.

Profit Maximization

If we rearrange our first-order conditions in a different way:

$$P * MP_L = w \text{ and } P * MP_K = r$$

This tells us that the cost of an additional unit of capital or labor should equal the revenue that it generates.

At the optimal output level the price of the output good should equal the amount of money it costs to produce the last unit with either labor or capital. That is,

$$P = MC(Q)$$

Example

Suppose that a firm has the production function

$$Q(K, L) = 20K - K^2 + 16L - L^2 \text{ with } MP_L = 16 - 2L \text{ and } MP_K = 20 - 2K.$$

Our optimality condition says that

$$P = \frac{w}{16 - 2L} \text{ and } P = \frac{r}{20 - 2K}$$

Example

$$P = \frac{w}{16 - 2L} \text{ and } P = \frac{r}{20 - 2K}$$

We can solve for K in the second equation, and find that

$$20 - 2K = \frac{r}{P} \text{ or } K^*(P, w, r) = 10 - \frac{r}{2P}$$

Doing the same with the first equation, we find that

$$16 - 2L = \frac{w}{P} \text{ or } L^*(P, w, r) = 8 - \frac{w}{2P}$$

Profit Maximization

Once we've found the unconditional input demand curves, we can now solve for the optimal production function: $Q^*(P, w, r)$.

That is;

$$Q^*(P, w, r) = Q(K^*(P, w, r), L^*(P, w, r))$$

Try It Yourself

Suppose that a profit-maximizing firm has the production function $Q(K, L) = 30K - \frac{1}{2}K^2 + 40L - \frac{1}{2}L^2$, with $MP_L = 40 - L$ and $MP_K = 30 - K$. Find the firm's unconditional labor demand function.

Solution

The profit-maximization optimality condition is $P = \frac{w}{MP_L}$. Substituting $MP_L = 40 - L$:

$$P = \frac{w}{40 - L}$$

Solve for L :

$$40 - L = \frac{w}{P} \implies L^*(P, w, r) = 40 - \frac{w}{P}$$

Notice that, as expected, the unconditional labor demand function doesn't depend on r – only K^* would.

Short-Run Profit Maximization

Suppose that K is fixed ($K = \bar{K}$) in the short run.

Can solve profit maximization problem with one variable input.

$$\max_L \pi(P, r, w) = PQ(L, \bar{K}) - wL - r\bar{K}$$

FOC:

$$P * \frac{\partial Q(L, \bar{K})}{\partial L} - W = 0 \Rightarrow P = \frac{w}{MP_L}$$

Short-Run Profit Maximization

Suppose that K is fixed ($K = \bar{K}$) in the short run.

Can solve profit maximization problem with one variable input.

$$\max_L \pi(P, r, w) = PQ(L, \bar{K}) - wL - r\bar{K}$$

FOC:

$$P * \frac{\partial Q(L, \bar{K})}{\partial L} - W = 0 \Rightarrow P = \frac{w}{MP_L} \Rightarrow P = SMC(Q, w)$$

No cost minimization condition, because there is only one way to produce Q units.

Deriving the Firm's Supply Curve

The solution to the firm's problem gives us the firm's supply function:

$$Q^*(P, w, r)$$

If w and r are fixed, can plot firm's supply curve in P-Q space.

Recall the profit maximizing condition under perfect competition in the short run: $P = SMC(Q)$

- Inverse supply curve is the short-run marginal cost curve (you give me Q , I give you $SMC(Q)$)

Deriving the Firm's Supply Curve

There is an exception though.

Suppose the firm has large fixed costs that could be recovered if firm sets $Q = 0$ (e.g. electricity costs)

Perhaps even at profit-maximizing quantity, profits do not offset total fixed costs (TFC).

- Then $Q = 0$ is optimal.
- Will occur if market price of final good is low enough.

Deriving the Firm's Supply Curve

If firm is making positive profit when $Q > 0$, must be making positive *per-unit* profit

- Because all units sell at price P , that means that average variable cost is below the price $P > ATC(Q)$.
- If $P < \min(ATC(Q))$, firm will produce $Q = 0$ in the long-run.

The price at which $P = \min(ATC(Q))$ is called the **break-even price**

Deriving the Firm's Supply Curve

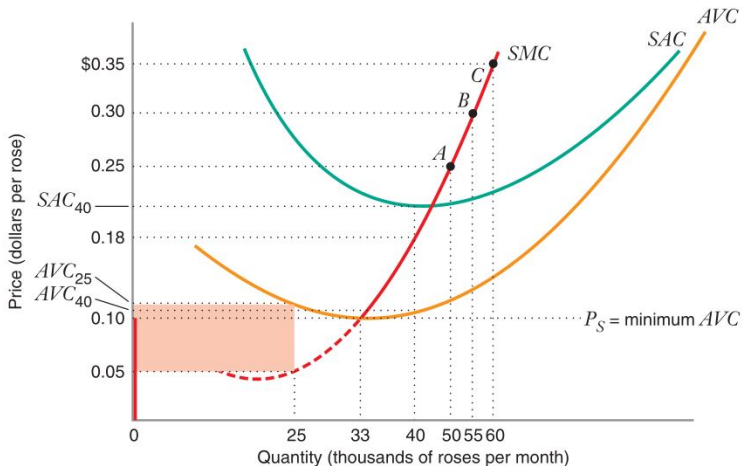
In the short-run, the fixed costs are already paid for (sunk costs). Therefore, 'short-run profit' is the difference between market price and *average variable cost*.

The reason is if price is above AVC , then you can pay off some of your sunk costs.

- Because all units sell at price P , that means that average variable cost is below the price $P > AVC(Q)$.
- If $P < \min(AVC(Q))$, firm will produce $Q = 0$ in the short-run.

The price at which $P = \min(AVC(Q))$ is called the **shut-down price**

Deriving the Firm's Supply Curve



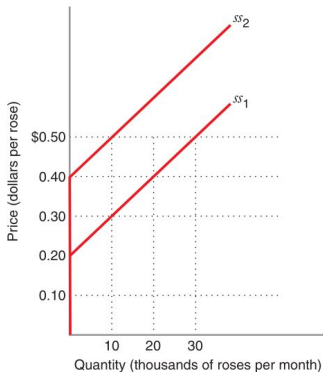
Short-Run Market Supply

Once we have each firm's supply curve, we can find the market supply curve. The market supply curve tells us the quantity supplied by *all firms* at any given price.

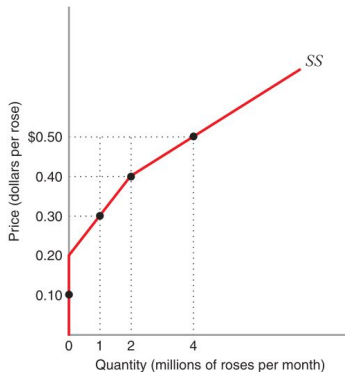
As we saw in chapter 2, the market supply curve is found by horizontally summing every firm's supply curve.

- Only works if input prices are constant (don't depend on level of output).
- Might be true for some inputs (such as unskilled labor), but not necessarily for others (such as very-skilled labor like PhD economists)

Short-Run Market Supply



(a)

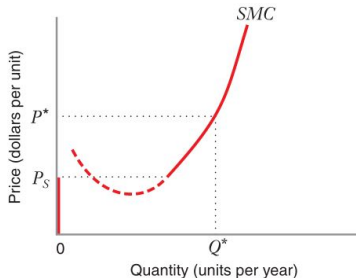


(b)

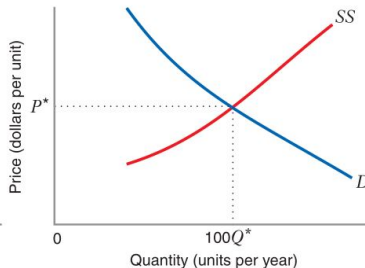
Short-Run Perfectly Competitive Equilibrium

Now that we have a market supply curve, we can find the **short-run perfectly competitive equilibrium**.

- That is, the price at which quantity supplied equals quantity demanded.



(a) Typical firm



(b) Market

Short-Run Perfectly Competitive Equilibrium

Try It Yourself

Suppose that a market is made up of 300 identical firms, each with short-run total cost curve $STC(Q) = 0.1 + 150Q^2$, and corresponding marginal cost $SMC(Q) = 300Q$ and average variable cost $AVC(Q) = 150Q$. If market demand for the good is $D(P) = 60 - P$, find the market equilibrium, assuming that all firms will supply at any price.

Solution

Each firm's supply curve comes from $P = SMC(Q)$:

$$P = 300Q \implies Q_{firm}(P) = \frac{P}{300}$$

With 300 identical firms, the market supply curve is:

$$Q_S(P) = 300 \times \frac{P}{300} = P$$

Set market supply equal to market demand:

$$Q_S(P) = Q_D(P) \implies P = 60 - P \implies P^* = 30$$

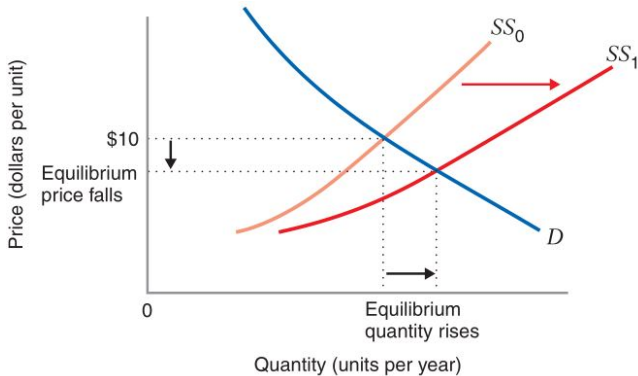
$$Q^* = 60 - 30 = 30 \qquad Q^* = 30 \implies 30 = 0.1$$

Comparative Statics Analysis

Suppose that the number of firms in a market increases (for example, suppose a large number of new oil refineries were built).

- This will shift the supply curve to the right.
- In chapter 2 we learned that if this happens, equilibrium *price* will fall, and equilibrium quantity will rise.

Comparative Statics Analysis



Comparative Statics Analysis

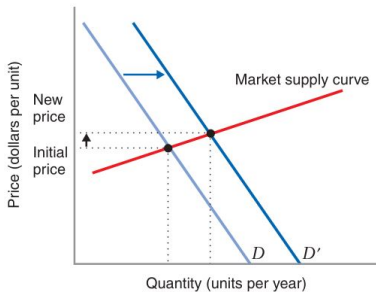
Or suppose that demand increases (maybe an economic boom means higher incomes for consumers).

- Then the market price will rise, and more units will be bought and sold.

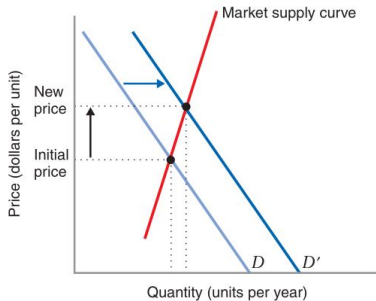
The magnitude of these changes depends on how responsive supply is to a change in price.

- If suppliers can't change their output level easily (inelastic supply), then price might rise by a lot (e.g. corn farmers).
- If suppliers can easily change their output level (elastic supply), then price might not rise by much, and quantity sold increases (e.g. hair salon).

Comparative Statics Analysis



(a) Effect of shift in demand:
Supply is relatively elastic



(b) Effect of shift in demand:
Supply is relatively inelastic

Comparative Statics Analysis

Try It Yourself

Suppose that in the Denver pizza market, the market supply curve is perfectly elastic. If people decide that pizza causes cancer, and demand falls, what will be the short-run effect on the equilibrium price and quantity?

- A) Price increases, quantity decreases
- B) Price decreases, quantity decreases
- C) Price unchanged, quantity decreases
- D) Price unchanged, quantity increases

Solution A **perfectly elastic** supply curve is horizontal – suppliers are

willing to provide any quantity at exactly that price, but none at a lower price.

When demand falls, the demand curve shifts left. Since the supply curve is horizontal, the new intersection point stays at the *same* price, but the equilibrium quantity falls – fewer units are bought and sold at that unchanged price.

Answer: C, price unchanged, quantity decreases.

Long-Run Equilibrium

In the short-run, new firms can't easily enter or exit the market.

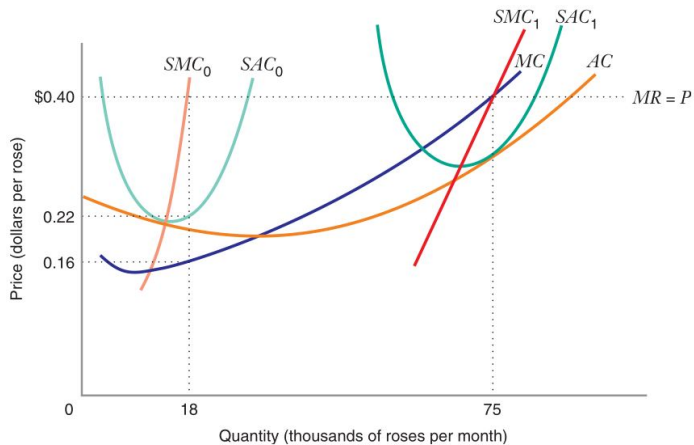
- So profit might be positive or negative.

In the long-run, however, firms can enter or exit, which drives profit to zero.

- If per-unit profits are positive, firms enter the market and lower the price and hence the per-unit profit margin
- If per-unit profits are negative, firms exit the market and raise the price and hence the per-unit profit margin

Firms in the long-run can also adjust plant size (and other fixed inputs), leading to new short-run cost curves.

Long-Run Equilibrium



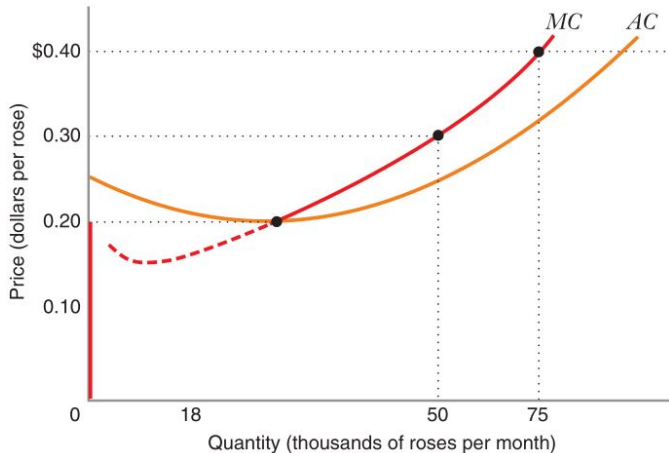
The Firm's Long-Run Supply Curve

Similar to the short run, the firm's long-run supply curve is equal to its LRMC curve, for most prices.

- But only if the firm is earning positive profit overall ($\pi \geq 0$).
- That is, only if $LRAC \geq P$

If the firm isn't earning positive profit, it will exit the market in the long run (raising the market price).

The Firm's Long-Run Supply Curve

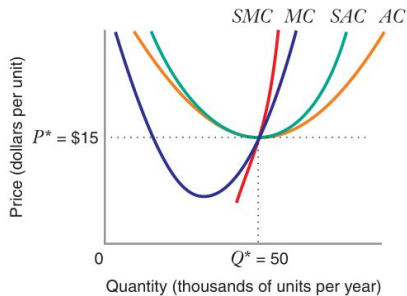


Long-Run Market Equilibrium

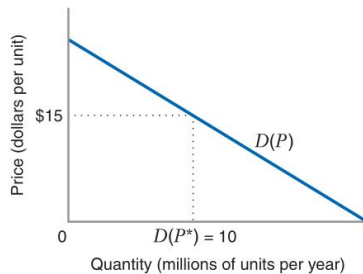
- If per-unit profits are positive, $P^* \geq AC(Q^*)$, firms enter the market and lower the price and hence the per-unit profit margin
- If per-unit profits are negative, $P^* \leq AC(Q^*)$, firms exit the market and raise the price and hence the per-unit profit margin

This drives long-run profits toward zero where $P^* = AC(Q^*)$.

Long-Run Market Equilibrium



(a) Typical firm



(b) Market

Long-Run Market Equilibrium

We can define the long-run perfectly competitive equilibrium as follows:

- Each firm chooses Q^* to maximize long-run profit, or $P^* = MC(Q^*)$ (zero profit).
- Each firm's economic profit is zero, or $P^* = AC(Q^*)$
- Market demand equals market supply at the equilibrium price, or $Q_D^*(P^*) = Q_S^*(P^*)$.

Example

Suppose that a market is composed of identical firms. Each firm has long-run cost curve $AC(Q) = 40 - Q + 0.01Q^2$ and corresponding marginal cost curve $MC(Q) = 40 - 2Q + 0.03Q^2$.

The market demand curve is given by $D(P) = 25,000 - 1000P$. Find the long-run equilibrium quantity and market price.

The first two equilibrium conditions, $P^* = MC(Q)$ and $P^* = AC(Q)$, imply that $MC(Q) = AC(Q)$. That is,

$$40 - Q + 0.01Q^2 = 40 - 2Q + 0.03Q^2 \implies Q^* = 50$$

Example

We can plug the firm's optimal quantity in to find their MC , and thus, the market price.

$$P^* = MC(Q^*) = 40 - 2 * 50 + 0.03 * 50^2 = 40 - 100 + 2,500 * 0.03 = 15$$

At a price of \$15, consumer demand is given by

$$D^*(P^*) = 25,000 - 1000 * 15 = 10,000$$

Example

Finally, we need to find the equilibrium number of firms needed to supply 10,000 units. That is,

$$Q_D^*(P^*) = n^* Q_{firm}^*(P^*) \implies n^* = 10,000/50 = 200.$$

Long-Run Market Equilibrium

Therefore, in this market, the long-run equilibrium can be defined as follows:

$$P^* = 15$$

$$Q_{firm}^* = 50$$

$$n^* = 200$$

Long-Run Market Equilibrium

Try It Yourself

Suppose that consumer demand has increased, such that $D(P^*) = 30,000 - 1,000P^*$. Find the new long-run equilibrium number of firms.

Solution

The firm's cost curves $AC(Q)$ and $MC(Q)$ haven't changed, so the zero-profit condition $MC(Q) = AC(Q)$ still gives the same firm-level solution as before:

$$Q_{firm}^* = 50 \qquad P^* = 15$$

Only the market demand has changed. At the (unchanged) equilibrium price of \$15:

$$D^*(P^*) = 30,000 - 1,000(15) = 15,000$$

Find the new number of firms:

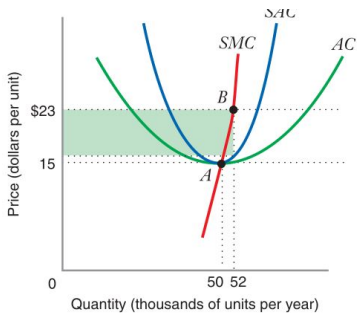
$$n^* = \frac{D^*(P^*)}{Q_{firm}^*} = \frac{15,000}{50} = 300$$

Long-Run Market Supply Curve

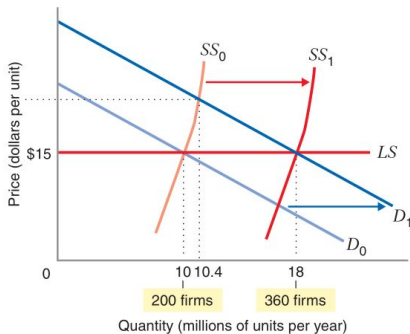
The **long-run market supply (LS) curve** tells us the total quantity of output that will be supplied at every price, assuming that all long-run adjustments have taken place.

- In other words, the long-run market supply curve traces the equilibria that result from increases in demand.

Long-Run Market Supply Curve



(a) Typical firm



(b) Market

Constant-, Increasing-, and Decreasing-Cost Industries

In the previous section, we assumed that new entry doesn't affect the price of inputs.

This may be the case when an industry's demand for an input is a small part of overall demand for that input (i.e. unskilled labor, electricity, etc...)

- When changes in an industry's output have no effect on input prices, we call it a **constant-cost industry**.

However, when expansion of an industry increases the price of an input, it is an **increasing-cost industry**.

- If an industry uses **industry-specific inputs**, they are likely to be an increasing-cost industry (e.g pharmacists).

Constant-, Increasing-, and Decreasing-Cost Industries

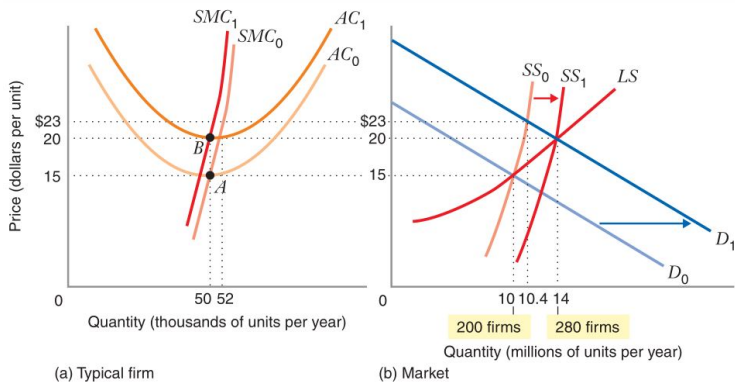
When more firms enter in an increasing-cost industry, marginal costs increase for all firms in the industry.

- It costs a firm more to produce a given quantity than before.
- Thus, minimum average cost increases.

As before, firms continue to enter until $P = \min(AC)$.

- But now $\min(AC)$ is greater than before, so the long-run equilibrium price has gone up.
- =Thus, the long-run supply curve is upward sloping.

Constant-, Increasing-, and Decreasing-Cost Industries



Constant-, Increasing-, and Decreasing-Cost Industries

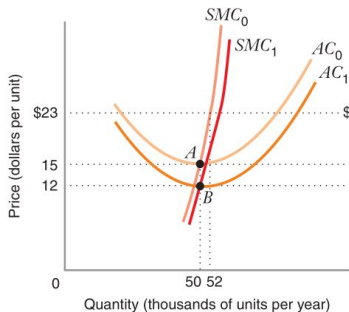
For other industries, input prices may decrease as industry output increases.

- This can be the case if the input producers can take advantage of economies of scale (computer chips, for example).

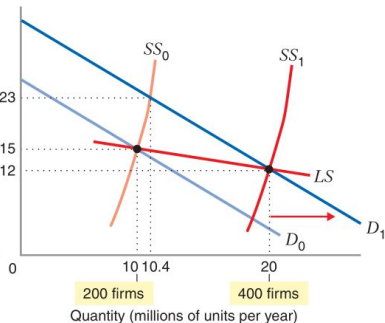
In this case, as firms enter, marginal costs fall for individual firms.

- And in the long-run, the price settles at the new, lower minimum average cost.
- Thus, the long-run supply curve is downward-sloping.

Constant-, Increasing-, and Decreasing-Cost Industries



(a) Typical firm



(b) Market

Producer Surplus

When a firm sells a product, they often sell some of the units at a higher price than they would have been *willing to* sell them at.

This difference is called **producer surplus**.

- Graphically, it is the area of the supply and demand graph below the market price, but above the firm's supply curve.
- It is analogous to consumer surplus in consumer theory.

Producer Surplus

What if a firm faces a fixed cost in the long-run (e.g. rent and machinery)?

- Then below some price, they may not be willing to supply any quantity below a certain price.
- Below this shutdown price, consumer surplus would be zero (as the firm would not supply any of the good).

Some markets don't exist even though there is potential demand for it(!)

Producer Surplus

Try It Yourself

Suppose that the market supply curve for milk is given by $Q = 60P$ (assume that there are no fixed costs), where Q is the quantity of milk sold per month (measured in thousands of gallons) when the price is P dollars per gallon. What is the producer surplus when the price of milk is \$3 per gallon?

Solution

Write the supply curve in inverse form: $Q = 60P \implies P = \frac{Q}{60}$.

At $P = 3$: $Q = 60(3) = 180$ (thousand gallons).

Since there are no fixed costs, the supply curve passes through the origin – at $Q = 0$, the minimum price a supplier would accept is \$0.

Producer surplus is the area of the triangle between the market price and the supply curve, from $Q = 0$ to $Q = 180$:

$$PS = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(180)(3 - 0) = 270$$

Producer surplus = 270 (thousand dollars), i.e. \$270,000 per month.

Lecture 10 - Competitive Markets: Applications

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Overview

In the previous chapter, we defined a perfectly competitive market, and learned how to calculate short-run and long-run equilibria

- We discussed the long-run equilibrium outcome (zero profit)

In this chapter, we will look at the impact of government interventions in perfectly competitive markets

- These include taxes and subsidies, prices floors and ceilings, and production quotas

Introduction

In this chapter, we will be considering the impacts of government interventions in a specific market.

- However, we will ignore spillover effects of these interventions into other markets

That is, we will be considering the partial equilibrium effects, as opposed to the general equilibrium effects

- A **partial equilibrium** approach focuses only on a single market (for example, the housing market).
- A **general equilibrium** approach considers how changes in that market might affect other markets.

We will also be ignoring any potential externalities.

An **externality** is a cost or benefit borne on a third party, which is not reflected in the price of the product in that market.

- For example, pollution from cars, or herd effects of vaccination.

The Invisible Hand

Why is a competitive market economically efficient (surplus maximizing)?

When the demand curve is above the supply curve, there is a consumer who is willing to pay more for the good than it costs to produce.

- If additional units are sold, consumer and/or producer surplus will increase.

The Invisible Hand

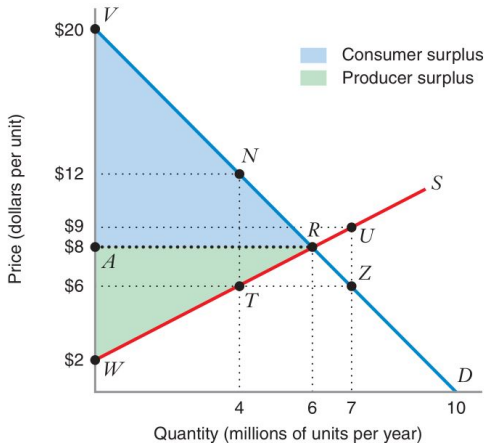
But if the demand curve is below the supply curve, additional consumers receive less benefit from the good than it costs to produce.

- If additional units are sold, consumer and/or producer surplus will decrease.

In a competitive market, there is no central planner arranging transactions.

- Yet decentralized decision-making by individual consumers and producers maximizes total surplus...
- As if the economy is guided by an *invisible hand*.

The Invisible Hand



- Notice that all trades where Willingness to Pay $>$ Marginal Cost occur
- And no sale where $WTP < MC$ occurs
- Thus, the market is maximizing total surplus

Excise Taxes

An **excise tax** is a tax on a specific commodity, such as gasoline, tobacco, or tea.

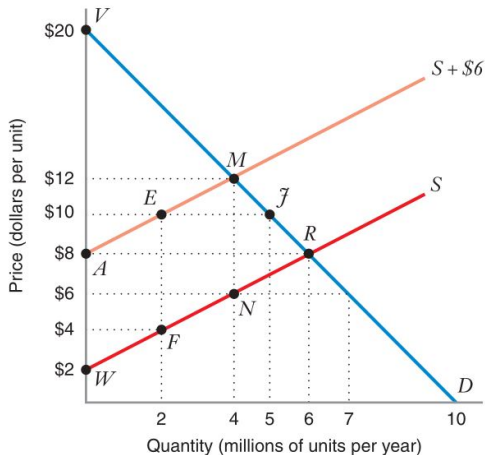
- In the absence of a tax, the price paid for the good by the consumer will equal the price received by the producer.
- That is, $Q_d(P) = Q_s(P)$.

However, when a tax is imposed, it creates a “tax wedge” between the price paid and the price received.

For example, if the government imposes a \$6 tax on gasoline, then

$$P_s = P_d - 6 \text{ or } P_s + 6 = P_d$$

Excise Tax



- The shifted supply curve intersects demand at M with $Q = 4$ and a price of $P_d = \$12$.
- However, the seller only receives $P_s = \$6$ at the point N because of the tax

Excise Tax

In this case, the market will underproduce relative to the efficient level.

- Consumer surplus will be lower from a higher P_d
- Producer surplus will be lower from a lower P_s
- The government will receive some revenue from the tax, \$6 per unit

Overall, the total surplus before the tax will be larger than the total surplus + tax revenue after the tax

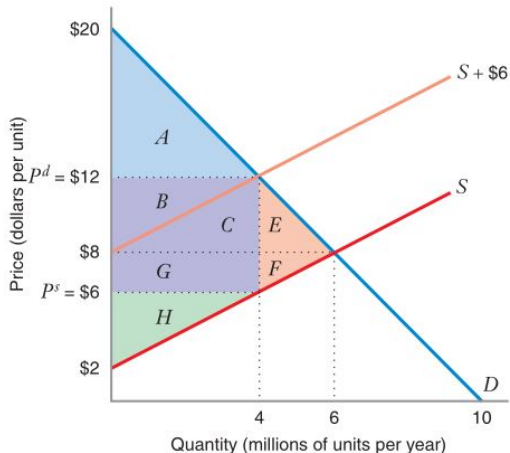
- A **deadweight loss** (in total surplus) will be created

Excise Tax

The deadweight loss is a result of the fact that the amount of gov't revenue gained through the tax is less than the consumer and producer surplus lost as a result of the tax.

Economically, the deadweight loss from an excise tax is a result of forgone market transactions where the marginal benefit gained to the consumer would exceed the marginal cost to the producer.

Excise Tax



Before:

Total Surplus

$$= A + B + C + E + F + G + H$$

After:

Consumer Surplus = A

Producer Surplus = H

Tax Revenue = $B + C + G$

Loss in Surplus:

Deadweight Loss = $E + F$

Excise Tax

Let's look at an example of the impact of an excise tax on the equilibrium quantity and price in a perfectly competitive market.

Suppose $Q_d = 10 - 0.5P_d$ and $Q_s = P_s - 2$.

Without the tax, the market equilibrium is where $Q_s = Q_d$, and $P_s = P_d$.

$$10 - 0.5P = P - 2 \Rightarrow 12 = 1.5P \Rightarrow P^* = 8$$

At the price $P^* = 8$, we have $Q_s(8) = Q_d(8) = 6$ units.

Excise Tax

Try It Yourself

From the previous example, what is the producer surplus in the market, absent the tax?

Solution

Write the inverse supply curve: $Q_s = P_s - 2 \implies P_s = Q_s + 2$.

At $Q = 0$, the minimum price a seller would accept is $P = 2$ (the y -intercept of the supply curve).

Producer surplus is the area of the triangle between the market price and the supply curve, from $Q = 0$ to $Q^* = 6$:

$$PS = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(6)(8 - 2) = 18$$

Producer surplus = \$18.

Excise Tax

Now suppose that the gov't imposes a tax of \$6. The market equilibrium is still where $Q_s = Q_d$, but $P_s = P_d - 6$

$$10 - 0.5P_d = (P_d - 6) - 2 \Rightarrow 18 = 1.5P_d \Rightarrow P_d^* = 12$$

That is the price buyers pay. To calculate the price sellers receive, $P_s^* = P_d^* - 6 = 12 - 6 = 6$. Finally, $Q_d = Q_s = P_s - 2 = 4$ units.

Excise Tax

Try It Yourself

From the previous example, what is the **change in** producer surplus resulting from the tax?

Solution

After the tax: $P_s^* = 6$, $Q^* = 4$. Using the same supply curve ($P_s = Q_s + 2$, y -intercept of 2):

$$PS_{after} = \frac{1}{2}(4)(6 - 2) = 8$$

Before the tax (from the previous slide): $PS_{before} = 18$.

Change in producer surplus:

$$\Delta PS = PS_{after} - PS_{before} = 8 - 18 = -10$$

Producer surplus falls by \$10 as a result of the tax.

Note that with the imposition of the tax, 2 fewer units were sold.

- This means that there were two consumers whose WTP exceeded the marginal cost...
- ...that were no longer willing to buy the good after the tax.

The deadweight loss is due to these forgone transactions.

Incidence of a Tax

Note that in the previous example, we didn't pay attention to who was paying the tax. It turns out, it doesn't matter who you tax, the result is the same!

And although the tax was \$6,

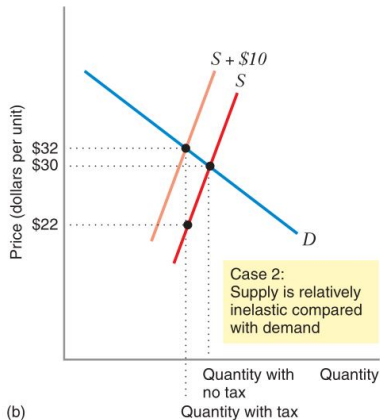
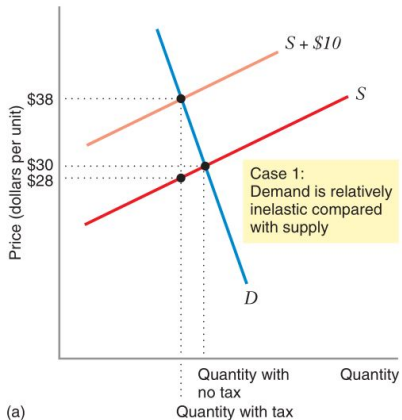
- The price paid by consumers only increased by \$4
- The price received by sellers fell by \$2

In some sense, the consumers paid more of the tax than the sellers.

The **incidence of the tax** refers to the effect that the tax has on the prices that consumers pay and that sellers receive.

Incidence of a Tax

Who has the larger incidence in each example?



Incidence of a Tax

If demand is relatively *inelastic* compared to supply, then consumers are less sensitive to a price change

- the consumers will bear more of the burden of the tax

On the other hand, if demand is relatively *elastic* compared to supply

- the producers will bear more of the burden

The incidence of a tax does not depend on who the tax is levied on!

Incidence of a Tax

Try It Yourself

Suppose that demand for sugary beverages is **perfectly inelastic**, and that supply is **relatively elastic**. What will the effect be of a \$3 tax on the price paid by consumers (P_d), and the price received by sellers (P_s)? (Hint: Draw the supply and demand curves)

- A) P_d will increase by \$3, and P_s will not change
- B) P_d will not change, and P_s will decrease by \$3
- C) P_d will increase by \$1.50, and P_s decrease by \$1.50
- D) P_d will not change, and P_s will increase by \$3

Solution

Perfectly inelastic demand is a *vertical* demand curve – consumers buy the same quantity no matter the price they pay.

Since the equilibrium quantity is pinned down entirely by demand, the tax cannot change Q^* at all. And if Q^* doesn't change, then the price along the (unchanged) supply curve at that same quantity doesn't change either – so P_s stays exactly where it was.

Since $P_d = P_s + 3$ and P_s is unchanged, the entire \$3 tax wedge shows up as an increase in P_d .

Answer: A. P_d increases by the full \$3, and P_s doesn't change – consumers bear 100% of the tax, exactly as we'd expect when demand is perfectly inelastic relative to supply.

Subsidies

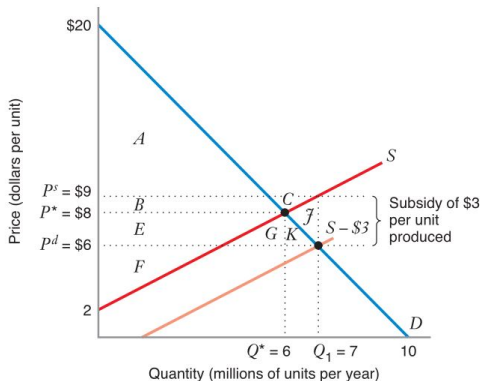
A subsidy can be thought of as a negative tax.

- When a subsidy of $\$T$ is offered by the government, the effect is basically the opposite of a tax.

The new market equilibrium will be such that $P_s = P_d + \$T$

- The market will overproduce the good.
- Consumer surplus will be higher.
- Producer surplus will be higher.
- The government must spend money on the subsidy.
- A deadweight loss will be created.

Subsidies



- The subsidies increases quantity to $Q_1 = 7$
- Consumers pay \$2 less and producers receive \$1 more, so they're both happy
- The government pays \$3 per unit which outweighs the increase in surplus

Subsidies

In this case, the deadweight loss is a result of the fact that gov't expenditures on the subsidy exceed the increase in total surplus

Economically, some consumers are encouraged to buy when the marginal cost to producers actually exceeds their marginal benefit, leading to negative total surplus.

Price Ceilings

A **price ceiling** mandates that the price of a commodity cannot rise above a given level (e.g. rent controls).

- In order to be binding, the price ceiling must be below the market price.

In this case, the lower price results in lower quantity supplied, and higher quantity demanded

- And thus, a shortage is created (excess demand)

Price Ceilings

As a result of a price ceiling, producer surplus will be lower because the

- the market price is lower
- and some producers will exit the market

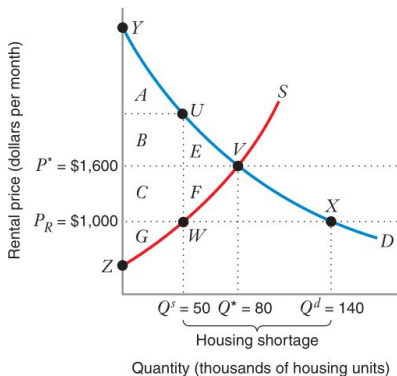
Some of that producer surplus will be transferred to consumer surplus (due to the lower price).

- However, some consumer surplus may be lost due to the lack of available supply (magnitude depends on who gets the existing supply).

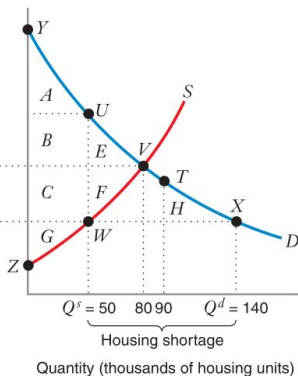
Finally, there will be deadweight loss due to foregone transactions.

Price Ceilings

50 units are sold, but you don't know which consumers can buy it, hence consumer surplus is ambiguous

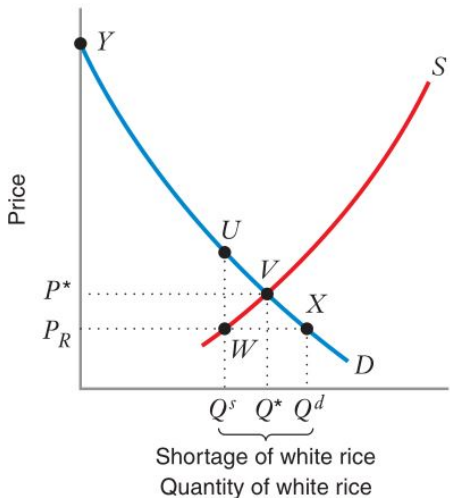


Case 1: Maximum consumer surplus



Case 2: Minimum consumer surplus

Food Shortages in Venezuela



In 2003, Venezuela's president imposed price ceilings on many household goods as a result of inflation.

- By 2009, 400 food items had mandated price ceilings.
- Today, Venezuela is still plagued by food shortages as a result.

Price Floors

A **price floor** mandates that the price of a commodity cannot fall below a given level.

- An example is a minimum wage.
- A price floor must be above the market price in order to be binding.

The result of a price floor is that quantity demanded falls, and quantity supplied rises.

- Yielding a surplus of the commodity (excess supply).

Price Floors

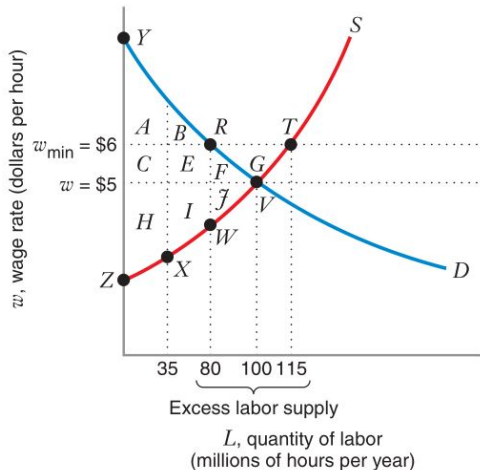
A price floor will lower consumer surplus:

- Some consumers will be forced out of the market
- **and** others will be paying a higher price than before.

Some of that will become producer surplus, as those producers are paid more.

But for those customers who are pushed out of the market, their consumer surplus disappears creating deadweight loss

Price Floors



- At the minimum wage of \$6, 115 units of labor will be supplied but only 80 units will be demanded (excess supply)
- Therefore, only 80 units will be bought at a price of \$6
- The deadweight loss
= $F + J$

Price Floors

Try It Yourself

Which of the following statements is false?

- A) With a price floor, the market will not clear.
- B) With a price floor, consumers will buy less of the good than they would in a free market.
- C) With a price floor, producer surplus will always increase.
- D) With a price floor there will be excess supply.

Solution

- A) True. A binding price floor creates excess supply, so the market does not clear.
- B) True. The higher price reduces quantity demanded.
- C) **False.** Producer surplus does *not* always increase. A higher price raises revenue per unit, but fewer units are sold, so the overall effect is **ambiguous**.
- D) True. A binding price floor creates excess supply ($Q_S > Q_D$).

Answer: C

Conclusion

In the previous few chapters, we defined a perfectly competitive market, and analyzed the implications of our assumptions.

- We also looked at a few applications of perfectly competitive markets

In the next few chapters, we will relax some of those assumptions, and look at markets that are not perfectly competitive.

- For example, monopolies and oligopolies.

Time permitting, we will also consider scenarios where perfectly competitive markets fail to maximize social welfare (total surplus) due to externalities.

Lecture 11 - Monopoly and Market Power

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

Up until now, we have assumed that there are a large number of producers and consumers in the market

- Producers are assumed to have very little market power
- They are unable to individually impact the market price (price takers)
- A result of this assumption is that total welfare is maximized

In this chapter, we will look at the other extreme.

- A market where only one firm produces the good
- And that firm controls the market
- They have complete market power
- And as we will see, they set the market price so as to maximize their own profit

Monopoly Examples

DeBeers

- Controlled 80% of the diamond market by 2000.

Microsoft Windows (in 1990s)

- Accounted for over 90% of the market for operating systems.

Google

- Ruled to be a monopoly in France, providing about 90% of web searches in France.

Profit Maximization

A perfectly competitive firm maximizes profit by setting

$$MR(Q) = P = MC(Q)$$

- That is, a firm will continue producing until the revenue generated by an additional unit of the good is equal to the additional cost of producing it.
- A monopolist will follow the same logic, except for the fact that P is not constant

Profit Maximization

For a monopoly, price has to go down as you sell more units (because of market demand):

Q (million ounces)	P (\$/oz.)	TR (\$ million)	TC (\$ million)	Profit (\$ million)
0	12	0	0	0
1	11	11.00	0.50	10.50
2	10	20.00	2.00	18.00
3	9	27.00	4.50	22.50
4	8	32.00	8.00	24.00
5	7	35.00	12.50	22.50
6	6	36.00	18.00	18.00
7	5	35.00	24.50	10.50
8	4	32.00	32.00	0
9	3	27.00	40.50	-13.50
10	2	20.00	50.00	-30.00

Profit Maximization

As the monopolist produces more, they push the price down.

- For a while, producing more leads to higher revenue (while MR is positive)
- But after a while, raising the price pushes away too many customers (when MR becomes negative)

Meanwhile, total cost is constantly increasing

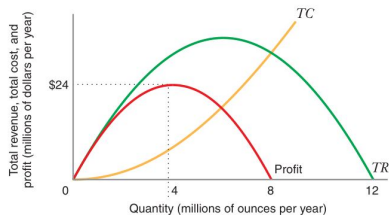
Profit Maximization

The firm will keep increasing output until the additional revenue generated by the last unit is equal to the additional cost incurred in its production.

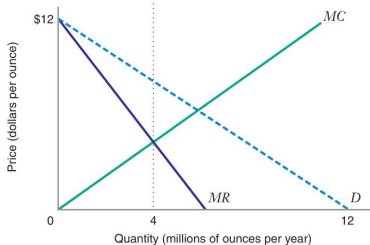
That is, until $MR(Q) = MC(Q)$

This is the **profit-maximization condition for a monopolist**.

Profit Maximization



(a)



(b)

- Total Revenue is no longer linear (since you have to lower price to sell more units)
- The maximizing profit is when $MR(Q) = MC(Q)$

Profit Maximization

Try It Yourself

If the monopolist is producing where marginal revenue exceeds marginal cost, then the monopolist should _____ to maximize profits.

- A) produce more
- B) produce less
- C) stop producing
- D) raise the price

Solution

If $MR(Q) > MC(Q)$, the last unit produced generates more revenue than it costs to produce.

That means producing one more unit would add even more profit – the firm hasn't yet reached the point where $MR(Q) = MC(Q)$.

Answer: A, produce more. The firm should keep expanding output until $MR(Q) = MC(Q)$.

Marginal Revenue Review

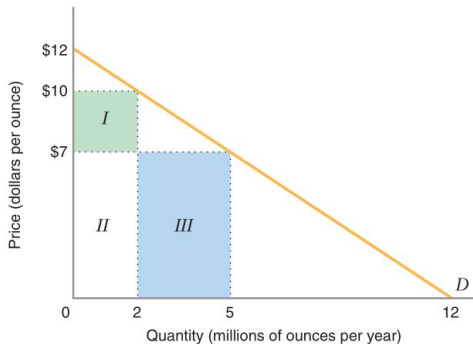
For a perfectly competitive firm, *marginal revenue equals the market price*

- If a P.C. firm sells an additional unit, the price does not change, so
$$MR(Q) = P$$

For a monopolist, the relationship does not hold

- In order to sell more units, the monopolist must lower the price on all units
- Their revenue increases by the market price P , but falls by a small amount for every unit they were previously selling.

Marginal Revenue



Original Revenue: I + II

New Revenue: II + III

- Revenue grows because you sell 3 more units (+ III)
- Revenue shrinks because you sell the original 2 units for a lower price (- I)

Marginal Revenue

In the graph, area *III* represents the revenue gained from selling additional units by lowering the price

- This is called the **quantity effect**

Area *I* represents revenue lost on the **inframarginal units**, or the units that were previously sold at the higher price.

- This is called the **price effect**

Marginal Revenue

The price and quantity effects move marginal revenue in opposite directions:

- Quantity effect (area *III*) = Price $\times$ change in quantity = $P\Delta Q$
- Price effect (area *I*) = Quantity $\times$ change in price = $Q\Delta P$

The change in total revenue is given by:

$$\Delta TR = \text{Price effect} + \text{Quantity Effect} = P\Delta Q + Q\Delta P$$

Marginal Revenue

If we divide this change in total revenue by the change in quantity, we get the marginal revenue:

$$MR = \frac{\Delta TR}{\Delta Q} = \frac{P\Delta Q + Q\Delta P}{\Delta Q} = P + Q\frac{\Delta P}{\Delta Q}$$

The first part corresponds to the quantity effect

- Another unit is sold at price P

The second part is the price effect.

- The price of Q previous units falls by $\frac{\Delta P}{\Delta Q}$

Monopolist's Profit Maximization Problem

Let's show that the monopolist's profit maximizing optimality condition holds, using the tools we've seen so far.

Because a monopolist's output impacts the price, price is no longer constant, but instead a function of Q . The monopolist's problem becomes

$$\max_Q \pi = P(Q)Q - TC(Q),$$

where now P depends on Q

Monopolist's Profit Maximization Problem

To find the profit maximizing quantity, take the derivative w.r.t Q , and set equal to zero

$$\frac{d\pi}{dQ} = Q \underbrace{\frac{dP(Q)}{dQ}}_{MR(Q)} + P(Q) - \underbrace{\frac{dTC(Q)}{dQ}}_{MC(Q)} = 0$$

Rearranging:

$$MR(Q^*) = MC(Q^*)$$

Monopolist's Profit Maximization Problem

Example: Suppose that a monopolist faces the market demand curve $P(Q) = 40 - Q$, and total cost function $TC(Q) = Q^2$.

Remember that the monopolist's optimality condition is $MR(Q) = MC(Q)$. Let's find MR and MC.

$$\begin{aligned}MR(Q) &= \frac{dTR(Q)}{dQ} \\&= \frac{d}{dQ}((40 - Q) * Q) \\&= \frac{d}{dQ}(40Q - Q^2) \\&= 40 - 2Q\end{aligned}$$

Monopolist's Profit Maximization Problem

Calculating marginal cost:

$$MC(Q) = \frac{\partial TC(Q)}{\partial Q} = 2Q$$

From our optimality condition, $MR(Q) = MC(Q)$:

$$40 - 2Q = 2Q \implies Q^* = 10$$

Finally, the market price will be the price at which the firm sells exactly Q^* units:

$$P(10) = 40 - 10 = 30$$

Monopolist's Profit Maximization Problem

Try It Yourself

Suppose that market demand in a given market can be written as $P(Q) = 20 - \frac{1}{3}Q^2$. What is the marginal revenue function for a monopolist in this market? The firm faces a marginal cost function given by $MC(Q) = 4Q^2$, find the monopolist's profit-maximizing quantity of output.

Solution

Find total revenue, then take its derivative:

$$TR(Q) = P(Q) \cdot Q = \left(20 - \frac{1}{3}Q^2\right) Q = 20Q - \frac{1}{3}Q^3$$

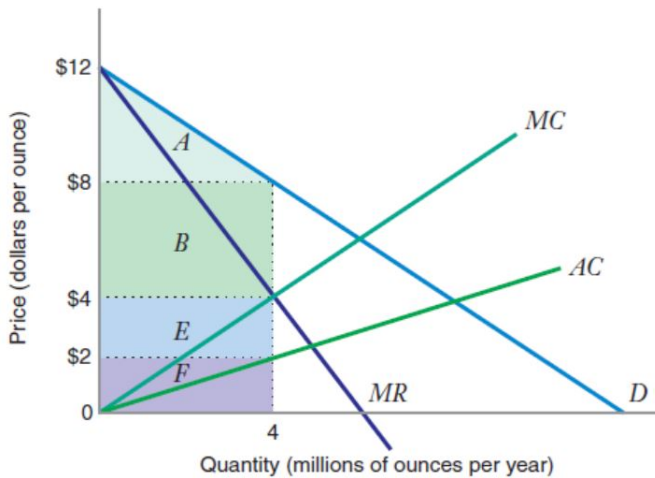
$$MR(Q) = \frac{dTR}{dQ} = 20 - Q^2$$

Set $MR(Q) = MC(Q)$:

$$20 - Q^2 = 4Q^2 \implies 20 = 5Q^2 \implies Q^2 = 4 \implies Q^* = 2$$

The profit-maximizing quantity is $Q^* = 2$.

Monopolist's Profit Maximization Problem



Three important points about the equilibrium:

1. The profit-maximizing price exceeds marginal cost of the last unit supplied.
2. Economic profits are not zero in the long run. **(why?)**
3. Consumers still receive some consumer surplus.

Implications

Try It Yourself

For a monopolist,

- A) selling price is greater than marginal revenue.
- B) selling price is equal to marginal revenue.
- C) selling price is less than marginal revenue.
- D) selling price may be above or below marginal revenue; it depends on the price buyers are willing to pay.

Solution

Recall $MR = P \left(1 + \frac{1}{\epsilon_{Q,P}} \right)$, where $\epsilon_{Q,P} < 0$ since demand slopes downward.

Since $\frac{1}{\epsilon_{Q,P}} < 0$, the term in parentheses is less than 1, so $MR < P$ – selling price always exceeds marginal revenue for a monopolist (as long as it's producing on the elastic portion of demand, which it must be at a profit-maximizing quantity with positive marginal cost).

Answer: A, selling price is greater than marginal revenue.

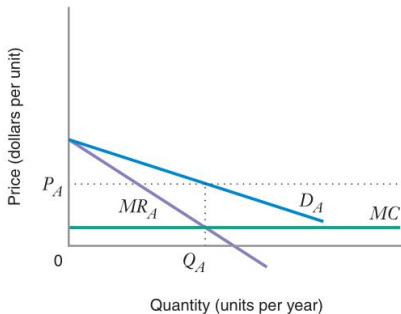
Price Elasticity of Demand and the Profit-Maximizing Price

The gap between the price that a firm charges, and their marginal cost, will depend on how sensitive consumers are to a change in price.

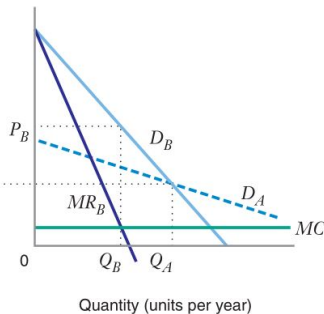
- If consumers are very price sensitive, the monopolist will set the price relatively low (the price effect dominates)
- If the opposite is true, the monopolist will set the price higher (the quantity effect dominates)

Intuitively, if a firm faces more competition from an outside industry, consumers will be more price-sensitive and the monopolist's price will be lower.

Price Elasticity of Demand and the Profit-Maximizing Price



(a) Market A



(b) Market B

Marginal Revenue and Price Elasticity of Demand

Formally, we can use the equation for marginal revenue to express the relationship between MR and price:

$$\begin{aligned}MR &= P + Q \frac{\Delta P}{\Delta Q} \\&= P + Q \frac{\Delta P}{\Delta Q} * \frac{P}{P} \\&= P + P \frac{\Delta P}{\Delta Q} * \frac{Q}{P} \\&= P \left(1 + \frac{\Delta P}{\Delta Q} * \frac{Q}{P} \right)\end{aligned}$$

Marginal Revenue and Price Elasticity of Demand

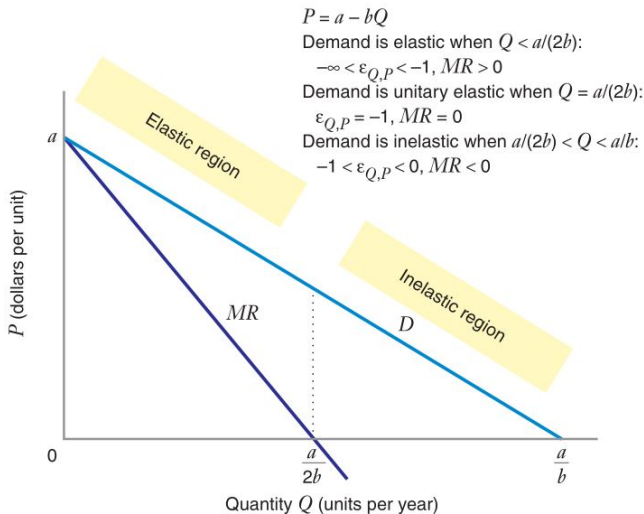
Remember that price elasticity of demand is given by

$$\epsilon_{Q,P} = \frac{\Delta Q}{Q} / \frac{\Delta P}{P} \text{ or } \frac{\Delta Q}{\Delta P} * \frac{P}{Q} \quad (1)$$

Therefore, marginal revenue can be expressed as

$$MR = P \left(1 + \frac{1}{\epsilon_{Q,P}} \right)$$

Marginal Revenue and Price Elasticity of Demand



Marginal Cost and Price Elasticity of Demand

Remember that, at Q^* , $MR(Q^*) = MC(Q^*)$.

We can therefore rewrite the previous equation as

$$MC(Q^*) = P \left(1 + \frac{1}{\epsilon_{Q,P}} \right)$$

Rearranging:

$$P = MC(Q^*) \frac{1}{1 + \frac{1}{\epsilon_{Q,P}}}$$

Since $(1 + \frac{1}{\epsilon_{Q,P}}) < 1$, you will charge a price above your cost

Differentiated Products

Monopolists aren't the only firms that face a downward-sloping demand curve.

- Many firms sell products that are unique (so they do not operate in a perfectly competitive market), but they still face competitors

Think of breakfast cereals (Lucky Charms, Frosted Flakes, Cap'n Crunch).

- These firms still charge a price above their marginal cost

The Lerner Index

If a monopolist faces fewer competitors (few close substitutes), consumers will be less price sensitive and the firm's markup will be greater.

Intuitively, we can use the firm's markup as a measure of market power.

- This is known as the **Lerner Index**.
- Firms with greater market power, and thus a higher markup, will score higher on the Lerner Index.

The Lerner Index

Try It Yourself

Inverse demand for a monopolist's product is given by $P(Q) = 300 - 6Q$ while the monopolist's marginal cost is given by $MC(Q) = 3Q$.

What is the profit-maximizing quantity of output for this monopolist?

The Lerner Index

Try It Yourself

Inverse demand for a monopolist's product is given by $P(Q) = 300 - 6Q$ while the monopolist's marginal cost is given by $MC(Q) = 3Q$.

What is the profit-maximizing price for this monopolist?

The Lerner Index

Try It Yourself

Inverse demand for a monopolist's product is given by $P(Q) = 300 - 6Q$ while the monopolist's marginal cost is given by $MC(Q) = 3Q$.

At the profit-maximizing quantity of output, what is the firm's marginal cost of the final unit produced?

Solution

Revenue:

$$TR(Q) = (300 - 6Q)Q = 300Q - 6Q^2$$

Marginal revenue:

$$MR(Q) = 300 - 12Q$$

Set $MR = MC$:

$$300 - 12Q = 3Q \implies Q^* = 20$$

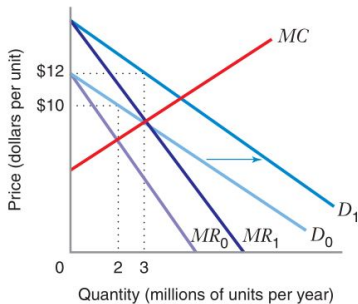
Then:

$$P^* = 300 - 6(20) = 180$$

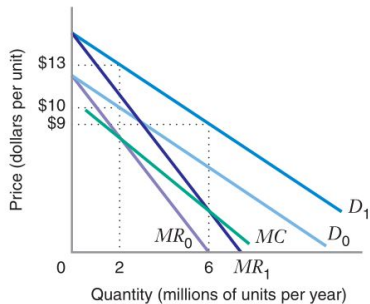
$$MC(Q^*) = 3(20) = 60$$

$$Q^* = 20, \quad P^* = 180, \quad MC = 60$$

Comparative Statics: Shifts in Market Demand

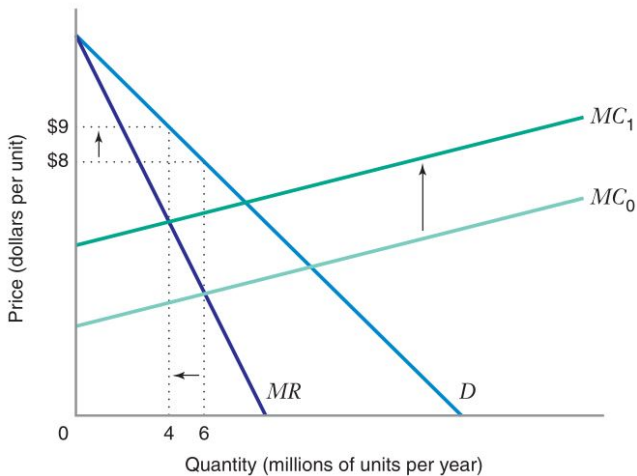


(a)



(b)

Comparative Statics: Shifts in Marginal Cost



Multi-plant Monopoly

Let's consider the case where a monopolist has two production plants. Think of a utility company with multiple power plants.

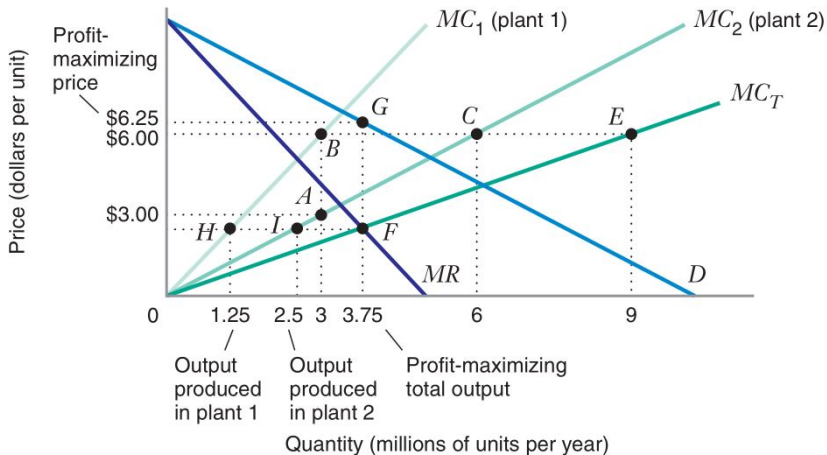
- Each plant has a separate marginal cost; MC_1 and MC_2 .
- How should the plant allocate production between the two plants?

Multi-plant Monopoly

Suppose the firm produces equal amounts at both plants:

- Because the marginal cost is different at each plant, the monopoly can produce fewer units at the higher-cost plant, and more units at the lower cost plant
- By doing so, marginal cost of the final unit is decreased and profit rises

Multi-plant Monopoly



Multi-plant Monopoly

This exercise illustrates the importance of the **multi-plant marginal cost curve**

- This curve denotes the marginal cost of producing a given quantity in the lowest-cost way
- Each additional unit will be produced at the plant where the marginal cost is the lowest

So the multi-plant marginal cost curve is found by simply summing the individual MC curves horizontally

Multi-plant Monopoly

The production decision is the same a single-plant monopolist by using the multi-plant marginal cost curve

- A multi-plant monopolist can find the profit maximizing quantity by setting marginal revenue equal to the multi-plant marginal cost
- The market price is the price that will allow the firm to sell all of its output from both plants

Multi-plant Monopoly

Try It Yourself

A monopolist owns two plants in which to produce product A. The marginal cost of producing A is increasing, but currently is lower in plant 1 than in plant 2. How should the monopolist allocate production?

- A) Produce all output in plant 1.
- B) Produce all output in plant 2.
- C) Produce 50 percent in plant 1 and 50 percent in plant 2.
- D) Produce in plant 1 up to the point where marginal costs are equated across the plants.

Solution

- **A)** False. Producing everything in plant 1 eventually makes its marginal cost too high.
- **B)** False. Plant 2 currently has the higher marginal cost.
- **C)** False. A 50/50 split is not generally cost-minimizing.
- **D) True.** Allocate production until

$$MC_1 = MC_2.$$

If one plant has a lower marginal cost, shift production to that plant.

Answer: D

Profit Maximization by a Cartel

A **cartel** is a group of producers that determine the price and output in a market through **collusion**

- An example is the Organization of Petroleum Exporting Countries (OPEC)
- This was a group of some of the world's largest oil producers, including Kuwait, Saudi Arabia, Iran and Venezuela

When a cartel works as intended, it acts as a monopoly firm with multiple plants

Profit Maximization by a Cartel

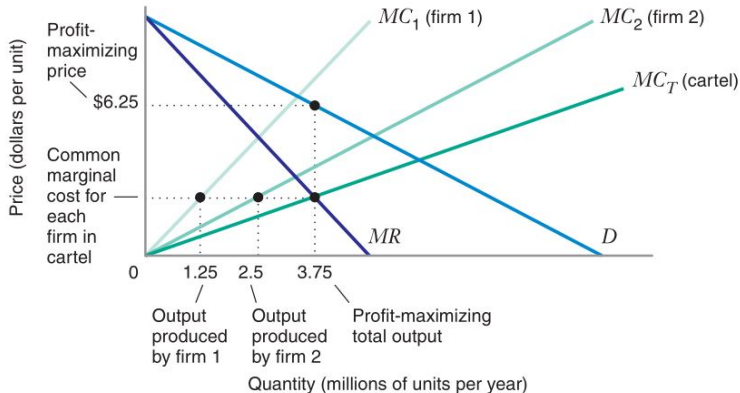
At the profit-maximizing solution, the cartel allocates production between the two firms such that marginal cost is equal.

For example, suppose that two firms collude.

Then for $Q^* = Q_1^* + Q_2^*$, the following should be true:

$$MR(Q^*) = MC_1(Q_1^*) \text{ and } MR(Q^*) = MC_2(Q_2^*)$$

Profit Maximization by a Cartel



Profit Maximization by a Cartel

So if cartels are so great for producers, why aren't there more of them?

- They are illegal (we will see why in a few slides).
- They are hard to keep together (one firm can lower the price and sell more units).

The Welfare Economics of Monopoly

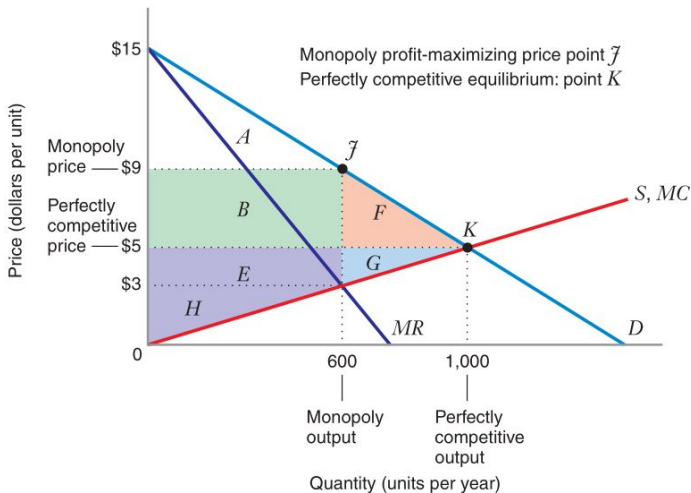
Suppose a perfectly competitive industry was monopolized

- For example, one company bought all of its competitors

Then the industry supply curve would become that firm's marginal cost curve...

- And that firm would set output such that $MC(Q^*) = MR(Q^*)$
- Quantity produced would fall, and price would rise

The Welfare Economics of Monopoly



The Welfare Economics of Monopoly

G and F are the **deadweight loss due to monopoly**

- The monopolist's surplus increases by area B (but they lose G)
- Consumer surplus decreases by B and F

By raising the price and lowering output, the firm has eliminated transactions that are welfare increasing.

- That is, transactions where the marginal benefit to consumers is greater than the marginal cost to the producer

The Welfare Economics of Monopoly

This additional producer surplus encourages firms to acquire monopoly power

- Activities aimed at creating or preserving monopoly power are called **rent-seeking activities**

For example, lobbying the government to regulate an industry in order to prevent entry (think of Uber in London)

- A firm would be willing to spend some of that additional surplus on rent-seeking activities in order to maintain its market power (through lobbying)

The Welfare Economics of Monopoly

This is why the government typically fights to keep firms from gaining too much market power.

- With some exceptions, the U.S government does not allow monopolies to exist (Microsoft, for example)
- Cartels are also illegal in the U.S.

Enforcement of this varies over time though ...

Why do Monopoly Markets Exist?

One reason may be that the total cost incurred by a single firm to produce output for the market might be lower than the combined total cost for two or more firms

An example is satellite television:

- Large up-front fixed costs (building a satellite and shooting it into space).
- One firm needs one satellite to serve the whole market.
- Two firms need two satellites.

This is known as a **natural monopoly**.

Why do Monopoly Markets Exist?

Another example is an electric company.

- Again, large up-front fixed costs (building transmission lines throughout a city)
- Would be very expensive for two firms to compete for customers in one city

Economies of scale imply it's cheaper for one firm to produce instead of two firms

Why do Monopoly Markets Exist?

A natural monopoly is an example of a **barrier to entry**.

- Barriers to entry allow an incumbent to earn a profit, while making entry unprofitable for newcomers
- Without barriers to entry, if an incumbent is earning a profit, more firms will enter
- Profit will go to zero

Why do Monopoly Markets Exist?

Examples

- Large fixed costs
- Social media platforms (Snapchat, Facebook, Instagram) have positive network externalities
- Government regulation (taxis)

Why do Monopoly Markets Exist?

Legal barriers to entry exist when an incumbent firm is legally protected against competitors.

Patents are the biggest example:

- Drug companies are given monopoly power by the government in order to recoup their R&D costs
- Once the patent expires, new entrants come to the market, and the price usually falls.

Why do Monopoly Markets Exist?

Strategic barriers to entry result when an incumbent firm takes explicit steps to deter entry.

An example is starting a price war if a new firm enters.

Why do Monopoly Markets Exist?

Try It Yourself

Economists consider monopolists:

- A) to be efficient, since they earn greater profits than perfect competitors.
- B) to be inefficient since all consumer surplus is transferred to the monopolist in the form of profits.
- C) to be inefficient since they earn less producers' surplus than all firms taken together in a competitive market.
- D) to be inefficient since the monopolist restricts output from the competitive level, thus creating dead-weight loss.

Solution

- **A)** False. Efficiency depends on total surplus, not profits.
- **B)** False. Some consumer surplus is transferred to the monopolist, but some is lost as deadweight loss.
- **C)** False. A monopolist typically earns *more*, not less, producer surplus than firms in perfect competition.
- **D) True.** A monopolist restricts output below the competitive level, creating deadweight loss.

Answer: D

Lecture 12 - Capturing Surplus (Not Covered)

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

In this chapter, we will discuss ways that firms can capture additional surplus.

- What information a firm needs to capture additional surplus
- What market characteristics are necessary to enable firms
- We will discuss price discrimination, bundling, and versioning

Price Discrimination

The act of charging different price to different consumers for the same good or service is known as **price discrimination**

- The goal is for the firm to increase their producer surplus.

Three Types of Price Discrimination

1. **First-degree price discrimination** - firms prices each unit of the good to the consumer's maximum WTP.
 - Example: auction - seller hopes to find the consumer with the highest WTP for the good.
2. **Second-degree price discrimination** - Firm offers customers quantity discounts.
 - Example: a grocery store offer to buy one and get the 2nd one at half price.
3. **Third-degree price discrimination** - Firm divides market into segments, and sets price for each segment by equating MR to MC for each group.
 - Example: movie ticket prices for students vs. adults.

Price Discrimination

There are three necessary market features for price discrimination:

1. Market power - demand curve is downward sloping
2. Information on consumer WTP
3. No opportunity for resale or arbitrage

Price Discrimination

Try It Yourself

The conditions for capturing more surplus through price discrimination **do not** include

- A) an ability to determine which groups of people have the greatest wealth.
- B) an ability to differentiate different market segments meaning that some groups of people are willing to pay more for a product than others.
- C) an ability to prevent resale of products.
- D) an imperfectly competitive industry.

Solution Recall the three necessary conditions for price discrimination:

(1) market power/downward-sloping demand, (2) information about consumer WTP, and (3) no resale.

B, **C**, and **D** map directly onto conditions (2), (3), and (1), respectively – all are necessary.

A is *not* a necessary condition. What matters is a group's *willingness to pay* for this specific good, not their overall wealth – a low-income group could have a high WTP for a particular product (or vice versa). Wealth and WTP for a given good aren't the same thing.

Answer: A

First-Degree Price Discrimination

Remember that we can think of a market demand curve as the willingness to pay curve:

- Consumers are lined up from highest WTP to lowest
- With first-degree price discrimination, firm is able to charge each customer their WTP (they need to be really smart to do this!!)

In this case, the firm will keep selling the good as long as the consumer's WTP is greater than their marginal cost.

- This is because the marginal revenue curve under first-degree price discrimination is the demand curve!

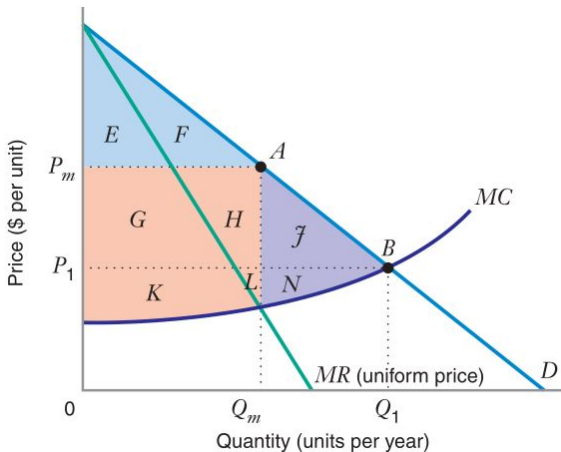
First-Degree Price Discrimination

What happens to surplus in this case:

1. Because each consumer pays their WTP, they receive no consumer surplus.
2. Producer surplus is the area above their MC curve and below the demand curve. In short, the firm captures all of the surplus.
3. And because they can charge different prices, they sell the socially-optimal quantity. There is no deadweight loss.

First-Degree Price Discrimination

With first-degree PD, the firm captures the total surplus in the market, and there is no deadweight loss.



First-Degree Price Discrimination

Would this work with a pair of designer sunglasses?

- Suppose the firm knew exactly how much each person that came into the store was willing to pay.
- And suppose that they charged every consumer their WTP.

Someone with a high WTP might instead just wait for someone with a low WTP to buy a pair and then offer them more than they paid

This illustrates why it can't be possible to resell the item. Otherwise, nobody would be willing to pay a higher price than anyone else.

First-Degree Price Discrimination

Try It Yourself

Suppose a firm has constant marginal cost $MC = 2$, and faces a market demand curve of $P = 20 - Q$.

What will be a single-price monopolist's producer surplus in the market?

Solution

$$TR(Q) = P \cdot Q = (20 - Q)Q = 20Q - Q^2$$

$$MR(Q) = \frac{dTR}{dQ} = 20 - 2Q$$

Set $MR = MC$:

$$20 - 2Q = 2$$

$$18 = 2Q$$

$$Q^* = 9$$

$$P^* = 20 - Q^* = 20 - 9 = 11$$

$$PS = (P^* - MC) \times Q^* = (11 - 2)(9) = (9)(9) = 81$$

Producer surplus: \$81

First-Degree Price Discrimination

Try It Yourself

Suppose that the firm can now perfectly price discriminate. What will be the firm's producer surplus?

Solution

Perfect price discrimination: produce until $P = MC$.

$$20 - Q = 2 \quad \Rightarrow \quad Q = 18$$

Producer surplus equals the entire area between demand and MC :

$$PS = \frac{1}{2}(18)(18) = 162$$

Producer Surplus: \$81 $\rightarrow$ \$162

First-Degree Price Discrimination

Higher education is a market where firms engage in first-degree PD. Even though universities are not monopolists, they still face a downward-sloping demand curve.

Question: How do colleges estimate your WTP?

- The price you pay depends on your expected family contribution (EFC) from FAFSA.
- Those with a higher ability to pay receive less money in need-based aid.

Second-Degree Price Discrimination

For many goods and services, consumers often buy more than one unit.

- And their individual demand curves are often downward sloping.
- In other words, their WTP decreases as they buy more.
- Sellers can capture additional surplus by charging customers a lower price for the second unit than the first.

This is **second-degree price discrimination**

Second-Degree Price Discrimination

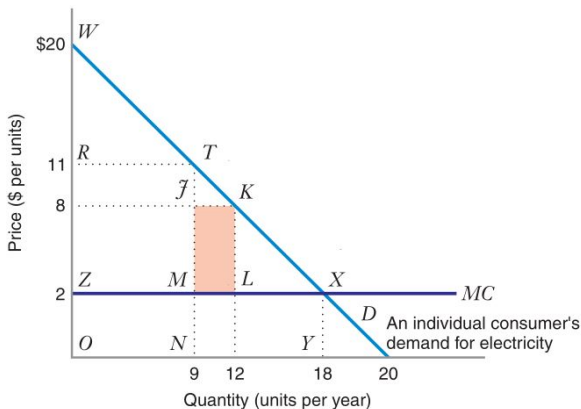
The first form of second-degree PD is block pricing.

For example, suppose there is only one consumer in the market for electricity.

- The electric company charges \$11 per unit for the first 9 megawatt-hour used and \$8 for every Mwh after that.
- The consumer is will buy more than if every unit simply cost \$11, but the additional revenue can still be extracted for the first 9 units.
- This type of pricing is known as a **block tariff**.

Second-Degree Price Discrimination

The firm captures additional producer surplus by selling additional units at a lower price.

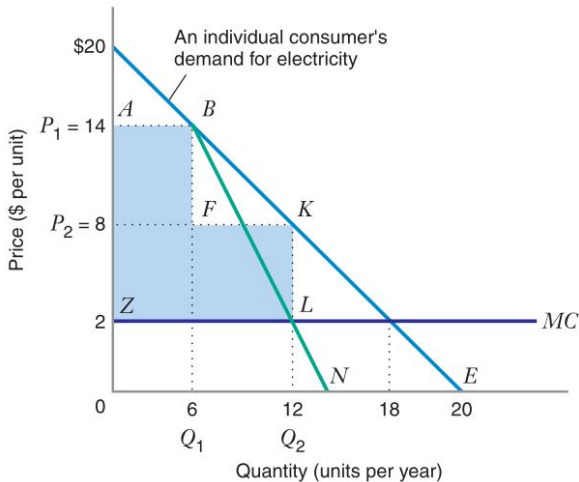


Second-Degree Price Discrimination

How do we find the optimal price and quantity for the first and second blocks?

1. Write producer surplus (or profit) in terms of Q_1 and Q_2 .
2. Maximize the profit function with respect to both Q_1 and Q_2 . Find two equations.
3. Solve the system of two equations for Q_1 and Q_2

Second-Degree Price Discrimination



Second-Degree Price Discrimination

Example: Suppose demand is given by $P = 20 - Q$, and the firm's total cost is given by $TC(Q) = 2Q$, such that $MC(Q) = 2$.

The firm's profit function is given by

$$\Pi = P_1 Q_1 + P_2 (Q_2 - Q_1) - 2 * Q_2$$

If we plug the demand function in for P_1 and P_2 , we find

$$\Pi = \underbrace{(20 - Q_1)}_{P_1=P(Q_1)} Q_1 + \underbrace{(20 - Q_2)}_{P_2=P(Q_2)} (Q_2 - Q_1) - 2 * Q_2$$

Second-Degree Price Discrimination

We are solving an unconstrained maximization problem, so we simply take derivatives w.r.t. Q_1 and Q_2 and set equal to zero:

$$\frac{\partial}{\partial Q_1} = (20 - 2Q_1) - 20 + Q_2 = 0$$

$$\frac{\partial}{\partial Q_2} = 20 - 2Q_2 + Q_1 - 2 = 0$$

Simplifying the first equation, we have $Q_2 = 2Q_1$. Plugging that into the second equation:

$$20 - 4Q_1 + Q_1 - 2 = 0 \implies Q_1^* = 6$$

Second-Degree Price Discrimination

Plugging $Q_1^* = 6$ into the simplified first equation:

$$Q_2^* = 12$$

Prices are found by plugging each quantity into the demand function:

$$P_1^* = 20 - Q_1^* = 14 \quad \text{and} \quad P_2^* = 20 - Q_2^* = 8$$

Second-Degree Price Discrimination

The second type of second-degree price discrimination is a two-part tariff.

- We discussed this from the consumer's perspective in chapter 5.

A **two-part tariff** is when the firm charges a subscription fee, and then a per-unit price.

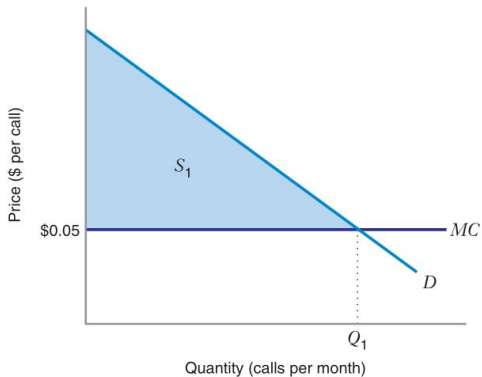
- Examples: Costco, Amazon Prime, Zipcar.

Second-Degree Price Discrimination

For example, suppose that a telephone company has a per-unit cost of \$0.05.

- If the provider sets the per-unit price to \$0.05, there will be no deadweight loss
- Then the firm can set the subscription cost to whatever price is needed to capture all of the surplus (e.g. \$20).

Second-Degree Price Discrimination



Second-Degree Price Discrimination

More generally, the firm should set the per-unit price such that $D = MC$ (as opposed to $MR = MC$).

- That's because with the subscription fee, they don't have to give up producer surplus on previous units.
- They just capture it back with the subscription fee.
- The subscription fee can be set as high as consumer surplus, and the consumer will be indifferent.

Second-Degree Price Discrimination

If there is a lot of heterogeneity in demand curves, then a firm can offer a menu of subscription packages.

- Think of a ski resort, or a cell phone service provider.

Then the customers “sort” themselves according to their demand.

- The firm can extract more surplus from each group.

Third-Degree Price Discrimination

Third-degree price discrimination means charging different prices to different groups of customers based on some observable characteristic. For example,

- Student prices at a movie theater
- Coupons and rebates
- Video game prices declining over time

These allow the firm to charge higher prices to those with higher average willingness to pay.

Third-Degree Price Discrimination

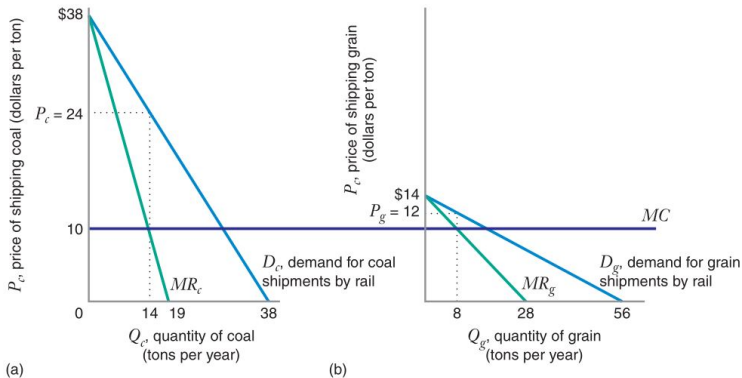
Consider rail transportation in the early 20th century:

- It cost the railroad about the same amount of money to transport coal and grain.
- But railroads faced more competition from barges and trucks when carrying grain (available substitutes)
- Thus, the railroad would charge 2-3 times the price to ship coal even though MC was the same

Again, market power matters!

Third-Degree Price Discrimination

The market demand for rail shipping is flatter for grain since there are better substitutes



Third-Degree Price Discrimination

How does a firm decide the profit maximizing prices to charge?

- If MC is constant, firm just sets $MR(Q^*) = MC(Q^*)$ in each market (or for each group), and then finds P such that $D(P) = Q^*$ (just like single-price monopolist).

If MC isn't constant, firm has to take into account that selling another unit in market A increases the cost in market B . Firm's problem is:

$$\max_{Q_A, Q_B} P_A Q_A + P_B Q_B + TC(Q_A + Q_B)$$

Third-Degree Price Discrimination

Example: Suppose a railroad faces demand curves for transporting grain and coal given by $P_C = 38 - Q_C$ and $P_G = 14 - \frac{1}{4}Q_G$. Marginal cost is given by $MC = 10$.

For grain, the firm sets $MR(Q_G^*) = MC(Q_G^*)$. $MR(Q_G^*)$ is given by

$$\frac{\partial [P(Q_G)Q_G]}{\partial Q_G} = \frac{\partial [14Q_G - \frac{1}{4}Q_G^2]}{\partial Q_G} = 14 - \frac{1}{2}Q_G$$

Third-Degree Price Discrimination

The optimal quantity of grain is given by:

$$14 - \frac{1}{2}Q_G = 10 \implies Q_G^* = 8.$$

Market price is

$$P_G^* = 14 - \frac{1}{4} * 8 = 12$$

Third-Degree Price Discrimination

For coal, the firm sets $MR(Q_C^*) = MC(Q_C^*)$. $MR(Q_C^*)$ is given by

$$\frac{\partial [P(Q_C)Q_C]}{\partial Q_C} = \frac{\partial [38Q_C - Q_C^2]}{\partial Q_C} = 38 - 2Q_C$$

The optimal quantity of coal is given by:

$$38 - 2Q_C = 10 \implies Q_C^* = 14$$

Market price is

$$P_C^* = 38 - 14 = 24$$

Third-Degree Price Discrimination

Try It Yourself

Which of the following statements regarding a monopoly's first-degree price discrimination is correct?

- A) With first-degree price discrimination, consumer surplus is small, yet still greater than zero.
- B) With first-degree price discrimination, producer surplus is lower than with uniform pricing.
- C) With first-degree price discrimination, deadweight loss is large.
- D) With first-degree price discrimination, total surplus is greater than when the monopoly charges a uniform price.

Solution

- A)** False – consumer surplus is exactly **zero** under perfect (first-degree) price discrimination, since every consumer pays exactly their WTP.
- B)** False – producer surplus is *higher* under first-degree PD than under uniform pricing, since the firm captures surplus on every unit instead of just the units sold at a single price.
- C)** False – deadweight loss is **zero** under first-degree PD, since the firm produces the socially efficient quantity (selling until $P = MC$).
- D) True.** Since first-degree PD eliminates deadweight loss entirely, total surplus reaches the maximum possible (efficient) level – strictly greater than under uniform monopoly pricing, which always leaves some deadweight loss on the table.

Answer: D

Since it's so hard to know consumers' WTP, firms sometimes use observable characteristics to sort people into groups with high or low expected WTP.

This is known as **screening**.

- For example, movie theaters figure that, on average, students have lower WTP.
- They can screen customers based on whether they are a student or not.

Intertemporal Price Discrimination

Intertemporal price discrimination is when a firm sells a good at different prices at different times of the day (or week, or year) to capture additional surplus.

- For example, movies and video games often sell at higher prices initially, and then the price falls over time.
- Airline tickets may be cheaper 3 months out, and very expensive one week out.

Note that prices declining over time does not always reflect price discrimination:

- Sometimes it's a result of decreasing marginal costs (think of flat-screen TVs)
- Sometimes it's a result of increasing competition in the market

Coupons and Rebates

Firms use coupons and rebates to price discriminate.

- People who are more price sensitive (lower WTP) are more likely to clip coupons
- They pay a lower net price as a result.
- Those who are less price sensitive may not bother with coupons.

Capacity Constraints

Some firms are constrained in the number of units of a good they can sell of a given product.

- Examples: Airlines, movie theaters, cruise lines.

In that case, the firm may not be able to set Q_1 and Q_2 such that $MR(Q_1) = MC(Q)$ and $MR(Q_2) = MC(Q)$ where $Q = Q_1 + Q_2$.

Instead, suppose that $MR(Q_1) = MC(Q)$ but $MR(Q_2) > MC(Q)$ for two segments of some market.

- This implies $MR_2 > MR_1$

Capacity Constraints

If $MR_2 > MR_1$, what should the firm do?

- Since MC is the same for both market segments, we can focus only on MR.

Recall that marginal revenue is the additional revenue from selling one more unit.

- Then if the firm sells one less unit in market 1 and one more unit market 2, total revenue will go up.
- Alternatively, if $MR_1 > MR_2$, the firm should sell more to market 1 and less to market 2.

Capacity Constraints

Therefore, the optimal allocation to segments 1 and 2 is such that $MR_1 = MR_2$.

- Because of the capacity constraint, the firm can't produce enough to where $MR = MC$ for either segment.

The result is that the firm must solve a system of two equations: $MR_1 = MR_2$ and $Q_1 + Q_2 = C$, where C is the firm's capacity.

Capacity Constraints

Do firms actually think about their problem in this way? Might seem somewhat unrealistic, but firms face this decision all the time

- Airlines have to decide where to set their prices so that they can sell some seats at a low price, and some at a higher price.
- Movie theaters have to decide how to set student and general admission prices to maximize profit.

Capacity Constraints

Suppose that a firm sells their good in two different markets, 1 and 2, and faces demand functions $Q_1 = 200 - 2P_1$ and $Q_2 = 250 - P_2$. Marginal cost is \$10 per unit, and the firm's capacity is 150 units. What is the optimal price and quantity in each market?

The firm should set Q_1 and Q_2 such that $MR(Q_1) = MR(Q_2)$. Therefore, we need to find the marginal revenue curves for both segments.

First, write the demand functions such that P is a function of Q .

$$P_1 = 100 - \frac{1}{2}Q_1 \quad \text{and} \quad P_2 = 250 - Q_2$$

Capacity Constraints

Next, find MR_1 and MR_2 :

$$TR(Q_1) = P_1 * Q_1 = 100Q_1 - (1/2)Q_1^2 \implies MR(Q_1) = 100 - Q_1$$

$$TR(Q_2) = P_2 * Q_2 = 250Q_2 - Q_2^2 \implies MR(Q_2) = 250 - 2Q_2$$

Finally, equate MR_1 to MR_2 , gives us $100 - Q_1 = 250 - 2Q_2$. Our capacity constraint is given by $Q_1 + Q_2 = 150$

Capacity Constraints

Finally, we can solve this system of two equations. First, plug the second equation into the first:

$$100 - (150 - Q_2) = 250 - 2Q_2 \implies Q_2^* = 100$$

Then

$$Q_1^* = 150 - Q_2^* = 150 - 100 = 50$$

Finally, we can find prices using the segment demand functions:

$$P_1^* = 100 - \frac{1}{2}50 = 75 \quad \text{and} \quad P_2^* = 250 - 100 = 150$$

Capacity Constraints

Try It Yourself

Let a monopolist face consumer group A with inverse demand $P_A = 100 - 2Q_A$ and consumer group B with inverse demand $P_B = 80 - Q_B$. The monopolist can conduct third degree price discrimination, but faces a capacity constraint that $Q_A + Q_B \leq 50$. What will be the amount supplied to each of the customer groups?

Solution

With a binding capacity constraint,

$$MR_A = MR_B, \quad Q_A + Q_B = 50.$$

$$MR_A = 100 - 4Q_A, \quad MR_B = 80 - 2Q_B.$$

Using $Q_B = 50 - Q_A$,

$$100 - 4Q_A = 80 - 2(50 - Q_A)$$

$$= -20 + 2Q_A$$

$$120 = 6Q_A$$

$$Q_A = 20$$

$$Q_B = 50 - 20 = 30$$

$$P_A = 60, \quad P_B = 50.$$

$$Q_A = 20, Q_B = 30, P_A = \$60, P_B = \$50$$

Versioning

Another way that firms capture additional surplus is by versioning.

Versioning is the strategy of selling two or more versions of a product with different qualities.

- For example, laptop manufacturers offer low-quality and high-quality versions of their products.
- The hope is that less price-sensitive consumers will be willing to pay a larger markup for the item with additional features, or quality.
- Likewise, the firm hopes that they can attract price sensitive consumers with a lower-quality product, without causing the price-insensitive consumers to switch.

Damaged Goods Strategy

An interesting type of versioning is known as the **damaged goods strategy**.

- A firm deliberately “damages” a product by removing features or adjusting performance and sells that good to the low-WTP consumers.
- Ironically, damaging the product may create an additional cost, so that the damaged good has a higher MC.

Lecture 13 - Oligopoly

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

So far we've discussed markets with either one firm or with a large number of firms.

- In this chapter, we will discuss markets with a small number of firms.
- We will look at a couple of different models of competition with two firms and discuss the resulting equilibria in each.

Market Structure

Below is an illustration of different market structures:

Product Differentiation	Number of Firms			
	Many	Few	One Dominant	One
<i>Firms produce identical products</i>	<i>Perfect competition</i> (Chapter 9) Example: fresh-cut rose market	<i>Homogeneous products oligopoly</i> Example: U.S. glass container market	<i>Dominant firm</i> Example: U.S. light bulb market	<i>Monopoly</i> (Chapter 11) Example: Internet domain name registration ^a
<i>Firm produce differentiated products</i>	<i>Monopolistic competition</i> Example: local physicians markets	<i>Differentiated products oligopoly</i> Example: breakfast cereal market	No applicable theory	

Market Structure

Aside from the two that we've discussed in depth, there are four more that we will talk about:

1. A **homogeneous products oligopoly market** is one in which there are few sellers, who all sell a similar product.
 - An example is the U.S. glass container industry. Three firms account for 82% of sales.
 - In the market for titanium dioxide, several large firms sell almost the exact same product.

2. In a **dominant firm market**, one large firm possesses a large share of the market but competes against many small firms.

- An example is the U.S market for light-bulbs.
- Many small firms compete, but GE alone counts for 50% of sales.

3. In **differentiated products oligopoly markets**, a small number of firms sell products that are substitutes, but differ in some way.
- Coke and Pepsi are a good example. These two firms dominate the market, but face competition from numerous smaller companies.
 - In the U.S. breakfast cereal market, Kellogg, General Mills, Post and Quaker Oats sell more than 85% of all cereal.
 - In Japan, Asahi, Kirin, Sapporo, and Suntory account for nearly 100% of Japanese beer sales.

4. Finally, **monopolistic competition** refers to a market in which many small firms produce differentiated products.
- Examples include dry cleaners, physicians, and restaurants.

Market Structure

Economists use various metrics to describe market structure. The **four-firm concentration ratio (4CR)** calculates the share of industry sales accounted for by the four largest firms.

- For example, if the four largest firms have 3%, 2%, 2% and 1% of sales, then the market would have a 4CR of 8.

Another metric is the HHI, which is the sum of square market shares for all firms.

- If one firm has all of the market share, then $HHI = 100^2 = 10,000$. This is the maximum possible value.
- On the other hand, if 100 identical firms each have 1% of market share, $HHI = 100 * (1^2) = 100$.

Cournot Oligopoly Model

An important feature of oligopoly models is competitive interdependence.

- The decisions of every firm significantly impact the profits of competitors.

One model of oligopoly is the **Cournot model**.

- We will consider two firms in the market, but it can be extended.

Cournot Oligopoly Model

Firms choose output quantity simultaneously and noncooperatively (without colluding). Once quantities are chosen, price instantly adjusts.

- Firm's output choice depends on market price, and market price depends on both firms' output levels.
- Firms don't know competitor's output choice, so they make their choice based on expectation.
- In other words, firm A makes output choice based on what it expects firm B to produce, and vice versa.

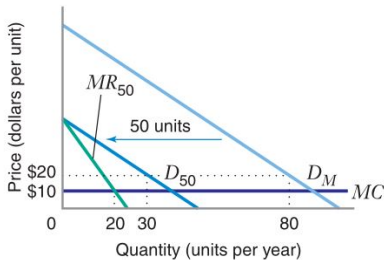
Cournot Oligopoly Model

A firm's **residual demand curve** is the demand curve that the firm faces, given its competitor's decision.

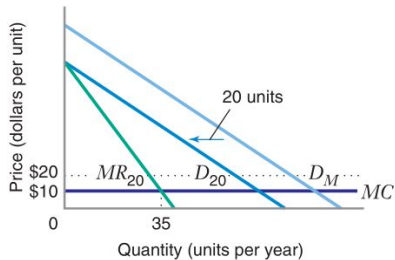
For example, suppose firms A and B face the demand curve $P = 100 - Q$, where $Q = Q_1 + Q_2$.

- If firm B decides to produce 50 units, for example, then firm A faces the residual demand curve $P = 100 - Q_1 - 50$ or $P = 50 - Q_1$.
- Likewise, if firm B decides to produce 20 units, then firm A faces the residual demand curve $P = 80 - Q_1$.

Cournot Oligopoly Model



(a) Samsung's profit-maximization problem when LG produces 50



(b) Samsung's profit-maximization problem when LG produces 20

Cournot Oligopoly Model

Each firm acts as a monopolist relative to its residual demand curve.

- In other words, it sets $MR(Q_i) = MC(Q_i)$, where MR is determined from their residual demand curve

The resulting optimal quantity will depend on the other firm's decision.

- For firm A, for example, we will find $Q_A^*(Q_B)$

These functions for the optimal quantity are the firm's **best response functions**, or **reaction functions**

- They indicate the *best response* of one firm to another firm's level of output.

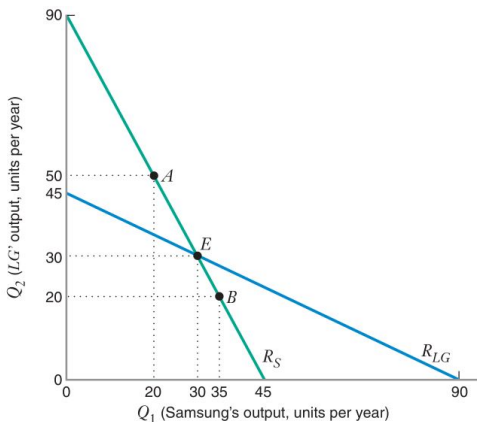
Cournot Oligopoly Model

The **Cournot Equilibrium** is the point where the two curves intersect:

- At this intersection, neither firm has an incentive to change their quantity.
- That's because their quantity is the *best response* to the other firm's output choice.

Cournot Oligopoly Model

Below are the best response functions of two firms in a duopoly. The intersection of their curves is the Cournot Equilibrium



Cournot Oligopoly Model

Try It Yourself

In the Cournot model, the firm chooses

- A) its optimal price, holding the price of its competitors constant.
- B) its best response to the price changes of the competitor firm.
- C) its optimal level of output, holding the output of the other firm constant.
- D) the level of output that would optimize profits for all firms.

Solution

In the Cournot model, firms compete by choosing **quantity** (not price), simultaneously and independently.

Each firm treats its rival's output as *fixed* when choosing its own profit-maximizing output – this is exactly what a “best response function” like $Q_A^*(Q_B)$ represents.

Options A and B describe price-setting behavior (that's the Bertrand model, not Cournot). Option D describes collusion/cartel behavior, which the Cournot model explicitly rules out (firms act noncooperatively).

Answer: C, its optimal level of output, holding the output of the other firm constant.

Cournot Oligopoly Model

Example: Suppose that market demand is given by $P = 100 - Q_1 - Q_2$, where Q_1 and Q_2 are the amount of output produced by firms 1 and 2, respectively. Marginal cost for each firm is \$10.

Firm 1's marginal revenue function can be found by taking the derivative of $P * Q$:

$$MR(Q_1) = \frac{\partial(100Q_1 - Q_1^2 - Q_1Q_2)}{\partial Q_1} = 100 - 2Q_1 - Q_2$$

Likewise for firm 2:

$$MR(Q_2) = \frac{\partial(100Q_1 - Q_1Q_2 - Q_2^2)}{\partial Q_2} = 100 - Q_1 - 2Q_2$$

Cournot Oligopoly Model

Firm 1's best response function is found by setting

$$MR(Q_1) = MC(Q_1):$$

$$100 - 2Q_1 - Q_2 = 10 \implies Q_1^* = 45 - \frac{1}{2}Q_2$$

Likewise for firm 2:

$$100 - Q_1 - 2Q_2 = 10 \implies Q_2^* = 45 - \frac{1}{2}Q_1$$

Cournot Oligopoly Model

To find the Cournot equilibrium, we simply solve the system of two equations. Let's plug firm 2's BR function in to firm 1's BR function.

$$Q_1 = 45 - \frac{1}{2}(45 - \frac{1}{2}Q_1) = 22.5 + \frac{1}{4}Q_1$$

Solving for Q_1^* :

$$\frac{3}{4}Q_1 = 22.5 \implies Q_1 = 30 \implies Q_2 = 45 - \frac{1}{2} * 30 = 30$$

Finally, plugging this into the demand curve gives us the price

$$P^* = 100 - Q_1 - Q_2 = 100 - 30 - 30 = 40.$$

Cournot Oligopoly Model

Note that the equilibrium price is above marginal cost, so this does not correspond to the perfectly competitive equilibrium. However, as the number of firms increases in this model, total profit falls.

Number of Firms	Price	Market Quantity	Per-Firm Profit	Total Profit
1 (monopoly)	\$55.0	45.0	\$2.025	\$2.025
2	\$40.0	60.0	\$ 900	\$1,800
3	\$32.5	67.5	\$ 506	\$1,519
5	\$25.0	75.0	\$ 225	\$1,125
10	\$18.2	81.8	\$ 67	\$ 669
100	\$10.9	89.1	<\$ 1	\$ 79
∞ (perfect competition)	\$10.0	90.0	0	0

Cournot Oligopoly Model

Try It Yourself

Suppose that firms A and B are Cournot duopolists in the salt industry. The market demand curve can be specified as $P = 200 - Q_A - Q_B$. The marginal cost to each firm is \$40. What is firm B's profit-maximizing quantity when firm A produces an arbitrary output Q_A ?

Solution

Treat Q_A as fixed.

$$TR_B = (200 - Q_A - Q_B)Q_B = 200Q_B - Q_AQ_B - Q_B^2$$

$$MR_B = \frac{\partial TR_B}{\partial Q_B} = 200 - Q_A - 2Q_B$$

Set $MR_B = MC = 40$:

$$200 - Q_A - 2Q_B = 40$$

$$Q_B^*(Q_A) = 80 - \frac{1}{2}Q_A$$

Best response function:

$$Q_B^*(Q_A) = 80 - \frac{1}{2}Q_A$$

Bertrand Model of Oligopoly

The **Bertrand model of oligopoly** assumes that firms each select a selling price.

- Consumers know both prices, and will purchase from whomever has the lowest price.
- Once a firm sets their price, they are able to supply as many units as consumers want.
- If one firm sets their price lower than the other, they will capture all of the demand in the market.

Bertrand Model of Oligopoly

A **Bertrand equilibrium** occurs when each firm chooses their profit-maximizing price, given the price set by the other firm.

Going back to our previous example, suppose two firms compete in a duopoly market. Both firms face a marginal cost of \$10.

- Suppose that both firms initially charge \$40 for a unit of the good.
- If firm A decides to undercut firm B by \$1, they will capture the whole market.

Bertrand Model of Oligopoly

But then firm B will have an incentive to lower their price.

- They would keep competing in this way until the price was equal to their marginal cost (it's turtles all the way down).
- Beyond that point, if they lowered their price more, profit would be negative.

The result is that both firms charge their marginal cost, and neither firm makes a profit.

- Compare this to the Cournot model, where the firms both charged a price above MC.

Comparing Bertrand and Cournot

Why are the results of these two oligopoly models so different?

- Cournot can be thought of as taking place over a longer time period.
- Firms choose their capacity and then compete as price setters.
- In contrast, Bertrand can be thought of as short-run, where both firms have enough capacity to satisfy market demand.

Comparing Bertrand and Cournot

Cournot oligopolists expect competitors to adjust prices immediately to keep sales constant.

- So one firm cannot quickly undercut the other to steal their sales.

In Bertrand, the opposite is true.

- Each firm knows it's competitors may undercut its price, so they behave more aggressively by keeping their price lower.